



BASE24 Release 6.0 Version 4 Implementation Guide

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Version Record

11/2003, Version 4.0 First Printing

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Preface

Document Control

A description of the document revisions is shown below. Any changes of text from one version to another are indicated with the symbol “|” in the outside margin.

Version 4.0 (D. Barton, J. Dargy, H. Earle, R. Krogstrand, C. McGeorge, C. Rink, A. Peek, R. Schieuer, T. Shah, P. Spanke, J. Steffes, J. Urbanec)-02/2003

Fourth Version

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Section 1

Introduction

The ***BASE24 Release 6.0 Version 4 Implementation Guide*** is the primary reference document for information needed to upgrade existing BASE24 Release 6.0 Version 3 systems to BASE24 Release 6.0 Version 4.

This section provides an overview of the planning requirements for BASE24 users migrating to release 6.0 version 4. It also provides guidance for using this manual.

Also included in this section is a listing of other documents that a BASE24 user should obtain for implementation planning, the general software requirements for release 6.0 version 4, and a glossary of terms used throughout the guide.

Implementation Planning

BASE24 users must develop their own project plans for migrating their BASE24 systems to release 6.0 version 4. Project plans should address such issues as staffing and training requirements, host external message changes, device programming enhancements, hardware and software requirements, and the migration method to be used.

ACI has provided BASE24 users with the ***BASE24 Release 6.0 Version 4 Implementation Guide*** to assist in preparing project plans. This guide includes a list of services available to the BASE24 user and information that existing BASE24 users will need to upgrade to release 6.0 version 4.

BASE24 Release 6.0 products include the following:

- BASE24-atm Release 6.0
- BASE24-pos Release 6.0
- BASE24-teller Release 6.0
- BASE24 Kernel/IEP Release 6.0
- BASE24-telebanking Release 6.0
- BASE24-billpay Release 6.0
- BASE24-customer service Release 6.0
- BASE24 Interchange Interfaces Release 6.0
- BASE24 Value-Added Products:
 - BASE24-Infobase
 - BASE24 Retail Products
 - BASE24-card
 - ACI Host Link
 - ACI Plastic Card Control System
 - ACI Claims Manager
 - ACI Proactive Risk Manager
 - ACI Settlement Manager
 - ACI Payments Manager

Who should use this guide

There are two audiences for this guide: those who plan and manage the migration process and those who implement the migration to release 6.0 version 4. In either case, the reader should be experienced with BASE24 software, HP NSK products, and the BASE24 user's current configuration. This guide is not directed at new BASE24 users who will be installing a BASE24 application for the first time. New users should refer to the formal BASE24 publications listed under the Publications heading in Section 3.

To migrate to release 6.0 version 4, users must be on release 6.0 version 3. Refer to the ***BASE24 Release 6.0 Implementation Guide (Version 3)*** to assist in planning the tasks necessary to get to release 6.0 version 3.

Using this guide, you can do the following

- Determine what ACI services are available to assist in the migration to release 6.0 version 4.
- Develop a project plan to migrate to release 6.0 version 4.
- Determine staffing and training needs.
- Gain a detailed understanding of the release 6.0 version 4 enhancements.

Software Requirements

HP NSK Software Requirements

ACI recommends that the BASE24 user be on a current HP NSK-supported operating system to obtain maximum performance and support. However, BASE24 release 6.0 is currently compatible with the HP NSK Guardian D45 Operating System. Customers choosing to implement Format 2 File support are required to be on a minimum Operating System level of D46. In addition, BASE24 release 6.0 requires TMF for several files.

XPNET

ACI requires that the BASE24 user be on the XPNET 3.0 system.

Conversion Planning

BASE24 users must develop their own project plans for upgrading their systems to release 6.0 version 4. Project plans should address such issues as staffing and training requirements, host external message changes, device programming enhancements, hardware and software requirements, and the migration method to be used.

In general, all changes for release 6.0 version 4 can be obtained by requesting the latest fixes from ACI and then retrofitting all fixes to the user's release 6.0 version 3 modules.

Conversion Program Guidelines

Users of previous releases of BASE24 and previous versions of BASE24 Release 6.0 must run file conversion programs in order to migrate to the latest version of release 6.0. Information about specific conversion programs can be found in subsequent sections of this document under the applicable enhancements. Following are general conversion program guidelines.

The *UC04 subvolumes (e.g., AT60UC04, BA60UC04, PS60UC04) contain the conversion programs for users migrating from release 5.3 to the latest version of release 6.0. The conversion programs on these subvolumes are kept up-to-date with each new version of release 6.0.

The *UC05 subvolumes (e.g., AT60UC05, BA60UC05, PS60UC05) contain the conversion programs for users migrating from release 6.0 version 1 to release 6.0 version 2. No changes are made to these programs other than necessary fixes. Also residing on these subvolumes are the Transaction Security Services (TSS) conversion programs, which are used to move key data to the new TSS database. The TSS conversion programs are required only if the user is implementing TSS.

The *UC06 subvolumes (e.g., AT60UC06, BA60UC06, PS60UC06) contain the conversion program for users migrating from release 6.0 version 2 to release 6.0 version 3. No changes are made to these programs other than necessary fixes.

The *UC07 subvolumes (e.g., AT60UC07, BA60UC07, PS60UC07) contain the conversion program for users migrating from release 6.0 version 3 to release 6.0 version 4. No changes are made to these programs other than necessary fixes.

If a release 5.3 user wants to convert to the latest version of release 6.0, the conversion programs on *UC04 should be run. If the user is implementing TSS, the TSS-specific conversion programs on *UC05 should also be run.

If a release 6.0 version 1 user wants to convert to release 6.0 version 2, the conversion programs on *UC05 should be run. If the user is implementing TSS, the TSS-specific conversion programs on *UC05 should also be run.

If a release 6.0 version 2 user wants to convert to release 6.0 version 3, the conversion programs on *UC06 should be run. If the user is implementing TSS, the TSS-specific conversion programs on *UC05 should also be run.

If a release 6.0 version 3 user wants to convert to release 6.0 version 4, the conversion programs on *UC07 should be run.

If a release 6.0 version 1 user wants to convert to release 6.0 version 4, the conversion programs on *UC05 should be run. If the user is implementing TSS, the TSS-specific conversion programs on *UC05 should also be run. Following this, the

user should run the conversion programs on *UC06. Finally, the user should run the conversion programs on *UC07.

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Other Documents

In addition to this guide, the reader can obtain other BASE24 and NET24-XPNET publications that cover operational and product-specific information required to fully understand and implement the release 6.0 version 4 software. Such manuals include the formal publications mentioned under the Publications heading in Section 3, the event message documentation located on the BA60LOGM and OK60LOGM subvolumes, and the following documents:

BASE24 Classic 6.0 Installation Guide (Informal) - BA-AE000-60

A primary reference document for information about the installation of a BASE24 release 6.0 system from scratch. This document is included as a Microsoft Word document and is shipped on the BASE24 6.0 CD-ROM.

The BASE24 Classic 6.0 Installation guide is also provided on the software tape with the code. The document is located on the BA60UD04 subvolume as a file called BASE24. On this subvolume there is also a readme file called AAREAD04, which details how to move the install document to your hard drive and open it with Microsoft Word.

BASE24 Release 6.0 Interchange Interface Implementation Guide (Informal) - SW-DS0000-60

Describes the function and purpose of ACI's Switch 6.0 interfaces. It explains how customers can implement Switch 6.0 interfaces in their existing networks as well as in a BASE24 6.0 network. It also explains how customers can implement their Fortify.1/Fortify.3 Interfaces in a 6.0 network. This document is intended to assist customers in configuring and controlling their Switch 6.0 Interchange Interfaces in all BASE24 release networks and their Fortify.1/Fortify.3 Interchange Interfaces in a BASE24 release 6.0 network. This document is included as a HP NSK TGAL document and is shipped with the release 6.0 Interchange Interface software. This document is named NSTLSW60 and is located on the SW60UD04 subvolume.

Note: This guide is to be used with BASE24 releases 4.0, 5.0, 5.1.2, 5.3, and 6.0.

User Bulletin for BASE24 Release 6.0 Version 4 (Informal)

A primary reference for information about all changes made to each BASE24 Release 6.0 Version 4 source module. This documentation is included as a set of HP NSK TGAL documents and is shipped with BASE24 Release 6.0 Version 4 software. This documentation consists of a number of files located on the BA60UB06 subvolume. All files ending in FH contain fix history sections for all changes made. All files ending in FP contain a list of all fixed Procedures for each change made to all modules, except those files that have been resequenced.

Release 6.0 Data File Impacts Document (Informal)

A primary reference document for information about the data file impacts and the methods that can be used to upgrade to release 6.0 version 4. This document is included as a HP NSK TGAL document and is shipped with the release 6.0 version 4 software. This document is named AAREAD06 and is located on the BA60UC06 subvolume.

Definition of Terminology

The following list has been defined to assist the reader in understanding the terms referred to within this document.

6.0 Interchange Interfaces

The BASE24 Interchange Interfaces developed in conjunction with release 6.0. These replace the Fortify.3 Interchange Interfaces and can be used with releases 4.0, 5.0, 5.1, 5.1.2, 5.3 and 6.0 of BASE24.

Accountholder

A person for whom a financial account has been established for some purpose. Accountholders may or may not use cards to initiate transactions on their accounts.

BASE24 User

An entity that has licensed BASE24 software products.

Cardholder

A person who has been issued a plastic card for the purpose of initiating transactions.

Consumer

A person who utilizes services offered by a BASE24 user who may or may not require a plastic card.

Conversion Program

A utility that reads the records from an input file, adds new fields and initializes appropriate fields and then writes the record to an output file. The input file (existing file) and output file (converted/initialized file) are different files. This method (versus the initialization program and online conversion) is required when the record structure changes (e.g., increasing the record length, adding/deleting fields from segments, etc.). This method can also be used to initialize existing user fields in a record.

This method requires system down time to convert the file (also requires an operator to manually rename the file that has been converted), move in the new objects of modules that access the file (e.g., device handler, auth, etc.) and then restart the objects.

This method impacts all BASE24 users who use the file being converted even if they are not taking advantage of the enhancement for which the modifications are being made.

Dynamic Key Management (DKM)

The automated distribution of encryption keys between BASE24 and an outside entity.

Extended Memory Table file (EMT)

A data file produced by the EMT Build utility program. Once the EMT has been built with the EMT Build utility, processes that use the EMT must be warmbooted (or started if not running).

Extended Memory Table Build utility (EMT Build)

The EMT Build utility is a stand-alone program used to build tables in extended memory from information contained in disk files. This utility can build hash tables, radix tables, or custom designed tables. Refer to the ***BASE24-atm Transaction Processing Manual***, ***BASE24-pos Transaction Processing Manual***, or the ***BASE24 ITS Kernel Configuration Manual*** for additional information about this utility.

Online (Inline) Conversion

A utility that reads and updates each record in a file to initialize user fields that have been redefined as new fields for use by an enhancement. The file is converted in place (i.e., a new file is not created). The record is read, updated and then rewritten to the same file. This method (versus the conversion program and online conversion) cannot be used when the record structure changes (e.g., increasing the record length, adding/deleting fields from segments, etc.).

The online conversion code will be executed each time a record is accessed (versus the conversion and initialization programs that are run once). When a record is read, the process will determine if the value of a new field is valid. If the value is not valid, the field will be initialized to the default value. The database is gradually converted as records are read and updated.

This method requires an operator to stop a process, roll in the new object and start processes that use the files that are impacted. This method does not require any additional steps to be performed by an operator (e.g., running conversion/initialization programs, renaming files).

Operator

A person who carries out network operations functions, such as starting and stopping the system, entering ONCF commands, controlling logging, dealing with event messages, performing refreshes and extracts. The term *operator* also can refer to persons using BASE24 screens for such purposes as file maintenance, perusal, etc.

Section 2

Functionality

This section contains an overview of the BASE24 Release 6.0 Version 4 enhancements.

Functionality has been organized by product, and within each product, according to whether the enhancement is a general enhancement or an optional enhancement.

- General enhancements are those enhancements that are inherent in the release 6.0 version 4 offerings. These enhancements affect all BASE24 users. BASE24 users must review the features that are part of these enhancements to ensure an understanding of the new processing and ensure proper configuration within the BASE24 environment.
- Optional enhancements are those enhancements that can be enabled or disabled in release 6.0 version 4. BASE24 users can implement these enhancements based on their own BASE24 processing and marketing requirements.

Enhancements are described in this section by product as follows:

- BASE24 Base (applicable to all products)
- BASE24-atm
- BASE24-pos
- BASE24-teller
- BASE24 Kernel/IEP (applicable to all ITS based products)
- BASE24-telebanking
- BASE24-billpay
- BASE24 Interchange Interfaces
- BASE24 Value-Added Products
 - BASE24-InfoBase
 - BASE24 Retail Products
 - BASE24-check auth
 - BASE24-frequent shopper
 - BASE24-refunds
 - BASE24-card
 - ACI Host Link
 - ACI Plastic Card Control System

- ACI Claims Manager
- ACI Proactive Risk Manager
- ACI Settlement Manager
- ACI Payments Manager

BASE24 users should acquaint themselves with the BASE24 Base section (Section 4) as well as the applicable product-specific sections in order to obtain an understanding of the overall release 6.0 version 4 offering.

BASE24 Base Enhancements

The BASE24 Base product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24 Base product, refer to Section 4.

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General Enhancements

The following enhancements are common to all products or provide the base for product-specific enhancements. As these enhancements are inherent in release 6.0 version 4, BASE24 users must review them to ensure an understanding of the new processing and proper configuration within the BASE24 environment.

General Changes

- Changes for all BASE24 Release 6.0 Version 4 enhancements affecting the DDL files and the BASE24 AFT Subsystem files (i.e., screens, requesters, servers) have been merged and cataloged under one SCN for each affected file. Applying changes for individual enhancements to the screen files does not allow BASE24 users to identify which changes go with a particular enhancement, as screen files do not have sequence numbers at the end of each line. Since the screen file changes usually involve new fields being added, all associated requester, server and DDL changes are required. Even though requester, server and DDL files do contain sequence numbers, all SCNs would need to be applied in order to accept the changes to the screen files. Therefore, all requester, server and DDL file changes were done under one SCN per file as well.
- Re-architecture changes were done to minimize MAKE dependencies for some BASE modules.
- Changed the following SCOBOL and COBOL modules for RMS compatibility. This was accomplished by changing the COPY statements where the file's define name started after column 35. The file's define name was moved to the next line.
 - CPF Requester/Server
 - KEYD Server
- Implemented new Make flags for preparation of the next version of BASE24.
- A new token was added for the PRM Release 7.0 product.
- The Card Prefix File (CPF) has been modified for the Self-Service Banking Check Enhancements to include four new fields in the CPF SSBC segment. All BASE24-atm users who implement the Self-Service Banking (SSB) Check add-on product, regardless of whether the version 4 Self-Service Banking Check Enhancements is being implemented, are required to run a CPF Conversion program due to the increased size of the CPF SSBC segment.

Thales Key Encryption Scheme

Thales HSM commands support double or triple keys encrypted using either the ANSI X9.17 key scheme or the variant key scheme. An alphabetic “tag” immediately preceding the key value itself specifies the scheme used to encrypt the key. BASE24 supports several Thales HSM commands in which double or triple length keys are passed. In some of these commands (typically older commands), BASE24 passes no tag at all (defaulting to the ANSI X9.17 method), and in other commands (typically newer commands), BASE24 passes a hard-coded “U” (the variant method).

When invoking the Thales commands in the following table, logic has been enhanced to ensure that all double length keys are immediately preceded with the character “U”.

Command	Function
CI	Translate a PIN from BDK Encryption to Interchange Key Encryption
KQ	ARQC (or TC/AAC) Verification and/or ARPC Generation
KU	Generate Secure Message with Integrity and Optional Confidentiality

To allow existing customers to migrate their key database from the ANSI X9.17 scheme to the new variant method, conversion programs will be provided. These programs will convert the keys specified in the table below from encryption using the ANSI X9.17 scheme to encryption using the variant scheme (using the Thales “B0” command).

File/Screen	Key Name	Key Usage
KEYD 1	Derivation Key	Translate a PIN from BDK Encryption to Interchange Key Encryption
KEYI 1	Cryptographic Keys	ARQC Verification and/or ARPC Generation
KEYI 1	Secure Messaging Keys (for Integrity)	Generate Secure Message with Integrity

This enhancement will resolve the inconsistencies surrounding the encryption of keys used by BASE24 in Thales HSM commands.

Optional Enhancements

The following BASE24 Base Release 6.0 Version 4 enhancements are optional. BASE24 users should review these changes to ensure an understanding of the new processing and proper configuration within the BASE24 environment before deciding to implement any of the changes.

Dual Site Processing

This enhancement is to support customers who are planning to implement a Dual Site production environment using GoldenGate to facilitate data replication between the two sites.

A Dual Site BASE24 production environment is where BASE24 is running on two separate NSK machines, with production transactions being processed on both machines. Standard BASE24 files will be held on both systems with updates replicated between the two systems by GoldenGate.

The TLF and PTLF files have been enhanced to enable the identification of individual records as “local” or “remote”, i.e. whether the transaction was authorized on this system or the alternate system. Records for all transactions will appear on both copies of the files, with one copy containing the remote token according to the location of the original transaction. As records on either system are added the log files, GoldenGate will replicate the records to the alternate system, add the token if not already present, and update the field.

Extract and Refresh have been enhanced to provide options relating to Dual Site Processing. Each module can be configured to process local only, remote only or all records in extracting or impacting respectively. Extract and Refresh have also been enhanced to support process specific TLF / PTLF assigns. Refresh has additionally been enhanced to support broadcast capability in notify lists.

For more details on using these enhancements and GoldenGate’s support of a dual site production environment, refer to the BASE24 Dual Site Processing Guide.

EMV Downgrade Enhancement

The EMV Downgrade Enhancement involves changes related to processing of EMV transactions as standard Magnetic Stripe transactions. This enhancement is to support issuers who have issued EMV cards but cannot yet support EMV transactions.

A ‘Downgrade’ transaction is one where an EMV card has been used at an EMV enabled terminal, but the transaction is processed as a magnetic stripe transaction.

The Card Prefix File and EMV Status Token have been enhanced to accept a new value for EMV processing, to specifically identify the “downgrade” situation

EMV M/Chip 4 Support

The EMV M/Chip 4 Support enhancement involves the changes required to support MasterCard's M/Chip Select 4 and M/Chip Lite 4 EMV applications.

BASE24 will have new fields at the card level and new values for fields at the card and card prefix level to identify an EMV M/Chip 4 card, and will have modified token data fields to accommodate new M/Chip 4 card data.

Migration of cards from M/Chip 2.1 to M/Chip4 will be supported by the ability to specify M/Chip 4 support at the card level.

Enhanced Expiration Date Checking

BASE24 is being enhanced to allow issuers to configure additional expiration date checks at the CPF level. The new expiration checks allow you to:

- Require an expiration date in the authorization message
- Set the maximum number of years allowed for an expiration date
- Compare the expiration date in the request message to the expiration date on the CAF record
- Set expiration date checking for recurring payments (POS only) independent of other transactions

Additionally, access to second card data fields on the CAF has been added to the CAF AFT screens. Note: Refresh and FHM were previously enhanced to allow update of the second card data on the CAF.

TSS Key Index Support

This enhancement allows BASE24 to specify an exact key be used by TSS in any encryption/decryption request. The key is identified to TSS by BASE24 sending a key index in TSS requests. This enhancement is required for Canadian BASE24 users to meet Interac's key management approach.

Changes were made to the Interac Interchange Interface, ATM Auth, Router/Auth, HISO, SPDH, HISWUTLS, SECUTILS and token processing modules to support the new TSS Key Index token. When the token is present in a transaction and these modules have been updated to support this enhancement, the key index from the token will be sent to TSS for encryption/decryption tasks.

BASE24-atm Enhancements

The BASE24-atm product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24-atm product, refer to Section 5.

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General Enhancements

The following BASE24-atm enhancements are inherent in the release 6.0 version 4 offerings. These enhancements must be reviewed by all BASE24-atm users to ensure an understanding of the new processing and proper configuration within the BASE24 environment.

There are no general enhancements inherent to BASE24-atm.

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Optional Enhancements

The following BASE24-atm Release 6.0 Version 4 enhancements are optional. BASE24-atm users should review these changes to ensure an understanding of the new processing and proper configuration within the BASE24 environment before deciding to implement any of the changes.

Acquirer Surcharging

BASE24-atm Surcharging has been enhanced to offer the following features:

- Card Prefix Level Configuration
- Minimum Transaction Amount
- Expanded Transaction Set
- Logging of Cardholder Declined Surcharges

Configuration at the card prefix level allows BASE24-atm Surcharging customers to refine their definition of cardholders and transactions that are eligible for surcharge assessments. Prior to this enhancement, configuration could be specified only at the card type level.

The introduction of a minimum transaction amount for which surcharge assessments can be applied brings BASE24 into compliance with interchange regulations. These regulations stipulate that cardholders cannot be assessed a surcharge if the transaction amount is less than a predefined amount.

The expanded transaction set now eligible for surcharges includes Deposit, Deposit with Cash Back, and Split Deposit. Prior to this enhancement, surcharge fees could only be assessed for withdrawal type transactions (i.e., Withdrawal, Fast Cash, Cash Advance). There are remaining transactions not eligible for surcharge assessments and these transactions include Administrative, Balance Inquiry, Check-Related, Log-Only, Payment, PIN Change, Transfers, and Statement Print.

Surcharge fees can now be assessed by transaction subtype in addition to transaction code. Transaction subtypes enable the BASE24 customer to distinguish between types of transactions that possess the identical transaction codes. Using transaction subtype, the BASE24 customer can configure multiple rate structures for a single transaction code.

Logging of transactions whereby the cardholder declines the surcharge is introduced with this enhancement. The intent is to provide details regarding the transaction, which might aid in determining whether terminals should be configured for surcharging.

Automated Key Distribution System

The Diebold 10XX/478X device handler has been enhanced to support the Automated Key Distribution System (AKDS) enhancement. AKDS provides a means of securely downloading a unique master key (A-key) from BASE24 to an ATM using the Rivest-Shamir-Adleman (RSA) public key method.

RSA is a highly secure cryptography method that utilizes public/private key pairs. The public key is transmitted between sender and recipient. The private key is kept secret by its owner and is never transmitted. There is a mathematical relationship between the public and private key, but it is virtually impossible to derive one from the other. These key pairs are used for data encryption. Data is encrypted under the recipient's public key, which can only be decrypted using the recipient's private key. These key pairs are also used for mutual authentication. A digital signature is created for the encrypted data using the sender's private key. The signature can only be authenticated using the sender's public key. (A signature is an encrypted hash of the data, used by the recipient to authenticate the identity of the sender and to verify that the original content of the data has not been altered.)

A BASE24 system has one public/private key pair, which is generated by an attached host security module (HSM) during the initial setup of AKDS and stored in the Transaction Security Services database. The BASE24 system's public key is submitted to the Certificate Authority (CA) to be verified and signed. Diebold uses a third-party CA, Digital Signature Trust Co. (DST). The Certificate Authority returns an X.509-format certificate containing the BASE24 system's public key and a signature of the public key. It also returns a certificate containing the CA's own public key. These certificates are entered into the Transaction Security Services database.

Each ATM has its own public/private key pair, which is generated and stored in the ATM's Encrypting PIN Pad (EPP) at the time of manufacture. Also stored in the EPP is the ATM's public key signature (provided by the Certificate Authority) and the CA's own public key. The public keys and signatures are contained within X.509-format certificates.

Once the initial keys and signatures are established in BASE24 and the ATM, an online dialogue can proceed. An operator at the Device Control Terminal (DCT) executes a LOAD PUBLIC KEY command, initiating an Authentication and Public Key Exchange sequence (alternatively, this command can be executed via NCS/NCPCOM). During this load sequence, BASE24 and the specified ATM authenticate each other and exchange public keys. These public keys are used for encryption and mutual authentication in all future master key downloads. This command is typically performed only once – when a new ATM with an AKDS-enabled EPP is initially deployed, or when an ATM already in the field is fitted with a new AKDS-enabled EPP.

After BASE24 and the ATM have authenticated each other and exchanged their

public keys, the master key can be downloaded to the ATM. An operator at the DCT executes a LOAD MASTER KEY command, initiating a Master Key Download sequence (alternatively, this command can be executed via NCS/NCPCOM). During this load sequence, BASE24 generates a new master key and downloads it to the specified ATM. This command can be performed as often as is necessary to preserve the integrity of the ATM's keys. In most cases, this is done relatively infrequently. After this command completes successfully, the DES encryption keys (e.g. PIN Encryption Key, MAC Key) may be generated and downloaded to the ATM, as per normal procedures prior to this enhancement.

Prerequisite: The Transaction Security Services enhancement must be installed prior to implementation of the Automated Key Distribution System. For more information, refer to the ***BASE24 Release 6.0 Implementation Guide (Version 2)***.

Diebold TCS Enhancements Support

The BASE24-atm Diebold 10XX/478X device handler has been enhanced to support the following TCS 4.5 – 5.3 Enhancements:

- The Double Expanded Hardware Configuration Message bytes 20 and 21. By using this information the device handler can limit the number of bills it can dispense based upon configuration information received from the ATM rather than having a hard coded limit for all devices/terminal types. (Requires TCS 5.0)
- The printing of a scanned check image to a thermal receipt printer. (Requires TCS 4.5)
- The printing of a scanned check image to a thermal statement printer. (Requires TCS 5.3)
- The printing of a scanned check amount to a thermal statement printer. (Requires TCS 5.3)
- The selection of a specific character set for printing. (Requires TCS 5.3)
- The displaying to EMS the contents of the MDS Status field when it is received in Solicited and Unsolicited Status messages.
- IDMI (Requires 5.0)
- IDMI (Requires TCS 5.3)

The BASE24-atm Diebold Entry Program has been enhanced to support the following TCS 4.5 – 5.3 Enhancements:

- The new Enhanced Set Language Bank State (@Y) (Requires TCS 4.5)
- The new Buffer Shift State (@N) (Requires TCS 5.2)

- The Copy Buffer State (@P) (Requires TCS 5.2)
- New values to the Card Reader Enhanced State (CR) to support Dip Smart Card Readers (Requires TCS 5.0)
- A new value to the Voice State (;) to control the beeping/voice coordination. (Requires TCS 5.3)
- The printing of a check image to a 8.5" thermal statement printer. (Requires TCS 5.3)
- The printing of a check amount to a 8.5" thermal statement printer. (Requires TCS 5.3)
- The addition of 251, 252, and 253 as valid state numbers. (Requires TCS 5.3)
- The sending of the MDS Status field in Solicited and Unsolicited Status messages.

Dual Site Support

This enhancement is to support customers who are planning to implement a Dual Site production environment using GoldenGate to facilitate data replication between the two sites.

A Dual Site BASE24 production environment is where BASE24 is running on two separate NSK machines, with production transactions being processed on both machines. Standard BASE24 files will be held on both systems with updates replicated between the two systems by GoldenGate.

The ATD file has been enhanced to enable the identification of individual terminals as "local" or "remote", i.e. whether the terminal is normally connected to this system or the alternate system. (Records for each terminal will appear on both copies of the file, with one set to local and one to remote according to the normal connectivity of the terminal.) As records on either system are updated, GoldenGate will replicate the changes to the alternate system and alter the setting of the local / remote flag.

The TLF has been enhanced to enable the identification of individual records as "local" or "remote", i.e. whether the transaction was authorized on this system or the alternate system. Local records are standard and remote records include a new field, SITE^IND, in the Multiple Logical Network token. (Records for all transactions will appear on both copies of the files, with one copy containing the remote token according to the location of the original transaction.) As records on either system are added, GoldenGate will replicate the records to the alternate system, add the token if not already present, and update the field.

Settlement, Perusal, ATD file access and DCT have been enhanced to support and provide options relating to Dual Site Support. Settlement can be configured to

process local only, remote only or all records and terminals during cutover processing and has been enhanced to support broadcast capability in notify lists and process specific TLF assign. TLF perusal has been enhanced to enable the identification of remote (replicated) transactions on the local log file. ATD file access supports a new field, DUAL^SITE^IND, indicating the normal connectivity of the terminal in question, and DCT enables transparent communication with an individual terminal regardless of its current connectivity.

For more details on using these enhancements and GoldenGate to support a Dual Site production environment, refer to the ***BASE24 Dual Site Processing Guide***.

Dynamic Key Management

Dynamic Key Management was initially implemented in BASE24-atm as an Americas Channel CSE for the Canadian market. With release 6.0 version 4, this functionality is available in global product. In Canada, this enhancement was known as Application Interface Layer (AIL). Generally speaking, this enhancement enables BASE24-atm to dynamically change the Personal Identification Number (PIN) and Message Authentication Code (MAC) keys based on transaction count, security error threshold, or by a key change interval.

In BASE24-atm a PIN or MAC key is changed when:

- The number of PIN verify or MAC verify/generate transactions exceeds a predetermined limit;
- The number of PIN verify or MAC verify/generate failures exceeds a predetermined limit;
- The number of MAC consecutive failures exceeds a predetermined limit.
- Twenty-four hours has passed since the last MAC or PIN key change.

Once any of the limits has been reached, the process to load new keys to the ATM is initiated. To minimize the time an ATM is down, if one key reload is required, both the PIN and the MAC key are reloaded, all of the counters are reset, and the key change timestamp is updated.

In BASE24-atm Release 6.0 Version 4, this enhancement was added to the NCR 5XXX Device Handler, the Diebold 10XX/478X Device Handler, and the Triton Device Handler.

Prerequisite: The ATD Infrastructure enhancement must be installed prior to implementing the Dynamic Key Management enhancement in the Diebold 10XX/478X and NCR 5XXX Device Handlers.

EMV M/Chip 4 Support

The EMV M/Chip 4 Support enhancement involves the changes required to support MasterCard's M/Chip Select 4 and M/Chip Lite 4 EMV applications.

BASE24-atm EMV processing has been enhanced to accommodate new Issuer Application Format and M/Chip 4 Status Field Checking processing for M/Chip 4 EMV cards.

Forced Cutover Audit

This billable enhancement provides an audit record when a forced cutover occurs at a Diebold 10xx/478x or NCR 5xxx device. At the time of forced cutover, the ATM will print a subtotals receipt on the audit/journal printer. This receipt will contain the same information as a receipt printed for the card balancing procedure.

Fujitsu 7000 Device Handler

BASE24-atm release 6.0 has been enhanced to support Fujitsu 7000 ATMs. The BASE24-atm Fujitsu 7000 Device Handler module provides the interface between the device and HP NSK. The Fujitsu 7000 ATMs are defined in the BASE24-atm Terminal Data File (TDF). This device handler was first introduced in BASE24-atm Release 5.1.

Fujitsu 7000 Triple DES

The standard Fujitsu 7000 device handler BASE24-atm module has been modified to support Triple DES encryption.

Prerequisite: The Transaction Security Services enhancement must be installed prior to implementation of the Fujitsu Triple DES enhancement. For more information refer to the ***BASE24 Release 6.0 Implementation Guide (Version 2)***.

Interactive Response Notification Fee Processing and Negative Surcharge Support

The BASE24-atm Tidel Device Handler has now been enhanced to support Interactive Response Notification Fee Processing and Negative Surcharge Support.

BASE24-atm fee processing has been enhanced to support interactive response notification of fees at an ATM. This differs from the original interactive **request** notification surcharge processing, in that the original interactive request notification surcharge processing performed an interactive request prior to the transaction going to the issuer. The cardholder account was impacted after the cardholder was notified of, and accepted, the fee.

The interactive **response** notification of fees, however, routes the transactions for authorization without notifying the cardholder first. The cardholder account is impacted before the cardholder accepts the fee, and the transaction may need to be reversed if the cardholder does not accept the fee displayed at the ATM.

The interactive response notification of fees can include both issuer fees and acquirer fees (i.e., surcharges). BASE24 maintains information in the Surcharge File (SURF) for only acquirer fees, and does not maintain information in its database on issuer fees. The setting of issuer fees must be done at a host system or an interchange whose interchange interface on BASE24 has been enhanced to carry the issuer fee information in a new BASE24 Issuer Fee/Rebate token.

Standard product interactive request notification surcharge processing has been enhanced to allow a negative fee amount to provide a cardholder with an incentive to use the ATM device. This type of incentive is also available with the interactive response notification fee processing.

This enhancement is compliant with the UK LINK Switch Interchange Standard (LIS5), Version 5.1, and can be used by the following UK product module:

- UK LINK ISO Interchange (LIS5) Interface

Multiple Currency Dispense Exchange Rate Display

The Multiple Currency Dispense enhancement allows cardholders to select the currency to dispense cash in for standard BASE24-atm cash withdrawal transactions (withdrawal from checking account, withdrawal from savings account, withdrawal from credit account). A customer may configure from two to four different currencies that a cardholder can select to withdraw cash in, but only a single currency may be selected and dispensed per withdrawal transaction. This enhancement was added in Release 6.0 Version 3. The MCD Exchange Rate Notification enhancement expands the Multiple Currency Dispense logic to display the exchange rate to the cardholder who can accept or deny the transaction based on the information presented.

The MCD Exchange Rate Notification enhancement is supported only in the Diebold 10XX/478X and NCR 5XXX device.

Prerequisite: The Multiple Currency Dispense and ATD Infrastructure enhancements must be installed prior to implementation of the MCD Exchange Rate Notification enhancement. For more information on these previous enhancements refer to the ***BASE24 Release 6.0 Implementation Guide (Version 3)***.

NCR EMV Dialup

The BASE24-atm NCR 5xxx Device Handler has been enhanced to support EMV transactions from an NCR 5xxx ATM connected to BASE24 directly via a dial-up connection.

Prerequisite: The NCR 5XXX Device Handler EMV and NCR 5xxx Device Handler Dial-up Support enhancements. For more information on these previous enhancements refer to the ***BASE24 Release 6.0 Implementation Guide (Version 2)***.

Self-Service Banking Check Enhancements

The BASE24-atm Self-Service Banking product currently supports two sets of limits and percentages for deposit credit allowed with deposit transactions for three types of media. Envelopes and checks use one set of limits/percentages and an additional set of limits/percentages. This enhancement allows checks, registered checks, and cash to have a unique set of limits/percentages for deposit credit. The following new fields are being added to the existing SSBC CPF segment for Registered and Non-Registered Check Deposits:

- Check Deposit Credit Percentage and Maximum Credit per Check Deposit for the non-registered check deposits
- Registered Check Deposit Credit Percentage and Maximum Credit per Registered Check Deposit for the registered check deposits.

Triton Triple DES

The Triton device handler has been upgraded to support triple DES. Additionally, the Triton device handler now supports single and triple DES MACing. This enhancement requires BASE24-es TSS release 03.1or greater.

Prerequisite: The Transaction Security Services enhancement must be installed prior to implementation of the Triton Triple DES enhancement. For more information refer to the ***BASE24 Release 6.0 Implementation Guide (Version 2)***.

BASE24-pos Enhancements

The BASE24-pos product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24-pos product, refer to Section 6.

ACI Worldwide Inc.

General Enhancements

The following BASE24-pos enhancements are inherent in the release 6.0 version 4 offerings. These enhancements must be reviewed by all BASE24-pos users to ensure an understanding of the new processing and proper configuration within the BASE24 environment.

General Changes

- Support was added to the ACI Standard POS Device Handler (SPDH) module for the ISO 87 standard for the ISO Response Code. This includes formatting FID 'X' in a response message to the device and mapping a BASE24 response code to an ISO response code. The module previously only supported the ISO 93 Standard for the ISO response code. A new LCONF param was added to control which ISO response code format is support for the device handler process.
- The ACI Standard POS Device Handler (SPDH) was enhanced to map an ISO response code to a BASE24 response code based on account type.

Enhanced Approval/Referral Code Generation

BASE24-pos is being modified to enhance its approval/referral code generation. Now, when an approval code generated results in all zeroes a new one will be generated until it is not all zeroes. If a referral code generated results in a duplicate key to the PRF, a new referral code will be generated until a unique record can be added to the PRF.

General CRT Authorization Enhancements

The BASE24-pos CRT Authorization module has been enhanced to display a new Fraud Prone Outlet Flag on the PRF screen, and will format and add a new date and time token when a reversal is generated from the OLT screen. Issuer Interface processes can utilize the reversal information in setting fields in reversals to switch networks.

SPDH Additional Data Fields

In recent years, a number of new data fields were introduced to BASE24-pos to support industry growth and development, new products, and new processes. For the most part these data fields were added to various tokens appended to the BASE24-pos Standard Internal Message (PSTM). However, their use was not extended to the ACI Standard POS Device Handler (SPDH). This enhancement adds support for a number of these data fields to the SPDH and the BASE24-pos file maintenance system.

Prerequisite: Users will need to modify their SPDH firmware to utilize these additional data fields in SPDH message processing.

Optional Enhancements

The following BASE24-pos Release 6.0 Version 4 enhancements are optional. BASE24-pos users should review these changes to ensure an understanding of the new processing and proper configuration within the BASE24 environment before deciding to implement any of the changes.

Configurable Approval Code Length

This enhancement allows the generation of an approval code with the approval code length being the lesser of the value sent from the terminal and the value specified at the card prefix level. The approval code length must be a value between 2 and 6 inclusive.

BASE24-pos Router/Authorization processing has been enhanced to compare the approval code length sent from the terminal (in the PSTM) to the length of the approval code configured in the CPF record. The length of the approval code generated will be the lesser of the two values.

Dual Site Support

This enhancement is to support customers who are planning to implement a Dual Site production environment using GoldenGate to facilitate data replication between the two sites.

A Dual Site BASE24 production environment is where BASE24 is running on two separate NSK machines, with production transactions being processed on both machines. Standard BASE24 files will be held on both systems with updates replicated between the two systems by GoldenGate.

The PTD file has been enhanced to enable the identification of individual terminals as “local” or “remote”, i.e. whether the terminal is normally connected to this system or the alternate system. (Records for each terminal will appear on both copies of the file, with one set to local and one to remote according to the normal connectivity of the terminal.) As records on either system are updated, GoldenGate will replicate the changes to the alternate system and alter the setting of the local / remote flag.

The PTLF has been enhanced to enable the identification of individual records as “local” or “remote”, i.e. whether the transaction was authorized on this system or the alternate system. Local records are standard and remote records include a new field, SITE^IND, in the Multiple Logical Network token. (Records for all transactions will appear on both copies of the files, with one copy containing the remote token according to the location of the original transaction.) As records on either system are added, GoldenGate will replicate the records to the alternate system, add the token if not already present, and update the field.

Settlement, Perusal and PTD file access have been enhanced to support and provide options relating to Dual Site Support. Settlement can be configured to process local only, remote only or all records and terminals during cutover processing and has been enhanced to support broadcast capability in notify lists and module specific PTLF assign. PTLF perusal has been enhanced to enable the identification of remote (replicated) transactions on the local log file. PTD file access supports a new field, DUAL^SITE^IND, indicating the normal connectivity of the terminal in question.

For more details on using these enhancements and GoldenGate to support a Dual Site production environment, refer to the ***BASE24 Dual Site Processing Guide***.

DUKPT MACing

The BASE24-pos SPDH has been enhanced to support DUKPT MACing for transactions originating at the terminals directly connected to BASE24-pos.

Using the DUKPT method, each terminal security module (TSM), or PIN pad, uses a derived unique key for each transaction. However, the TSM never contains any information that could lead to the determination of any past MAC keys used by the device, or of any past or future keys used by any other TSM. The originating TSM must derive the current MAC key from an initial key, loaded into the TSM on initialization. The receiving BASE24-pos security module (Atalla or Thales) then determines the current MAC key by using a key held on BASE24-pos, and non-secret information contained in the transaction message.

The Standard POS Device Handler (SPDH) message definition can optionally include FID 6 sub-FID T, which is used to transmit the Key Serial Number (KSN) and KSN descriptor from the terminal to BASE24. This KSN is used for PIN verification using DUKPT but it also is used for DUKPT MACing.

Prerequisites: The Transaction Security Services release 04.1, Derived Unique Key Per Transaction (DUKPT) and SPDH DUKPT Support via TSS enhancements must be installed prior to implementation of the DUKPT MACing enhancement. For more information on the Transaction Security Services and DUKPT enhancements refer to the ***BASE24 Release 6.0 Implementation Guide (Version 2)***. For more information on the SPDH DUKPT Support via TSS enhancement refer to the SPDH DUKPT Support via TSS section in this document.

EMV M/Chip 4 Support Enhancement

The EMV M/Chip 4 Support enhancement involves the changes required to support MasterCard's M/Chip Select 4 and M/Chip Lite 4 EMV applications.

BASE24-pos EMV processing has been enhanced to accommodate new Issuer Application Format and M/Chip 4 Status Field Checking processing for M/Chip 4 EMV cards.

Optional CRT Authorization Enhancements

BASE24-pos CRT Authorization has been enhanced to support Purchase with Cash Back transactions, with the exception of Adjustment and Force Post Adjustment transactions. CRT Auth can now also allow entering of an Override Approval Code in the POS Referral File (PRF).

POS Balance Inquiry Service

Visa supports prepaid cards for various uses such as gift, mall, payroll and incentive cards. Unlike Visa debit and credit cards, the cardholder needs to know how many dollars are available after a purchase. Visa mandated support of returning balances to the merchant for all prepaid cards effective October 2003. This enhancement allows an issuer to return balance information to the network.

Configuration at the card prefix level will allow BASE24 issuers to specify whether balances are to be returned on the purchase transaction. Balance information, using the POS Balance Token (P-63), can be returned on the following POS transactions whether transactions are approved or denied:

- Purchase
- Purchase with Cash Back
- Pre-Auth Purchase
- Balance Inquiry

The SPDH device handler will also be enhanced to allow balance information to be returned to the merchant using FID J.

Secure Internet Validation

The BASE24-pos Secure Internet Validation module provides product support for today's most current e-commerce consumer payment validation schemes. It allows issuers to seamlessly integrate BASE24 with internet authentication product offerings from the ACI Secure Commerce Suite. Alternatively, the Secure Internet Validation module also provides a BASE24 message interface for third party e-commerce authentication vendors. The Secure Internet Validation module supports the following authentication schemes:

- Visa's 3-D Secure CAVV; marketed to consumers and merchants as Verified by Visa
- MasterCard's SecureCode Universal Cardholder Authentication Field (UCAF)
- Virtual Primary Account Numbers (VPANs)

MasterCard and Visa have instituted new consumer authentication programs to secure public network (Internet and mobile) electronic commerce transactions. Visa's

3-D Secure technology and MasterCard's three SecureCode UCAF payment schemes are approved authentication methods for validating the identity of cardholders during the online purchase process.

The ACI Secure Commerce Suite provides issuers with solutions for consumer enrollment and registration, Verified by Visa and/or SecureCode cardholder authentication, and the generation of Cardholder Authentication Validation Value (CAVV) and UCAF authentication values. The Secure Internet Validation module provides a message interface to obtain the CAVV, the UCAF, and additional merchant and consumer information from the Secure Commerce Suite. The Secure Internet Validation module also provides the ability to validate this authentication data prior to authorizing the financial transaction. Note that the preauthorization processing for a lesser amount is not supported in the Secure Internet Validation module.

Virtual Primary Account Numbers (VPANs) protect both issuers and their cardholders by using a proxy credit card number that is linked to the real card number. The ACI Secure Commerce Suite contains products, which generate and provide VPANs to authenticated consumers. Using the same message interface and processing logic developed for support of SecureCode and Verified by Visa, the Secure Internet Validation module provides the ability to validate the virtual card number while maintaining a single set of limits for the real card number.

SecureCode, Verified by Visa, and VPANs have been introduced to reduce risk/fraud, decrease chargebacks, and increase consumer confidence with Internet purchases. The following is additional information on Secure Internet Validation support of Visa 3-D Secure, MasterCard SecureCode, and VPANs.

Visa 3-D Secure

BASE24-pos can participate in the initial consumer enrollment/registration authentication via a card verification transaction and the subsequent financial transactions (i.e., purchase, preauthorization request, or preauthorization completion transaction). The ACI Secure Commerce Suite generates the CAVV and optional TransStain number during the consumer authentication request from the consumer. These new data elements are sent to BASE24-pos immediately via the ISO Host acquiring interface using a card verification message. The necessary card specific and 3-D Secure data will be placed in a new file called the E-Commerce Authentication File (EAF) for later validation during authorization. 3-D Secure's CAVV can optionally be calculated from the information included in the authorization message and does not have to be provided to BASE24-pos for comparison of the e-commerce data using the EAF.

If the customer is making a 3-D Secure Internet purchase, the merchant will have captured all of the data necessary to complete the transaction during the authentication transaction. The merchant then sends the authorization request with

the necessary e-commerce data to the acquirer. The acquirer sends the purchase/pre-authorization purchase transaction through the VisaNet network using the VisaNet ISO or Visa DPS ISO Interface for issuer validation and authorization.

BASE24-pos Router/Authorization interrogates the authorization message to determine if the transaction originated as an e-commerce transaction (i.e., PS51TKN.E-COM-FLG), the CAVV is present (i.e., TRANS-STAIN-XID-TKN, token ID equals 'C6'), and the CAVV should be verified based on the new fields in the CPF (i.e., CPF.POS.CAVV-AVV-DISP-FLD, CPF.POS.CAVV-AVV-CHK). If the data indicates an e-commerce transaction, Visa has not already verified the CAVV, and the CAVV should be verified by BASE24, Router/Authorization must either verify the CAVV using the saved value in the EAF or using the CVV keys from the KEYA and data from the message by making a call to the Security Module. If the CAVV is validated, Router/Authorization will set the CAVV/AAV status field in the PS51 Token accordingly. Note that a new Secure Internet Validation (SIV) KEYA group has been added to the CPF for this authentication method. If the CAVV cannot be validated, Router/Authorization determines whether to deny the transaction or continue transaction processing based on a new field in the CPF. In either case, the CAVV/AAV status field in the PS51 Token is set accordingly.

MasterCard SecureCode

If the customer is making a SecureCode Internet purchase, the ACI Secure Commerce Suite assigns the Accountholder Authentication Value (AAV) during the purchase request from the consumer. This e-commerce data is sent to BASE24-pos immediately via the acquiring ISO Host Interface using a card verification message. The necessary card specific and SecureCode data is placed in the EAF for later validation during authorization. The merchant sends the authorization request with the necessary e-commerce data to their acquirer. The acquirer sends the purchase/pre-authorization purchase transaction through the MasterCard network using the Banknet ISO or MMDS ISO Interface for issuer validation and authorization.

The BASE24-pos Router/Authorization process interrogates the authorization message to determine if the transaction originated as an e-commerce transaction (i.e., PS51TKN.E-COM-FLG), the AAV is present (i.e., AUTHN-DATA-TKN, token ID equals 'CE'), and the AVV should be verified. Validation of the AAV is performed based on the new fields in the CPF (i.e., CPF.POS.CAVV-AVV-DISP-FLD, CPF.POS.CAVV-AVV-CHK). If the data indicates an e-commerce transaction and the AVV should be verified by BASE24, the Router/Authorization process attempts to locate the transaction in the EAF file. If the AAV is validated (using a compare against the EAF data), BASE24 sets the CAVV/AAV status field in the PS51 Token. The AAV result is not passed back to MasterCard, but for trouble shooting the issuer could use it. The transaction is marked as used in the EAF provided it is not a split payment transaction. If the transaction is a split payment, the transaction is marked in the EAF to allow further authorizations to complete with the same UCAF. If the AAV cannot be validated, Router/Authorization determines whether to deny the

transaction or continue transaction processing based on a new field in the CPF. In either case, the CAVV/AAV status field in the PS51 Token is set accordingly.

Virtual Primary Account Numbers

If the customer is making a Virtual PAN (VPAN) internet purchase, transaction processing is similar to MasterCard SecureCode. The consumer requests a VPAN from the ACI Secure Commerce Suite, which is then used by the consumer in a subsequent internet purchase. The ACI Secure Commerce Suite product sends a card verification request containing the normal verification information with the real card number (i.e., real PAN), the VPAN and its expiration date to BASE24-pos for later validation during authorization. The merchant passes the authorization request using the VPAN with the necessary e-commerce data to their acquirer. The acquirer sends the transaction through the appropriate network for issuer validation and authorization. This data is passed in the purchase/pre-authorization purchase transaction to be verified by the issuer institution.

SPDH DUKPT Support via TSS

The SPDH supports DUKPT for the encryption of PIN blocks for transactions originating at the terminals directly connected to BASE24-pos. DUKPT support was added in Release 6.0 Version 2. This enhancement expands support for DUKPT in three ways:

- through Transaction Security Services
- using double-length or triple-length keys
- performing additional security checks - comparing the KSN IDs and transaction counters from two consecutive transactions.

Prerequisites: The Derived Unique Key Per Transaction (DUKPT) and Transaction Security Services enhancements must be installed prior to implementation of the SPDH DUKPT Support via TSS enhancement. For more information on these previous enhancements refer to the ***BASE24 Release 6.0 Implementation Guide (Version 2)***.

SPDH EMV Additional Support

The SPDH EMV Additional Support enhancement involves changes related to processing of:

- Authorization Response Codes
- Pre Auth Request Data
- CVM Results Data
- Additional Discretionary Data Elements

BASE24 SPDH has been enhanced so that a separate Authorization Response Code (BASE24 generated) is always sent to the Terminal in extended sub-FID 6Q.

At present EMV pre-authorization transactions include a cryptogram, which is calculated using the amount field plus other EMV Request Data fields sent in the request message. This cryptogram is verified successfully within BASE24. The completion message may include the original cryptogram, but if the amount sent in the message is different (in most cases they would be different) from the original pre-authorization message, the cryptogram verification fails. BASE24 SPDH has been enhanced so that the transaction currency code and amount used to create the cryptogram are new mandatory fields in pre-authorization and completion request messages received from the Terminal. This will allow BASE24 to include the cryptogram in both the request messages sent to issuers. The new fields, cryptogram amount and cryptogram currency code should be included in the extended subFID 6O sent from the terminal to the SPDH Device Handler.

The CVM Results field is not one of the 13 mandatory data elements in EMV request messages. The advent of offline PIN processing means that for these transactions, acquirers and issuers cannot determine whether offline PIN verification was performed (and successful) unless the CVM Results data element is included in the POS transaction request. BASE24 SPDH has been enhanced so that the CVM Results data element (6 bytes) can be included in the extended subFID 6P from the terminal to the SPDH. (The BASE24-pos APACS 30/40 Device Handler has also been enhanced to add CVM results data to the EMV Status token.)

Within BASE24 the EMV Discretionary Token contains a number of fields that cannot be used for messages received in the SPDH. BASE24 SPDH has been enhanced so that the Application Version Number and Dedicated File Name which are fields in EMV Discretionary Token, can be included in the extended subFID 6P from the terminal.

SPDH Enhanced Multi-Threading

The ACI Standard POS Device Handler (SPDH) currently supports multi-threaded transaction processing. That is, the SPDH is able to process transactions from multiple terminals concurrently. The terminals may be stand-alone or may belong to a group of terminals at a controller-based site. A controller-based site has multiple terminals connected to a single controller, with the controller managing the message traffic and communication with the SPDH. Examples of controller-based sites are multi-lane department stores and pay-at-the-pump gas stations.

The SPDH Enhanced Multi-Threading enhancement supplements the multi-threaded transaction processing capabilities of the SPDH with regard to controller-based sites. Currently, in order for the SPDH to process transactions from a controller-based site in a multi-threaded manner, one record must be defined in the BASE24-pos Terminal Data File (PTD) for each terminal at the site. This requirement is still in effect with this enhancement. However, this enhancement adds three new functions to the PTD file maintenance screen to make the management of these records somewhat easier. This enhancement also adds the ability to maintain aggregate transaction totals for

the site.

PTD File Maintenance Screen Changes

Two new fields, Site ID and Record Type, have been added to PTD file maintenance screen 1. Site ID contains a user-defined identifier for the site, used to group the PTD records of the terminals at the site. Record Type further defines the site's PTD records as Primary (P) or Secondary (S). If this enhancement is not implemented, these fields will be set to blanks.

A site can have only one primary PTD record. The primary record does not define an actual terminal. Rather, it serves two purposes:

- The primary record is used to update "inheritable" fields in all of the site's secondary PTD records. Inheritable fields are the user-modifiable fields in the PTD Static (PTDS1) file, with the exception of Terminal ID and Terminal Type. With the stroke of a function key, the contents of these inheritable fields can be copied from the primary record to every secondary record with the same Site ID.
- The primary record is also used to accumulate the site's aggregate transaction totals. These totals include the service, batch, shift, daily and current network totals.

A site can have multiple secondary PTD records. The secondary records define the physical terminals at a site. The SPDH uses the secondary records to process transactions just as it did prior to this enhancement.

The new function keys added to the PTD file maintenance screen are as follows:

- SF2 – Read Site Primary Record: This function reads a site's primary record. Enter a valid value in Site ID. When SF2 is pressed, the site's primary record (Record Type = "P") is read.
- SF5 – Update Site Secondary Records: This function updates all of a site's secondary records with inheritable data from the site's primary record. First, read the site's primary record (SF2). If you wish to modify any fields in the primary record, you must make those changes and update (F5) the primary record before you can use SF5 to update the secondary records. When SF5 is pressed, every secondary record (Record Type = "S") with the same Site ID is updated with inheritable data from the primary record. Inheritable data is the user-modifiable data in the PTD Static (PTDS1) file, with the exception of Terminal ID and Terminal Type. Dynamic data (PTDD1) is not inheritable.

Note: In a secondary record, any inheritable field that contains unique data will be overwritten and will have to be manually reset to its unique value, if so desired.

- SF6 – Read Next Record by Site ID: This function reads the next record by Site ID. When SF6 is pressed, the next record is read according to the Site ID.

Site's Aggregate Transaction Totals

During the processing of a financial transaction, the SPDH currently updates the terminal's transaction totals in the terminal's PTD record. With this enhancement, after the terminal's transaction totals are updated, the SPDH tests the value of Record Type in the terminal's PTD record.

If the Record Type is "S" (Secondary), the SPDH assumes this terminal is one of a group of terminals at a site. Using the Site ID from the terminal's PTD record and a Record Type of "P" (Primary), the SPDH reads the site's primary PTD record. The SPDH then increments (or decrements, if this is a reversal) the service, batch, shift, daily and current network totals in the site's primary PTD record. The aggregate transaction totals in this primary PTD record are incremented or decremented for every financial transaction from every terminal with the same Site ID and a Record Type of "S".

If the Record Type is blank (N/A), the SPDH does not attempt to update the aggregate transaction totals for a site. Processing continues as normal.

Triple DES for BASE24 SPDH

The Triple DES for BASE24 SPDH enhancement modified the BASE24-pos SPDH module to support Transaction Security Services (TSS) and triple DES encryption/decryption.

Prerequisite: The Transaction Security Services enhancement must be installed prior to implementation of the Triple DES for BASE24 SPDH enhancement. For more information refer to the ***BASE24 Release 6.0 Implementation Guide (Version 2)***.

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BASE24-teller Enhancements

The BASE24-teller product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24-teller product, refer to Section 7.

ACI Worldwide Inc.

General Enhancements

There are no general enhancements inherent to this product.

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Optional Enhancements

There are no optional enhancements inherent to this product.

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BASE24 Kernel/IEP Enhancements

The BASE24 ITS Kernel/IEP product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24 ITS Kernel/IEP product, refer to Section 8.

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General Enhancements

There are no general enhancements inherent to this product.

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Optional Enhancements

There are no optional enhancements inherent to this product.

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BASE24-telebanking Enhancements

The BASE24-telebanking product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24-telebanking product, refer to Section 9.

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General Enhancements

There are no general enhancements inherent to this product.

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Optional Enhancements

There are no optional enhancements inherent to this product.

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BASE24-billpay Enhancements

The BASE24-billpay product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24-billpay product, refer to Section 10.

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General Enhancements

There are no general enhancements inherent to this product.

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Optional Enhancements

There are no optional enhancements inherent to this product.

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BASE24-customer service Enhancements

The BASE24-customer service product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24-customer service product, refer to Section 11.

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General Enhancements

There are no general enhancements inherent to this product.

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Optional Enhancements

There are no optional enhancements inherent to this product.

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BASE24 Interchange Interface Enhancements

All BASE24 Interchange Interface product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24 Interchange Interfaces, refer to Section 12.

ACI Worldwide Inc.

General Enhancements

There are no general enhancements inherent to this product.

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Optional Enhancements

The following BASE24 Interchange Interface enhancements are optional. BASE24 Interchange Interface users converting to release 6.0 version 4 should review these changes to ensure an understanding of the new processing and proper configuration within the BASE24 environment before deciding to implement any of the changes.

- BIC ISO Interchange Interface Enhancements

BASE24 Interchange Interface users should refer to the documentation provided for each Interchange Interface for implementation instructions. This documentation can be found on the release 6.0 Interchange Interface subvolume and the Interchange Interface add-on product subvolume with a file name of 'XXXXMAN', where 'XXXX' is the Interchange Interface acronym name. All BASE24 Release 6.0 Interchange Interfaces that have been enhanced with the described general or optional enhancements are delivered on 'SW60XXXX' subvolumes. A formal manual is available on the ACI CD-ROM for the BIC ISO Interchange Interface, Europay EPS-Net Interchange Interface, LINK (LIS5) Interchange Interface and Switch Authorization Network Interface (SWANI).

BIC ISO Interchange Interface Enhancements

In the Republic of Ireland, electronic transactions (both ATM and POS) are passed between the acquiring and issuing banks using a message format based closely on the ISO 8583 (1987) Standard. The format is very similar to the BASE24 external message, and the BASE24 users amongst the Irish banks use the BIC ISO Interchange Interface module to support this message processing.

The following enhancements have been made to the BIC ISO Interchange Interface to support the Irish processing requirements, e.g. for the Irish "Chip and PIN" initiative. Although the changes have been applied predominantly to support the requirements of the Irish market, other BASE24 users of the BIC ISO Interchange Interface can take advantage of the enhanced functionality.

- Enhance response code mappings in BIC ISO for EMV-specific failure conditions.
- Allow BASE24-originated reconciliation messages to be logged to the Interchange Log File (ILF) by BIC ISO.
- Increase flexibility in the setting of institution ids in BIC ISO.
- Rationalize currency conversion processing in BIC ISO.

These enhancements are described in detail below.

New unique ISO response codes (from the range of ISO values "reserved for private use") will be used to identify specific EMV failure conditions, e.g. bad ARQC. The change will be configurable, because this enhancement will be controlled by a new LCONF parameter, and the processing can therefore be disabled.

The BIC ISO reconciliation message processing will be enhanced to write reconciliation totals to the ILF under additional conditions currently allowed. The impact of this enhancement on existing BASE24 customers will be minimal, i.e. a maximum of two additional records per day written to the ILF, so the implementation of this functionality will not be configurable.

In the 1987 version of the ISO 8583 Standard on which the BIC ISO message protocol is based), there are effectively three institution ids defined - the acquiring institution id, the forwarding institution id, and the receiving institution id. In the BIC ISO interface, these ids are mapped between the Standard External Message (SEM) and the BASE24-atm Standard Internal Message (STM) or the BASE24-pos Standard Internal Message (PSTM).

Some banks are members of different national card schemes, therefore require unique values for these institution ids depending on the cards being processed. BIC ISO will now utilize overrides when present for one or more of these institution ids. The change will be configurable, because this enhancement will be controlled by the new LCONF parameter, and the processing can therefore be disabled.

The BIC ISO module may perform currency conversion on amount fields in messages received from an interchange and/or messages sent to an interchange. This processing is performed only if the BASE24 Multiple Currencies product is not configured, but some of this currency conversion logic is incomplete, because it is not possible to pass certain amounts in the BASE24 Standard External Message (SEM) in more than one currency. Changes to the SEM layout to include additional amount fields will ensure that full currency data can be passed between BASE24 and an interchange. The change will be configurable, because this enhancement will be controlled by a new LCONF parameter, and the processing can therefore be disabled.

BASE24 Value-Added Products Enhancements

There are no enhancements inherent to these products.

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Section 3

Support Services

This section summarizes the various support services available from ACI. These support services are described as follows:

- Product consulting and project management
- Publications
- Education
- HELP24
- ASTech services

Other services, besides the traditional Installation service, such as go-live support, onsite conversion assistance, and software methodology and utility training are available or are being planned; please contact your ACI Customer Manager for more details of the services ACI can offer.

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Product Consulting and Project Management

No formal service programs have been created by ACI for upgrading customers to BASE24 Release 6.0 Version 4. Customers need to review this entire document and any referenced manuals, to determine what they will need to do to upgrade their BASE24 system.

After reviewing, customers should contact their ACI Customer Manager to discuss any services, such as project management, custom code upgrades, onsite staff, etc., required for upgrading, after reviewing all materials.

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Project Planning Guide

The following Project Planning Guide is designed to assist in the successful upgrade of a BASE24 system to release 6.0 version 4. Information contained in this guide provides insight into required planning and control of the entire project.

Following this project guide helps ensure the realization of the goals and business objectives of the BASE24 user.

1. IDENTIFY PROJECT TEAM

One of the initial tasks for the BASE24 upgrade is for the user to assign a project team who will be responsible for the success of the project. A project manager and team members will need to be assigned and specific responsibilities established.

2. DETERMINE HARDWARE REQUIREMENTS

An ACI sales representative can assist in determining whether additional hardware will be required for the BASE24 upgrade. It is the BASE24 user's responsibility to order the hardware and establish the delivery date with HP NSK. HP NSK will perform diagnostic testing on the hardware when it is installed at the site. Should hardware encryption be required, it will be necessary to determine the hardware requirements. Additional hardware should be ordered for installation before testing can occur.

If applicable, the BASE24 user should order any new device firmware/software required to implement the enhancements they have chosen to support. The delivery date for the device firmware/software should be agreed upon with the vendor, as this time frame will affect the entire project.

3. IDENTIFY COMMUNICATION REQUIREMENTS

If the BASE24 user has identified additional protocol requirements, the communications lines and modems should be ordered. The customer must coordinate this effort with the local telephone company to ensure that the lines are configured in the least costly manner. The lines and modems must be installed and tested prior to the scheduled software installation date.

4. UPDATE HP NSK SOFTWARE REQUIREMENTS

ACI recommends that the BASE24 user be on a current HP NSK supported operating system to obtain maximum performance and support. BASE24 Release 6.0 is currently compatible with the HP NSK Guardian D45 Operating System. Customers choosing to implement Format 2 File support are required to be on a minimum Operating System level of D46. In addition, BASE24 Release 6.0 requires TMF for several files.

5. UPGRADE TO XPNET 3.0

Customers should note that XPNET 3.0 is the minimum requirement for BASE24 Release 6.0.

6. PROJECT SCOPE MEETING

ACI does not believe this step is necessary for customers upgrading to BASE24 Release 6.0 Version 4.

7. MIGRATION PLANNING

Users will need to use this document, and any references within it, for detailed analysis when creating their migration plan.

8. DEVELOP PROJECT SCOPE DOCUMENT

ACI does not believe this step is necessary for customers upgrading to BASE24 Release 6.0 Version 4.

9. DEFINE REQUIREMENTS DEFINITION

The customer will be required to review existing CSMs to determine which ones will require retrofitting to the new version. With ACI's Fix Release methodology, BASE24 users should evaluate their CSMs to determine if they are handled as standard product in release 6.0 version 4 or if they are specific to the customer's environment and will need to be upgraded to the new version.

10. DEVELOP PROJECT PLAN

The customer's project manager is responsible for developing an internal project plan. A review of the ACI project guidelines will help this person identify job responsibilities and establish time frames. A tracking mechanism should be included in the plan to ensure that the project is proceeding on schedule.

11. IDENTIFY CUSTOMER SPECIFIC MODIFICATION DEVELOPMENT

If the customer elects to have ACI develop modifications identified as required for the new version, a request will be submitted to ACI Software Development for a cost estimate. The functional descriptions and estimates will be sent to the customer for review and approval. Resources will be assigned and the final delivery dates established upon receipt of the signed contract.

Should the customer decide to develop any modifications in-house, resources will need to be assigned and scheduled. Once delivery dates are established, impacts to system testing can be determined.

12. DEFINE CERTIFICATION REQUIREMENTS

Customers currently on Fortify.3 will need to verify with each of the interchanges they connect to, to determine if they need to re-certify with the Interchanges with release 6.0 version 4 code.

If re-certification is required, preliminary test times should be determined and a copy of the interchange test script secured.

The customer needs to set up test time with the interchange well in advance in order to insure time frames are met.

13. ESTABLISH TEST FACILITY

It is recommended that an area be designated specifically for testing the BASE24 Release 6.0 Version 4 software. The following items are recommended for the test area:

- 3 PCs
- Test devices
- Hard copy printer
- Test cards

14. INSTALL PRODUCTION HARDWARE

Installation of any additional computer hardware or peripherals is generally performed by the local hardware service organization or customer staff.

15. ATTEND BASE24 EDUCATION CLASSES

ACI offers a full suite of instructor led courses for BASE24. An enhancement course is also available upon request. The course descriptions can be found in the BASE24 Education Catalog available in hardcopy through your ACI Customer Manager or viewed on the ACI Worldwide web site at www.aciworldwide.com.

16. DEVELOP INTERNAL PROCEDURES

There are no new internal procedures that ACI is aware of for BASE24 Release 6.0 Version 4 enhancements. User/Operational impacts are noted in the detailed sections for each enhancement described within this document.

17. DEVELOP SYSTEM TEST PLAN

The customer is responsible for developing a system test plan. Included in the test plan should be a description of each step specifying the test card to be used, the account to be used, the transaction to be executed, and the expected results. A procedure for retaining the results of the transaction processing should also be

specified. Controlled testing and result verification procedures are key elements for successful system testing.

The execution of the system test plan will occur over a period of weeks to fully test all aspects of the system. BASE24 product functionality should be the first application tested. Any system modifications should then be thoroughly tested. The interface between the host and BASE24 should also be tested. This includes testing on-line authorization and the refresh and extract programs.

18. DEVELOP HOST SOFTWARE

During the course of a business day, BASE24 logs all completed and declined transactions to a transaction log file. After network settlement, the transaction log file can be extracted to tape and used as input into the customer's nightly processing cycle on the host. If the customer would like to take advantage of the new features in release 6.0 version 4, a host program will need to be developed to interpret the new token data that can be extracted with release 6.0 version 4.

After the nightly processing cycle on the host or periodically during the business day the host may need to update card and account files used by BASE24 for pre-authorization or stand-in processing. If the customer would like to take advantage of the new features in release 6.0 version 4 in these files, a host program will need to be developed to initialize the new data fields in BASE24 files.

19. INSTALL BASE24

Release 6.0 version 4 software will need to be installed at the customer site. A tape containing the licensed release 6.0 version 4 software modules will be sent to the customer upon request for installation on their system.

20. CONVERT DATA FILES

ACI provides database conversion programs for selected BASE24 files. These files are discussed in the detail sections for each enhancement described within this document.

21. CERTIFY INTERCHANGE

This task involves implementing the interchange test script and receiving notice of certification from the interchange. Current BASE24 users should not have to re-certify their Interchange Interfaces, unless they are implementing new functionality not previously tested.

22. PERFORM SYSTEM TESTING

Full business day cycle testing of the system as determined by the customer testing plan and procedures.

23. DEVELOP CONVERSION PLAN

The customer should develop an implementation plan for converting existing devices and interchange links to release 6.0 version 4. The installation of new devices and interchanges should also be included in this plan. A method of informing users of any changes to existing procedures should also be developed. The conversion schedule should be routed to all departments.

24. GO-LIVE

At the customer's request, an operational support analyst will visit the site one day prior to the "go-live" date to review the current software status and the BASE24 facilities. This individual will remain onsite to assist the customer with production cutover. The migration plan developed by the customer will be utilized in production cutover. On-site go-live support is a billable service.

25. COMPLETE POST IMPLEMENTATION REVIEW

In general, the purpose of this meeting is to ensure overall satisfaction of the customer's project.

26. TURN OVER TO CUSTOMER MANAGEMENT

If an ACI Project Manager has handled the BASE24 upgrade, a formal turnover meeting is held between the ACI Project Manager and the ACI Customer Manager to ensure a smooth transition in turnover of responsibilities for on-going customer support.

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Publications

The BASE24 formal publications set has been updated to include the release 6.0 version 4 enhancements. The updated BASE24 documentation is available via CD-ROM.

The BASE24 6.0 CD-ROM also contains the previous release (5.3/1.3) documentation. The 5.3/1.3 version of a manual can be considered the most current if the manual is not listed under the 6.0 documentation menus. The XPNET 3.0 documentation is available on a separate CD-ROM titled NET24-XPNET.

BASE24 customers can order a copy of the BASE24 6.0 CD-ROM and/or XPNET 3.0 CD-ROM from their ACI Customer Manager.

ACI has discontinued hard copy distribution of manuals that are available on CD-ROM. Hard copies, however, can be printed at your site from the CD-ROM as necessary and in accordance with your license agreement.

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Education

BASE24 Technical Education offers a complete set of one- to four-day modules covering the components of the BASE24 system. BASE24 courses are structured into a modular format that eliminates repetitious material, provides information specific to certain job functions, offers hands-on lab experience, and enhances retention of information taught.

A BASE24 Enhancement course is also available upon request for existing customers upgrading to BASE24 6.0 version 4.

The modular format enables BASE24 users to choose modules applicable to their employees' job responsibilities. This provides consolidated training time and offers opportunities for additional employees to receive education.

For a current education schedule, see the BASE24 Education Catalog available in hardcopy through your ACI Customer Manager or on the ACI Worldwide web site at [\\www.aciworldwide.com](http://www.aciworldwide.com).

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HELP24

Users migrating to BASE24 Release 6.0 Version 4 should use their existing procedures to report problems or ask questions. HELP24 will continue to use the same response and tracking methodology as they currently do today.

All HELP24 staff have received training in all methods of release upgrades. In addition, there is a complete library of migration documentation available for reference in quickly resolving upgrade issues.

HELP24 has implemented Remote Diagnostics or Access capabilities for assistance in troubleshooting complex problems. The service has grown because of the obvious benefits it offers. Information about Remote Diagnostics or Access is available from HELP24 or your ACI Customer Manager. HELP24 strongly recommends that customers start planning to take advantage of this service now, so it will be available and ready in a crisis situation.

Access to HELP24's bulletin board (in the EMEA this is provided through a web site) is still available as an additional method for communication of important support information. BASE24 users are assigned individual logons to use this facility and are able to peruse the information 24 hours a day. The system can be used to transfer files such as fix files, log files, audit files, and dump files to help expedite problem resolution.

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ASTech Services

ACI offers customer support services through our ASTech Division which allows the selection of a la carte support services to meet individual customer requirements. ASTech, ACI Support and Technical Services, is a service support team which provides the customer with a variety of on-site and remote technical support. ASTech provides customers with supplemental staffing, system performance analysis, review and/or upgrading of previous release customer-specific software development, and detailed operational technical training.

The following are example descriptions of the various services ASTech could undertake to assist you during the BASE24 upgrade. The number of staff weeks identified to accomplish these tasks are estimates and will vary depending on the amount of customer specific code to be developed and delivered by ACI.

Your ACI Customer Manager can provide you with details of services available in your region that have been tailored to the special needs of different areas of the world.

Customer Software Analysis

Effort: Two to four staff weeks

ASTech personnel will perform a complete review of the customer's BASE24 software system and determine what current modifications (CSM code) exist at the customer location. ASTech personnel will perform a detailed comparison of existing customer specific software modifications to release 6.0 version 4 functions. Upon completion of the review, the customer will be provided with a document that identifies all modules that contain code unique to their BASE24 system (file names, custom functions, custom function applicability, etc.).

Conversion Planning

Effort: One and one-half to three staff weeks depending on the size and current system configuration.

This service option provides the customer with comprehensive support during the system migration to the new version. ASTech assigns the customer a Technical Project Lead, or Tech Lead from ACI. The Tech Lead will be responsible for ensuring the success of the BASE24 upgrade project.

The Tech Lead will:

- Review the migration options and develop a detailed migration plan for the conversions project
- Review the Project Scope Document on behalf of the customer (if applicable)
- Review the customer's internal procedures to accommodate release 6.0 version 4 functions.

Upgrade Custom Code

Effort: The amount of time necessary for this task varies depending on the number of modules identified as having modifications unique to the customer. Systems with a great number of customizations may take up to sixteen staff weeks. Actual estimates would be provided upon completion of the Customer Software Analysis.

Prerequisite: The Customer Software Analysis to identify all code unique to the customer.

With this service, ASTech will upgrade all custom code to release 6.0 version 4. This includes unit testing of the code. Customizations that are no longer necessary will be removed. Customizations that must be carried forward can either be upgraded in their current form or be reengineered to take advantage of new functionality in release 6.0 version 4. Reengineering customizations will add time to the estimate.

System Test Planning

Effort: One to four staff weeks depending on the results of the Customer Software Analysis and the number of BASE24 products to be tested. Some of this time will be on-site to perform the System Test Plan.

Prerequisite: The Customer Software Analysis to identify all code unique to the customer.

ASTech personnel will develop and execute a BASE24 System test plan designed for the customer's specific needs. This service also includes the performance and documentation of the system acceptance testing.

Switch and Host Certification Support

Effort: Two to five staff weeks depending on the results of the Customer Software Analysis and the number of BASE24 products and interfaces to be certified. All of this time will be on-site time.

Prerequisite: The Customer Software Analysis to identify all code unique to the customer.

This service option allows ASTech personnel to provide complete support of the various switch and/or host communication certification testing. This includes

reviewing and refining the test scripts, performance of certification testing functions, evaluation of the testing results, and documentation and presentation of the results obtained during the certification testing.

Go-Live Support

Effort: One week (on-site).

ASTech personnel will be at the customer site assisting in all conversion tasks and being present for go-live support. This includes remaining on-site for a couple of days to ensure system stability.

Post Go-Live Support

Effort: Varies.

After a successful go-live, a Tech Lead will provide on-going support which may be required by the customer. Tech Lead support can be provided at the customer location and/or remotely through ACI.

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Section 4

BASE24 Base

Section 4 discusses the changes and enhancements made to the BASE24 Base modules in release 6.0 version 4, along with the resulting conversion and implementation requirements that existing BASE24 users must consider when upgrading from release 6.0 version 3.

Changes and enhancements to the BASE24 Base modules impact all users. Consequently, all users should review the material in this section to determine how it applies to them.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the Base modules changes that affect all BASE24 users, along with the associated conversion and implementation requirements that all BASE24 users must address.
- Optional enhancements describe the Base modules changes that affect only those BASE24 users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24 users must address to implement the enhancements.

Section 4 must be used in conjunction with Sections 5 through 13 to identify the changes that BASE24 users need to consider when upgrading to release 6.0 version 4.

Standard Information Format

A standard format is used in Sections 4 through 13 to highlight and describe the various changes, conversion requirements, and implementation requirements that BASE24 users need to consider when upgrading to release 6.0 version 4. This standard format is defined below.

In situations where the following information does not apply, the information is labeled as “Not applicable”

Software Impacts

Describes the software modules impacted for a given change or enhancement and the nature of the impacts. This information includes a high level list of the modules impacted, how the modules are impacted (inline changes, hooks, dummy procs, extensions) and which customers are affected (all, only those implementing the enhancement).

CUSTMACS Changes

Describes the CUSTMACS changes required to implement a given change or enhancement. This information includes Make defines that have been added, deleted, or modified.

LCONF Assign Changes

Describes the LCONF assign changes required to implement a given change or enhancement. This information includes assigns that have been added, deleted, or modified, along with an explanation of how these assigns can or must be set.

LCONF Param Changes

Describes the LCONF param changes required to implement a given change or enhancement. This information includes params that have been added, deleted, or modified, along with an explanation of how these params can or must be set.

Database Configuration

Describes the data file changes required to implement a given change or enhancement. This information includes file fields that have been added, deleted, or modified, along with an explanation of how these fields can or must be set. It also identifies file records and record segments that must be added, deleted, or modified.

New Data Files

Identifies new data files that must be created in order to implement a given change or enhancement.

Template Changes

Identifies file templates that must be created or modified in order to implement a given change or enhancement. It also identifies COBNAMES and Device Handler Names (xxxxNAMS) file modifications that must be made.

Device Configuration

Identifies device configuration changes that are required in order to implement a given change or enhancement.

Mega Table (MEGATBL) Changes

Identifies Mega Table changes that are required in order to implement a given change or enhancement.

PI-Table Changes

Identifies Product Indicator (PI) Table changes that are required in order to implement a given change or enhancement.

SEG-Table Changes

Identifies Segment Indicator (SEG) Table changes that are required in order to implement a given change or enhancement.

Security Table (SECTBL) Changes

Identifies Security Table changes that are required in order to implement a given change or enhancement.

PATHCONF Changes

Identifies PATHCONF changes that are required in order to implement a given change or enhancement.

CRT Access Security Changes

Identifies CRT Access security changes that are required in order to implement a given change or enhancement. This includes changes that need to be made to BASE24 operator security records.

Routing Changes

Identifies routing and configuration changes that are required in order to implement a given change or enhancement. This includes required changes to files such as SPROUTE and the BASE24-pos Authorization Selection Table (AST).

Network Configuration

Identifies configuration changes that are required in order to implement a given change or enhancement. This includes NEF changes.

File Conversions

Identifies the files that need to be converted in order to implement a given change or enhancement. This implies that a given file could be impacted by multiple changes and/or enhancements. BASE24 users need to consider all of the impacts to any one specific file before the actual conversion of a file.

Host/Co-Network Impacts

Identifies impacts to a user's host or co-network as a result of a given change or enhancement. This includes changes to the external messages used to communicate with hosts and co-networks as well as changes to refresh and extract tape formats.

User/Operational Impacts

Identifies potential changes to a user's internal BASE24 operating procedures. This includes changes to operations and file maintenance screens, operator-initiated commands, and reports.

Vendor Impacts

Identifies any issues that need to be addressed with vendors in order to implement a given change or enhancement. This includes hardware, firmware, and software changes that vendors need to make in order for their products to function with the given change or enhancement.

Reference Documents

Lists any outside sources used in the design of a given change or enhancement. BASE24 users can refer to these documents for additional information.

General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24 users must address in order to upgrade to release 6.0 version 4.

- General Changes
- Thales Key Encryption Scheme

These areas are summarized on the following pages.

Section 2 contains a functional overview of the general enhancements described here.

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General Changes

1. Changes for all BASE24 Release 6.0 Version 4 enhancements affecting the DDL files and the BASE24 AFT Subsystem files (i.e., screens, requesters, servers) have been merged and cataloged under one SCN for each affected file. Applying changes for individual enhancements to the screen files does not allow BASE24 users to identify which changes go with a particular enhancement, as screen files do not have sequence numbers at the end of each line. Since the screen file changes usually involve new fields being added, all associated requester, server and DDL changes are required. Even though requester, server and DDL files do contain sequence numbers, all SCNs would need to be applied in order to accept the changes to the screen files. Therefore, all requester, server and DDL file changes were done under one SCN per file as well.
2. Re-architecture changes were done to minimize MAKE dependencies for some BASE modules. The following changes were made:
 - BACOUTLS change to accommodate BASE24-atm reports (i.e., moved report utility to another location that is BASE24-atm specific).
 - COBNAMES change to remove tables associated only with reports (i.e., moved report table to another location that is BASE24-atm specific).
 - COBNAMES change to remove a Token table only used by the Token Requester
 - COBNAMES change to remove tables associated with ATM receipt processing
 - DDLBCNST change to remove EMV specific constants and SIV specific constants
3. Changed the following SCOBOL and COBOL modules for RMS compatibility. This was accomplished by changing the COPY statements where the file's define name started after column 35. The file's define name was moved to the next line.
 - CPF Requester/Server
 - KEYD Server
4. Implemented new Make flags for preparation of the next version of BASE24.
5. A new token was added for the PRM Release 7.0 – BASE24 changes.
6. The Card Prefix File (CPF) has been modified for the Self-Service Banking Check Enhancements to include four new fields in the CPF SSBC segment. All BASE24-atm users who implement the Self-Service Banking (SSB) Check add-on product, regardless of whether the version 4 Self-Service Banking Check Enhancements is being implemented, are required to run a CPF Conversion program due to the increased size of the CPF SSBC segment.

Software Impacts

All BASE24 customers:

- Inline changes to CUSTMACS and FLAGSDOC for new Make flags
- Inline changes to BACOUTLS, COBNAMES, DDLBCNST, DDLFCPF, RQCPFS, SVCDFS, SVKEYDS, RQTKNS, RQTKNM, BAAFTMM
- New file BA60AFT.COBTKN
- Inline changes to the Base Token Conversion utilities.

All BASE24-atm customers:

- Inline changes to the report modules and to the RCPT AFT modules
- New files AT60RPT.COBATRPT and AT60AFT.COBRCPPT

All BASE24-atm EMV customers:

- Inline changes to BA60IEMV.BAIEMVMM, AT60IATH.AUTHEMVM, AT60IATH.AUTHEMVG
- New file BA60IEMV.EMVG

All BASE24-pos EMV customers:

- Inline changes to BA60IEMV.BAIEMVMM, PS60IRTA.RTAUEMVM, PS60IRTA.RTAUEMVG
- New file BA60IEMV.EMVG

Only BASE24-pos EMV customers with the SPDH:

- Inline changes to PS60ISPD.SPDHEMVG, PS60ISPD.SPDHEMVM
- New file BA60IEMV.EMVG

CUSTMACS Changes

The following Make flags have been added to the CUSTMACS file for preparation for the next version of BASE24. Support of the following Make flags does not exist in the BASE24 release 6.0 version 4 code.

Set the following flag to TRUE if the DIAS Interchange Interface (Greece) is present. This flag must be TRUE if any of the following dias flags are set to TRUE.

- dias_on

Set the following flag to TRUE if the DIAS Interchange Interface will support BASE24-atm. The atm_on flag must also be set to TRUE.

- dias_atm_on

Set the following flag to TRUE if the DIAS Interchange Interface will support the EMV enhancement.

- dias_emv_on

Set the following flag to TRUE if the DIAS Interchange Interface will use the ICFE.

- dias_enhanced_mode_on

Set the following flag to TRUE if the EPS-NET Interchange Interface will support the extension required for U.K. Maestro cards.

- euro_uk_on

Set the following flag to TRUE if the NBGC Interchange Interface will support the EMV enhancement.

- nbgc_emv_on

Set the following flag to TRUE if the SAMA SPAN2 Interchange Interface (Saudi Arabia) is present.

- sama_span2_on

Set the following flag to TRUE if the SAMA SPAN2 Interchange Interface will support the EMV enhancement.

sama_span2_emv_on

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Token file (TKN)

The following new BASE24 Base token has been added for the PRM Release 7.0 – BASE24 changes:

- The PRM Real Time Token (Token ID “29”) contains real time scoring information and real time rules to assist in processing for PRM.

If the new token for the PRM Release 7.0 – BASE24 changes is not to be logged to the TLF, PTLF, or ILF, the appropriate token records must be configured.

If the new token for the PRM Release 7.0 – BASE24 changes is to be extracted with the TLF, PTLF, or ILF, sent to the host, or passed through BIC ISO Interchange Interface, the appropriate token records must be configured.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Card Prefix File (CPF)

The CPF conversion program must be run to convert the CPF. Running the conversion program is required for any self-service banking check (SSBC) user. Users must create the CPF file using the BADDLFUP file.

The CPF conversion program on the BA60UC04 subvolume can be used to convert the CPF records from the release 5.3 format to the release 6.0 format. The CPF conversion program on the BA60UC07 subvolume can be used to convert the CPF records from the release 6.0 version 3 format to the release 6.0 version 4 format. If modifications to the conversion program are necessary, the program can be modified by editing the CNVCPFS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#cpfcnv.t,name $cpfm,nowait/ cnvcpfm –  
force –list –listreplace
```

Release 5.3 users should update the CNVCPFR obey file on the BA60UC04 subvolume with the appropriate information. Release 6.0 version 3 users should update the CNVCPFR obey file on the BA60UC07 subvolume with the appropriate information. The CNVCPFR should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREADME file on the BA60UC04 subvolume for release 5.3 customers and the BA60UC07 subvolume for release 6.0 version 3 customers.

Release 6.0 version 1 and release 6.0 version 2 customers should refer to the **Conversion Program Guidelines** within Section 2 of this document.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

If a customer specific token is required to be displayed and processed by the TKN AFT, the new file COBTKN must be updated to contain the customer specific token information.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Thales Key Encryption Scheme

This section describes the implementation requirements associated with the enhancement to provide support for the new Thales key encryption scheme.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the enhancement to provide support for the new Thales key encryption scheme.

All customers who use the Derivation Key File (KEYD) or the ICC Key File (KEYI), and a Thales HSM without the Transaction Security Server (TSS) to perform BASE24 transaction security processing, must implement this change.

Thales HSM commands support double or triple keys encrypted using either the ANSI X9.17 key scheme or the variant key scheme. An alphabetic “tag” immediately preceding the key value itself specifies the scheme used to encrypt the key. BASE24 supports several Thales HSM commands in which double or triple length keys are passed. In some of these commands (typically older commands), BASE24 passes no tag at all (defaulting to the ANSI X9.17 method), and in other commands (typically newer commands), BASE24 passes a hard-coded “U” (the variant method).

When invoking the Thales commands in the following table, logic must be enhanced to ensure that all double length keys are immediately preceded with the character “U”.

Command	Function
CI	Translate a PIN from BDK Encryption to Interchange Key Encryption
KQ	ARQC (or TC/AAC) Verification and/or ARPC Generation
KU	Generate Secure Message with Integrity and Optional Confidentiality

To allow existing customers to migrate their key database from the ANSI X9.17 scheme to the new variant method, conversion programs will be provided. These programs will convert the keys specified in the table below from encryption using the ANSI X9.17 scheme to encryption using the variant scheme (using the Thales “B0” command).

File/Screen	Key Name	Key Usage
KEYD 1	Derivation Key	Translate a PIN from BDK Encryption to Interchange Key Encryption
KEYI 1	Cryptographic Keys	ARQC Verification and/or ARPC Generation
KEYI 1	Secure Messaging Keys (for Integrity)	Generate Secure Message with Integrity

This enhancement will resolve the inconsistencies surrounding the encryption of keys used by BASE24 in Thales HSM commands.

Software Impacts

All code changes have been applied inline.

All BASE24 customers:

- Changes to the BA60SRC.SECUTILS file to ensure that the double length key in the Thales HSM “CI” command is immediately preceded with the character “U”.

All customers with CAM Utilities (i.e. those customers who support EMV Issuer processing:

- Changes to the BA60SRC.CAMUTILS file to ensure that the double length keys in the Thales HSM “KQ” and “KU” commands are immediately preceded with the character “U”.

Only customers with Thales HSMs, who do not use TSS, and who support the Derived Unique Key Per Transaction (DUKPT) security scheme:

- A new conversion program (BA60UC07.CNVKEYDS) to translate the KEYD key from encryption under the ANSI X9.17 scheme to encryption under the variant scheme.

Only customers with Thales HSMS, who do not use TSS, and who support EMV issuer processing:

- A new conversion program (BA60UC07.CNVKEYIS) to translate the KEYI keys from encryption under the ANSI X9.17 scheme to encryption under the variant scheme.

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD07.SCNNTKES.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Derivation Key File (KEYD)

The conversion program will invoke the Thales “B0” command to translate the double-length Derivation Key (if present) from encryption using the ANSI X9.17 scheme to encryption using the variant scheme (using a key type of “009”).

ICC Key File (KEYI)

The conversion program will invoke the Thales “B0” command to perform the following processing:

- Translate up to 24 Cryptographic Keys from encryption using the ANSI X9.17 scheme to encryption using the variant scheme (using a key type of “109”).
- Translate the Secure Messaging for Integrity Key (if present) from encryption using the ANSI X9.17 scheme to encryption using the variant scheme (using a key type of “209”).

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Thales e-Security (formerly Racal) introduced longer key lengths in many commands (mainly to support Triple DES and EMV) in version 5.05 of the HSM software.

Reference Documents

- Thales e-Security - Host Security Module RG7000 Programmer's Manual, 1270A514 Issue 5.

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Optional Enhancements

This section identifies the Base modules enhancements that can be enabled or disabled with release 6.0 version 4, along with the changes, conversion requirements, and implementation requirements. These enhancements are the following:

- Dual Site Support
- EMV Downgrade
- EMV M/Chip 4 Support
- Enhanced Expiration Date Checking
- TSS Key Index Support

BASE24 users can choose whether or not to implement the above enhancements. Users who plan to implement these particular enhancements must address the corresponding release 6.0 version 4 changes, conversion requirements, and implementation requirements. Users who do not plan to implement these particular enhancements can review the corresponding release 6.0 version 4 changes, conversion requirements, and implementation requirements for future reference.

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Dual Site Support

This section describes the implementation requirements associated with Dual Site Support.

These changes and implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Dual Site Support Enhancement.

Sections 5 (ATM) and 6 (POS) contain product specific changes relating to the Dual Site Support enhancement.

Dual Site Support involves changes related to the management of a Dual Site BASE24 production environment. This enhancement is to support customers who are planning to implement a Dual Site production environment using GoldenGate to facilitate data replication between the two sites.

A Dual Site BASE24 production environment is where BASE24 is running on two separate NSK machines, with production transactions being processed on both machines. Standard BASE24 files will be held on both systems with updates replicated between the two systems by GoldenGate.

The TLF and PTLF files have been enhanced to enable the identification of individual records as “local” or “remote”, i.e. whether the transaction was authorized on this system or the alternate system. Local records are standard and remote records include a new field, SITE^IND, in the Multiple Logical Network token. (Records for all transactions will appear on both copies of the files, with one copy containing the remote token according to the location of the original transaction.) As records on either system are added, GoldenGate will replicate the records to the alternate system, add the token if not already present, and update the field.

Extract and Refresh have been enhanced to provide options relating to Dual Site Support. Each module can be configured to process local only, remote only or all records in extracting or impacting respectively. Extract and Refresh have also been enhanced to support process specific TLF / PTLF assigns. Refresh has additionally been enhanced to support broadcast capability in notify lists.

For more details on using these enhancements and GoldenGate to support a Dual Site production environment, refer to the ***BASE24 Dual Site Processing Guide***.

Software Impacts

All BASE24 customers:

- DDL changes to BA60DDL.DDLBATKN and BA60DDL.DDLBCNST

- Inline changes to BA60SRC.BACOUTLS
- Inline changes to Base Extract and Refresh

All BASE24-atm customers:

- DDL changes to AT60DDL.DDLFTDF, AT60DDL.DDLFATD, AT60DDL.DDLGATD
- Inline changes to the ATD AFT modules
- Inline changes to the TLF AFT modules
- Inline changes to the DCT server files
- Inline changes to TLF Extract modules
- Inline changes to ATM Refresh modules
- Inline changes to Settlement modules
- Inline changes to Utility modules
- Inline changes to the ATD conversion program

All BASE24-pos customers:

- DDL changes to PS60DDL.DDLFPTDF, PS60DDL.DDLFPTD, PS60DDL.DDLGPTD
- Inline changes to the PTD AFT modules
- Inline changes to the PTLF AFT modules
- Inline changes to PTLF Extract modules
- Inline changes to POS Refresh modules
- Inline changes to Settlement modules
- Inline changes to Utility modules
- Inline changes to the PTD conversion program

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD07.SCNDUALS.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

This section describes the Assign records that must be added to the LCONF to support this enhancement. These modifications are provided in the LCONFxx file located on the BA60MISC subvolume where xx is the product identifier of AT, BA, PS.

TLF

The fully qualified file name of the Transaction Log File (TLF). This assign is now supported as a process specific assign for the Super Extract process and the Refresh process.

POS-PTLF

The fully qualified file name of the POS Transaction Log File (PTLF). This assign is now supported as a process specific assign for the Super Extract process and the Refresh process.

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement. These modifications are provided in the LCONFBA file located on the BA60MISC subvolume.

DUAL-SITE-MODE

This param is used by the Super Extract process and the Refresh process. It specifies the dual site mode for the process. Valid values are:

L - Local records only to be processed by Extract/Refresh

R - Remote records only to be processed by Extract/Refresh

A - All records to be processed by Extract/Refresh

BROADCAST-NOTIFY

This param is used by the Refresh process. It defines whether the process uses the broadcast facility. Valid values are 0 - Broadcast facility not to be used by Refresh, 1 - Broadcast flag to be set to 1 (broadcast to all service providers) for notify messages sent by Refresh and 2 - Broadcast flag to be set to 2 (broadcast to running service providers) for notify messages sent by Refresh.

BROADCAST-DELAY

This param is used by the Refresh process. It defines the number of seconds to delay after broadcasting to the notify list before completion of impacting.

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Host/Co-Network impacts will be incurred by those BASE24 users implementing the Dual Site Support enhancement. Modifications to existing files and processing will result in impacts to the external interfaces.

External Interfaces

The host or co-network may require modification to recognize and process new data within the Multiple Logical Network Token. The new data will be passed in the ISO message from the BASE24-atm ISO Host Interface (HISO) or BASE24 Interchange (BIC) ISO Interface.

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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EMV Downgrade Enhancement

This section describes implementation requirements associated with the EMV Downgrade enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the EMV Downgrade enhancement.

The EMV Downgrade enhancement involves the changes related to processing of EMV transactions as standard magnetic stripe transactions. This enhancement is to support issuers who have issued EMV cards but cannot yet support EMV transactions.

A 'Downgrade' transaction is one where an EMV card has been used at an EMV enabled terminal, but the transaction is processed as a magnetic stripe transaction.

The Card Prefix File (CPF) has been enhanced so that the CAM check type field can cater for a new situation, which is "Downgrade from EMV to Magnetic stripe". Each prefix for an issuer, which cannot support EMV transactions, should have the CAM check type set to this new value.

The EMV Status Token has been enhanced so that the ARQC verify field can cater for a new situation which is "ARQC not verified – transaction "downgraded" to magnetic stripe". When the CPF has CAM check type set to "Downgrade from EMV to Magnetic stripe", the ARQC verify field is set to "ARQC not verified – transaction "downgraded" to magnetic stripe".

Software Impacts

All BASE24-atm EMV customers

- Inline changes to the AT60IATH.AUTHEMVS to support the EMV downgrade transactions.

All BASE24-pos EMV customers

- Inline changes to the PS60IRTA.RTAUEMVS to support the EMV downgrade transactions.

A complete list of files affected by this enhancement can be found in the SCN file on BA60UD07.SCNEMVDG

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes the new fields in the files that have been modified to support this enhancement.

Card Prefix File (CPF)

CPF screen 11 has been modified to allow a new value for the following existing field:

Field Name: CAM CHECK TYPE
Screen: CPF Screen 11
DDL Name: CPF.EMVCPF.CAM-CHK-TYP

This field is used to indicate the action to be performed if the ARQC has not been verified. An extra value of "9" indicates that the EMV transaction should be downgraded to magnetic stripe.

The full list of values for this field can be found in the ***BASE24-atm EMV Support Manual***.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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EMV M/Chip 4 Support

This section describes the implementation requirements associated with the EMV M/Chip 4 Support enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the EMV M/Chip 4 Support enhancement.

Section 5 contains BASE24-atm specific requirements associated with the EMV M/Chip 4 Support enhancement.

Section 6 contains BASE24-pos specific requirements associated with the EMV M/Chip 4 Support enhancement.

The EMV M/Chip 4 Support enhancement involves the changes required to support MasterCard's M/Chip Select 4 and M/Chip Lite 4 EMV applications.

Issuer Application Data Format

A new counters sub-field is now included in the Issuer Application Data in an M/Chip 4 message and will be used in generating the Application Cryptogram when the cryptogram version number is Hex 11 or 13. The Card Verification Results field is 6 bytes in an M/Chip 4 transaction. A new redefinition has been added to the EMV Request token to accommodate the M/Chip 4 format of the Issuer Application Data.

Identification of Issuer Application Data Format

The Issuer application Data contains proprietary issuer data to be sent to the Issuer in an On-line transaction. The EMV ISS APPL DATA FRMT on the CPF will specify where the proprietary data is to be found within the EMV Request Data Token. A new format of this data is now available, M/Chip 4. A value of 3 in the EMV ISS APPL DATA FRMT field of the CPF represents M/Chip 4 and points the user to retrieve the information from the MCHIP4-APPL-DATA of the EMV Request Data Token.

Migration of cards from M/Chip 2.1 to M/Chip 4 will be supported by the ability to specify M/Chip 4 support at the Card level. To facilitate this two new fields have been added to the Cardholder Authorization File, Issuer Application Data Format and Status Checking Action Index. The Issuer Application Data Format will support two values, one to specify M/Chip 4 support, the other to indicate that the CPF Issuer Application Data Format value will be used in processing. The Status Checking Action Index will support values 1 to 4 and will be invoked if the Issuer Application Data Format field on the CAF is set to M/Chip 4.

The data on the CAF file can be used as an override to data on the CPF. Therefore if a CAF record exists for a Card and the CAF IAD format field holds the value 3

(M/Chip 4) then this value will override the IAD format defined on the CPF. In addition the Status Checking Action Index value from the CAF will be used to specify the index to the Card Verification Results table.

Issuer Authentication Data

A new redefinition has been added to the EMV Response Token to reflect the format of the additional Issuer Authentication Data for M/Chip 4 and M/Chip 2.1.

Cryptogram Verification/Generation

The HSM Cryptogram generation interface (CAMUTILS) has been modified so all Cryptogram verification/generation messages include a configurable ARPC response code instead of the Issuer Authentication Response Code. For M/Chip 4 transactions the Cryptogram verification/generation messages also accommodate the increased size of the Card Verification Results field and include the counters parameter where the Cryptogram version number is Hex 11 or 13.

Software Impacts

All BASE24 customers:

- Base DDL changes (DDLBATKN, DDLFCAF, DDLFCPF, DDLGISO, DDLGTSS, DDLGCAFX)
- Inline changes in the Cardholder Authorization File (CAF) User Interface (AFT) module screen, requester, and server
- Inline changes in the Card Prefix File (CPF) User Interface (AFT) module screen, requester, and server
- Inline changes to Refresh
- Inline changes to OMF Extract

All BASE24-atm EMV customers:

- Inline changes to the Base Card Authentication Method Utilities (BA60IEMV.CAMUTILS)
- Inline changes to the Authorization EMV extension modules

All BASE24-pos EMV customers

- Inline changes to the Base Card Authentication Method Utilities (BA60IEMV.CAMUTILS)
- Inline changes to the Router/Authorization EMV Extension modules

All BASE24-fhm customers

- Inline changes to the FHM modules for ULF Extract support, FHM CAF support and FHM Reports

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD07.SCNMCHP4.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes the new and enhanced fields in the files that are required to support this enhancement.

Card Prefix File (CPF)

CPF screen 11 has been modified to allow a new value for the following existing field:

Field Name: EMV IAD FORMAT
Screen: CPF Screen 11
DDL Name: CPF.EMVCPF.EMV-ISS-APPL-DATA-FRMT

This field is used to identify the format of the Issuer Application Data for this card type. An extra value of “3” indicates that the EMV transaction is M/Chip 4 format.

The full list of values for this field can be found in the ***BASE24-atm EMV Support Manual***.

Cardholder Authorization File (CAF)

CAF screen 13 has been modified to include the following new fields:

Field Name: STATUS CHECKING ACTION INDEX
Screen: CAF Screen 13
DDL Name: CAF. ACTION-TABLE-INDEX

This field is used to identify which of the preset rules for status field processing are to be used for this Card scheme. The rules are defined in the Card Verification Results table (CVRTBL). Valid values are 1 – 4.

Field Name: EMV IAD FORMAT
Screen: CAF Screen 13
DDL Name: CAF. ISS-APPL-DATA-FRMT

This field indicates the format of the Issuer Application Data for this cardholder. Valid values are 0 - Use the value on the CPF (default value) and 3 - M/Chip 4 format

The full list of values for these fields can be found in the ***BASE24-atm EMV Support Manual***.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Host/Co-Network impacts will be incurred by those BASE24 users implementing the EMV M/Chip 4 Support enhancement. Modifications to existing files and processing will result in impacts to the From Host Maintenance and Refresh Process.

From Host Maintenance

Host code changes are required for all BASE24 users that utilize the Cardholder Authorization File (CAF) From Host Maintenance (FHM) process to update the CAF records that include the EMV segment. BASE24 users utilizing the 6.0 or the 5.x FHM message format can specify appropriate values for the STATUS CHECKING ACTION INDEX and EMV IAD FORMAT fields in the CAF.

Refresh Process

Host code changes are required for all BASE24 users that utilize the release 6.0 CAF Refresh file format to accept and process information in the CAF EMV segment.

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

- MasterCard Europe - M/Chip 4 card application specification, version 4.0, 16th May 2002
- EMVCo - EMV 2000 version 4.0, December 2000

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Enhanced Expiration Date Checking

This section describes the implementation requirements associated with the Enhanced Expiration Date Checking enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Enhanced Expiration Date Checking enhancement.

The Card Prefix File (CPF) and Cardholder Authorization File (CAF) have been modified to include the new configuration parameters required to support Enhanced Expiration Date Checking.

This enhancement involves the changes related to expiration date checking. There are a number of related changes:

- **Expiration Date Required in Transaction Request Messages**
This enhancement allows acquirers and issuers to configure whether a valid expiration date must be present in transaction request messages.
- **Card Validity Period**
This enhancement allows card issuers to configure the maximum validity period of a card. If the expiration date in the authorization request message is beyond this validity period, then the transaction will be declined.
- **Expiration Date Comparison**
This enhancement allows card issuers to configure an additional expiration date check. If the expiration date in the authorization request message does not match the expiration date held on the CAF, then the transaction will be declined. This will also allow expiration date checking to use the expiration date held on the CAF, if no expiration date is present in the authorization request message.
- **Second Card Data**
This enhancement allows a second card be configured on the CAF record to provide support for both cards. The enhancement also allows card issuers to utilize the “second” card data in authorization processing. The only way of determining which card is being used in a transaction is by comparing the expiration date in the authorization request message to the two expiration dates held on the CAF. Therefore, the Enhanced Expiration Date Checking enhancement is required to support “second” card data. Conversely, without the second card data, it would not be possible for some card issuers to implement the expiration date comparison enhancement.

In addition to the above changes, BASE24-pos customers will have the following additional change:

- **Recurring Payment Transactions**
This enhancement allows the expiration date checking performed on recurring payments to be configured separately to that performed on other transaction types. For example, issuers may decide not to check the expiration date provided by the merchant via the acquirer for recurring payments, and would therefore not decline the transaction because the expiration date quoted in the authorization message is incorrect.

Software Impacts

All BASE24 customers:

- Inline changes to BA60DDL.DDLFCPF
- Inline changes to the BA60AFT files (RQCAFS, RQCPFS, SVCAFS, SVCDFS, SCRNCAF and SCRNCDF) to support this enhancement
- Inline changes to BA60AFT.SECTBL to support CAF screen 7 and CPF screens 3 and 8
- Inline changes to BA60EXT.OMFXS to support the new fields

All BASE24-atm customers:

- Inline changes to AT60AUTH files (AUTHG, AUTHLIBS and AUTHS)

All BASE24-pos customers:

- Inline changes to PS60AUTH files (AUTHLIBS, ROUTERS, RTAUG)
- New Log (Event) Message Manuals for the Router/Authorization module with the new Enhanced Expiration Date Checking events.

All BASE24-fhm customers:

- Inline changes to FH60RPT.FHMRPTS

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD07.SCNEXP1

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes the new fields in the files that have been modified to support this enhancement.

Card Prefix File (CPF)

Field Name: EXPIRATION DATE PROCESSING FLAG
Screen: CPF Screen 3
DDL Name: CPF.BASE.EXP^DAT^REQ

This field specifies whether an expiration date is required in transaction requests. Valid values are 0 - No expiration date required (default value), 1 - Expiration date required for all transactions and 2 - Expiration date required for card-read transactions only

Field Name: EXPIRATION DATE COMPARISON
Screen: CPF Screen 3
DDL Name: CPF.BASE.EXP^DAT^CMP

This field specifies whether the expiration date on the card should be compared to the expiration date on the CAF. If this field is set to '1', and the expiration date on the card is not present in the transaction message, then the expiration date on the CAF will be used in the expiration date check. For recurring payment transactions, this field will be checked only if the RECUR-PMNT-EXP-CHK-IND field in the POS segment is set to '9'. Valid values are 0 - No expiration date comparison required (default value) and 1 - Expiration date comparison required.

Field Name: CARD VALIDITY PERIOD
Screen: CPF Screen 3
DDL Name: CPF.BASE.EXP^DAT^PRD

This field specifies the maximum validity period (in months) for a card. Valid values are 0 - no checks performed (default value) to 600.

Field Name: RECURRING PAYMENT EXP CHECK TYPE
Screen: CPF Screen 8
DDL Name: CPF.BASE.RECUR^PMNT^EXP^CHK^IND

This field specifies the type of card expiration date check to use for this

prefix for recurring payment transactions. Valid values are 0 - Do not check the expiration date, 1 - Check the expiration date on Track 1 or Track 2, 2 - Check the expiration date in the CAF record and 9 - Use EXP-CHK-IND field on the Base segment (Default value)

Card Authorization File (CAF)

A new screen 7 has been added to specify second card usage controls.

If these fields are related to a re-issued card the same logical card is used but a different physical card.

Field Name: SECOND CARD EXPIRATION DATE
Screen: CAF Screen 7
DDL Name: CAF.BASE.SCND^CRD^DATA.EXP^DAT^2

The expiration date of the second card. A transaction attempted after the month indicated is denied. 0206 indicates that the card is valid through June 2002. Valid values are four numeric characters in year and month format (default 0000).

Field Name: SECOND CARD STATUS
Screen: CAF Screen 7
DDL Name: CAF.BASE.SCND^CRD^DATA. CRD^STAT^2

This field gives the card status of the second card. Valid values are:

- 0 - (ISSUED NOT ACTIVE)
- 1 - (OPEN)
- 2 - (LOST CARD)
- 3 - (STOLEN CARD)
- 4 - (RESTRICTED)
- 5 - (VIP)
- 6 - (CARD NOT USED), Default
- 9 - (CLOSED)
- A - (REFERRAL)
- B - (MAYBE)
- C - (DENIAL)
- D - (SIGNATURE RESTRICTED)
- E - (COUNTRY CLUB)
- F - (EXPIRED)
- G - (COMMERCIAL)

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Entries added to the SECTBL to support screen 7 for the CAF requester/servers.

PATHCONF Changes

Not applicable

CRT Access Security Changes

Access to the new CAF screen 7 and CPF screens 3 and 8 must be added through the SEC requester for the appropriate users.

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Card Prefix File (CPF)

If the BASE24-pos user wishes to enable Enhanced Expiration Date Checking processing, the user must update the new fields in the CPF with the appropriate value. Otherwise, the new fields in the CPF will default to indicate that enhanced expiration date checking is not performed.

Host/Co-Network Impacts

Super Extract Process

Host code changes are required for all BASE24 users that utilize the OMF Extract Process to accept and process the new information in the CAF.

Host code changes are required for all BASE24 users that utilize the OMF Extract Process to accept and process information in the CPF.

User/Operational Impacts

Operational Impacts

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate remedy for a given message. Refer to the event message documentation files on \$<volume>.BA60LOGM.

Vendor Impacts

Not applicable

Reference Documents

- Visa CEMEA - Member Letter #27/99, July 21 1999.
- MasterCard International - UK and ROI Industry Bulletin, Edition 3, August 2002.
- Switch Card Services Limited - Inter-Member System Specification, Part 5, Authorisation Specification, Phase 12, version 2.0, June 18 2002.

TSS Key Index Support

This section describes the implementation requirements associated with the TSS Key Index Support enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the TSS Key Index Support enhancement.

The enhancement is needed to meet Interac's key management approach. Interac regulations require that the transaction acquirer/issuer determine which encryption keys (PIN, data and MAC) are to be used in a transaction request/response. BASE24 was modified to support sending the key index in requests to TSS and TSS was modified to perform encryption/decryption tasks only with the key corresponding to the key index passed. Changes were made to the Interac Interchange Interface, ATM Auth, Router/Auth, HISO, SPDH, HISWUTLS, SECUTILS and token processing modules to support a new TSS Key Index token. When the token is present in a transaction and these modules have been updated to support this enhancement, the key index from the token will be sent to TSS for encryption/decryption tasks.

Prerequisite: The Transaction Security Services enhancement Version 3.3 must be installed prior to implementation of the TSS Key Index Support. For more information refer to the ***BASE24 Release 6.0 Implementation Guide (Version 2)***.

Software Impacts

All BASE24 customers:

- Inline changes to the DDL files DDLFKEYF, DDLBATKN, DDLGTSS
- Inline changes to BA60AFT.COBTKN
- Inline changes to CUSTMACS and FLAGSDOC for a new Make flag
- Hooks in SECUTILS to TSS Key Index procedures
- Make file changes (BA60SRC.BASEMM, BA60SRC.BASEFM, BA60SRC.BASRCM, BA60SRC.SECUTILM)
- New files BA60SRC.SECUTILD, BA60SRC.SECUTLLM

All BASE24-atm customers:

- Make file changes (AT60SRC.ATMMM, AT60SRC.ATMFM, AT60AUTH.AUTHM, AT60HISO.ATHISOMM, AT60HISO.ATHISOM)
- Hooks in the Authorization files to TSS Key Index procedures
- Hooks in the HISO files to TSS Key Index procedures

All BASE24-pos customers:

- Make file changes (PS60SRC.POSMM, PS60SRC.POSFM, PS60RTAU.RTAUM, PS60HISO.PSHISOMM, PS60HISO.PSHISOM)
- Hooks in the Router/Authorization files to TSS Key Index procedures
- Hooks in the HISO files to TSS Key Index procedures

Only customers with ACI Standard POS Device Handler Data Encryption extension:

- New files in the PS60DSPD subvolume.

Only customers taking the TSS Key Index Support functionality:

- New files in the extension subvolumes containing TSS Key Index Support functionality (i.e., AT60YATH, AT60YHIS, BA60YSRC, PS60YHIS and PS60YRTA). The files on these subvolumes include source files, global files and Make files required for the new functionality.

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD07.SCNTKI.

CUSTMACS Changes

The following Make flags have been added to the CUSTMACS file.

Set the following flag to true to bind in the TSS Key Index Support subvolumes:

- tss_key_idx_on

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

BA60DDL.DDLGTSS has been modified to support new TSS commands and indexes.

Token File (TKN)

The following BASE24 token has been added for the TSS Key Index Support enhancement:

- The TSS Key Index Token (Token ID “BC”) contains the PIN, MAC and data encryption key indexes.

If the TSS Key Index token is not to be logged to the Transaction Log File (TLF/PTLF) or Interchange Log File (ILF), the appropriate token records must be configured in the TKN.

If the TSS Key Index token is to be extracted with the TLF/PTLF, sent to the host, or passed through the BIC ISO Interchange Interface process, the appropriate token records must be configured in the TKN.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

External Interfaces

The host or co-network must be modified to accept and process the TSS Key Index token in the ISO message from the BASE24 ISO Host Interface (HISO) or BASE24 BIC ISO process.

Super Extract Process

If the new TSS Key Index token is to be included in the TLF, PTLF or ILF Extracts, BASE24 users must consider this additional data in host programs that use the data from the log files.

User/Operational Impacts

Disk Impacts

The TLF, PTLF, and ILF will require additional disk space as the new TSS Key Index token can be included in the transaction records logged to these files. The total impact varies for each BASE24 user depending on the transaction volume in the BASE24 system.

If the TLF, PTLF and/or ILF is extracted to a disk file, the extract record size is increased. This can increase the disk/tape space required for the extract file.

System Limitations

BASE24 users installing this enhancement must be aware of their internal system limitations for messaging to the TLF, PTLF, and ILF. There are system limitations regarding the amount of data that can be passed through the BASE24 system, logged to a log file, and passed to the host or co-network in an ISO External Message. BASE24 users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the **BASE24 Tokens Manual** provides information on assessing the impacts of tokens on log file disk space requirements.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Section 5

BASE24-atm

Section 5 discusses the changes and enhancements made to BASE24-atm in release 6.0 version 4, along with the resulting conversion and implementation requirements that existing BASE24-atm users must consider when upgrading from release 6.0 version 3.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24-atm users, along with the associated conversion and implementation requirements that all BASE24-atm users must address.
- Optional enhancements describe the changes that affect only those BASE24-atm users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24-atm users must address to implement the enhancements.

Section 5 must be used in conjunction with Section 4 to identify the changes that BASE24-atm users need to consider when upgrading to release 6.0 version 4.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24-atm users must address in order to upgrade to release 6.0 version 4.

There are no general enhancements inherent to BASE24-atm.

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Optional Enhancements

This section identifies the BASE24-atm enhancements that can be enabled or disabled with release 6.0 version 4, along with the changes, conversion requirements, and implementation requirements associated with each. These enhancements include the following:

- Acquirer Surcharging
- Automated Key Distribution System
- Diebold TCS Enhancements Support
- Dual Site Support
- Dynamic Key Management
- EMV M/Chip 4 Support
- Forced Cutover Audit
- Fujitsu 7000 Device Handler
- Fujitsu 7000 Triple DES
- Interactive Response Notification Fee Processing and Negative Surcharge Support
- Multiple Currency Dispense Exchange Rate Notification
- NCR EMV Dial-Up
- Self-Service Banking Check
- Triton Triple DES

BASE24-atm users can choose whether or not to implement each of the above enhancements. Users who plan to implement a particular enhancement must address the corresponding release 6.0 version 4 changes, conversion requirements, and implementation requirements. Users who do not plan to implement a particular enhancement can simply review the corresponding release 6.0 version 4 changes, conversion requirements, and implementation requirements for future reference.

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Acquirer Surcharging

This section describes the implementation requirements associated with the BASE24-atm Acquirer Surcharging enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the BASE24-atm Acquirer Surcharging enhancement.

Standard BASE24-atm modules have been modified to support this enhancement. These modules include Authorization, the Diebold 10XX/478X device handler, the NCR 5XXX device handler, OMF Extract, and Settlement Reports.

Software Impacts

All BASE24 customers:

- Base DDL changes (i.e., DDLM, DDLMM, DDLBATKN, DDLFOMF, DDLFSURF, DDLFTSRF)
- Changes to BA60AFT.BAAFTM, BAAFTMM, and CCAFMLM to accommodate the new Transaction Code/Subtype Relationship File (TSRF) requester/server
- COBATRPT change to add new defines for reports
- Kernel AFM change to support a new transaction define
- MEGATBL, SECTBL, and SVMHLPS change to accommodate the TSRF requester/server
- OMF Extract change to accommodate the Surcharge File (SURF) changes and the introduction of the TSRF
- TCF Server change to support new functionality required by TSRF requester/server
- A new requester/server (i.e., TSRF) with associated global and Make files

All BASE24-atm customers:

- Inline code changes to Authorization (AUTHS), Authorization Library (AUTHLIBS), and Authorization Globals (AUTHG)
- New Log (Event) Message Manuals for Authorization
- Inline code change to TLF Requester
- Inline code changes to Settlement Reports (i.e., RP02B24S, RP03B24S, RP04B24S, RP06B24S, RP08B24S)
- Inline code change to ARCOUTLS
- Inline change to COBRCPT to allow an existing field type code on receipt

All customers with Diebold 10XX/478X or NCR 5XXX device handler:

- Inline changes to offer BASE24-atm Surcharging enhancements (i.e., AT601000.C1000S, AT60N50.N50S)
- Minor inline changes in the global files (i.e., AT601000.C1000G, AT60N50.N50G)
- New load groups in the configuration files (i.e., AT601000.CNFGIX, AT60BN50.CNFG5070)
- Changes to the documentation files (i.e., AT601000.CN100DOC, AT60N50.CNN50DOC)

All customers with Diebold 10XX/478X or NCR 5XXX device handler Dial Up Extensions:

- Inline changes to the Dial-up product files (i.e., AT60P100.C100DLUS, AT60PN50.N50DLUS)

All BASE24-atm Self Service Banking (SSB) customers:

- Inline code changes in the SSB Base, Check, and Device Handler packages (i.e., AT60B100.C100SSBS, AT60BN50.N50SSBS, AT60C100.C100SSCS, AT60CN50.N50SSCS, AT60D100.C100SSDS, AT60N50.N50SSDS).

All BASE24-atm Surcharging customers:

- A new file, COBSURF, which is comprised of the transaction codes eligible for surcharge assessment
- Inline code changes to the Surcharge File (SURF) requester/server
- Changes to the SURF requester/server Make files

A complete list on the files affected by this enhancement can be found in the SCN file on BA60UD07.SCNSRCH.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

TSRF

The location of the Transaction Code/Subtype Relationship File (TSRF).

LCONF Param Changes

Not applicable

Database Configuration

BASE24-atm Receipt File (RCPT)

The RCPT File can be used to display the surcharge fee and amount fields that reflect the surcharge fee on the cardholder's receipt. No new field type codes have been introduced with this enhancement; but, existing field type codes have been altered to handle surcharge fees associated with deposit, deposit with cash back, and split deposit transactions.

The A1 field type code is used for the transaction amount (STM.RQST.AMT-1). For a withdrawal transaction, A1 equals the transaction amount plus the surcharge fee. For deposit and deposit with cash back transactions, A1 equals transaction amount minus the surcharge fee. For a split deposit transaction, A1 equals the amount deposited into the first account associated with the transaction minus the surcharge fee.

The AM field type code is used for the transaction amount for split deposit transactions, not reflective of the surcharge fee. For a split deposit transaction with a surcharge, AM equals the amount deposited into the first account (STM.RQST.AMT-1) plus the amount deposited into the second account (STM.RQST.AMT-2) plus the surcharge fee. For other transactions, AM equals STM.RQST.AMT-1.

Additional information on specific field type codes may be found in the **BASE24-atm Diebold 10xx/478x** or **BASE24-atm NCR 5xxx Device Support Manual** and the **BASE24-atm Diebold 10xx/478x** or **BASE24-atm NCR 5xxx SSB Manual**. Information on use of the RCPT file may be found in the **BASE24-atm Files Maintenance Manual**.

Surcharge File (SURF)

The SURF contains the following new fields:

Field Name: CARD PREFIX
Screen: SURF screen 3
DDL Name: SURF.SURF-PRIKEY-SPROUTE

The card prefix to which this record information applies. For Sproute routing, this field allows surcharge fees to be assessed at a more refined level than card type alone.

Field Name: TRANSACTION CODE
Screen: SURF screens 2 and 3
DDL Name: SURF.ATM.TRAN.TRAN-CDE

The transaction code, prior to this enhancement, was entered by the user and the description was 12 bytes, hard-coded within the SURF Requester. With this enhancement, all transactions that are eligible for surcharge assessments are stored within the COBSURF. Each transaction code within the COBSURF is accompanied by a 24 byte description.

Field Name: SUB TYPE
Screen: SURF screens 2 and 3
DDL Name: SURF.ATM.TRAN.TRAN-SUBTYP

The transaction subtype to which this record information applies. The transaction subtype can either be wildcarded ("****") or can be specified in order to distinguish between transactions of the same type that should be surcharged based upon different fee structures.

Field Name: MIN TXN AMT
Screen: SURF screens 2 and 3
DDL Name: SURF.ATM.TRAN.MIN-TXN-AMT

The minimum transaction amount is used to determine whether a surcharge fee can be assessed. If the amount of a transaction falls below the minimum transaction amount, a surcharge fee cannot be assessed to the transaction being processed. If the amount of a transaction is greater than or equal to the minimum transaction amount, a surcharge fee can be assessed to the transaction being processed.

Token File (TKN)

Although no tokens were added for the Surcharge enhancement, the enhancement added values to the existing Surcharge Data Token. The Authorization Indicator field within the Surcharge Data Token has two new values, 'D' and 'M'. The value 'D' is placed in the token when the cardholder declines to accept the surcharge fee and the device handler is configured to log declines to the log file. The value 'M' is placed in the token when the surcharge fee calculated for a transaction is greater than the value specified in the BASE24-atm Terminal Data File (ATD) download surcharge amount field.

New Data Files

Transaction Code/Subtype Relationship File (TSRF)

The TSRF is an optional file and contains one record for each transaction subtype supported in the network. This file is used to assign an alphanumeric description to a transaction subtype as well as to associate the transaction subtype to one or more transaction codes.

The TSRF contains the following fields:

Field Name: TRANSACTION SUBTYPE
Screen: TSRF screen 1
DDL Name: TSRF.PRIKEY

Transaction Subtypes are four character values used to indicate that the processing associated with a particular transaction code should be altered. The transaction being processed should, in the broad sense, be processed according to its transaction code. However, characteristics exist which delineate some transactions with the same transaction code from others. In these cases, a Transaction Subtype and its associated processing logic can be added within BASE24. With the addition of Transaction Subtypes, BASE24 can effectively process new transactions without adding new transaction codes. It should be noted that cases may still arise in which a new transaction code is required.

Transaction Subtypes must be unique and must fall within the ranges defined below:

ATM Product	'A000' – 'AZZZ'
BASE Product	'B000' – 'BZZZ'
POS Product	'C000' – 'CZZZ'
GEN Card Product	'G000' – 'GZZZ'
Teller Product	'T000' – 'TZZZ'
CSM	'R000' – 'RZZZ'
CSM	'P000' – 'PZZZ'
CSM	'Q000' – 'QZZZ'
New Initiatives	'N000' – 'NZZZ'
America's Channel	'U000' – 'UZZZ'
EMEA Channel	'V000' – 'VZZZ'
Asia/Pacific Channel	'W000' – 'WZZZ'
Distributors	'X000' – 'XZZZ'
Distributors	'Y000' – 'YZZZ'
Distributors	'Z000' – 'ZZZZ'

Field Name: TRANSACTION SUBTYPE DESCRIPTION
Screen: TSRF screen 1
DDL Name: TSRF.TXN-SUBTYP-DESCR

The text description of the transaction subtype.

Field Name: TRANSACTION CODES
Screen: TSRF screen 1
DDL Name: TSRF.TXN-CDE

The transaction codes associated with the transaction subtype. A maximum of 50 transaction codes can be associated with a single transaction subtype.

Template Changes

Not applicable

Device Configuration

With this enhancement, the following files were modified to include new load groups:

Diebold 10XX/478X Configuration File (CNFGIX)

The following load groups were modified for the Surcharging enhancement:

- Load Group 304

Load group 304 is the Surcharge Group and was modified to present the Surcharge screen (screen 113) for deposit transactions in addition to withdrawal transactions. Load group 304 does not support logging of cancelled transactions when the cardholder declines the surcharge fee.

- Load Group 305

Load group 305 is the group required for dial-up ATM non-interactive surcharge processing. Load group 305 was modified to present the Surcharge screen for deposit transactions in addition to withdrawal transactions. Load group 305 does not support logging of cancelled transactions when the cardholder declines the surcharge fee.

The following load groups were added for the Surcharging enhancement:

- Load Group 314

Load group 314 is identical to load group 304 with the exception that it supports the logging of cancelled transactions when the cardholder declines the surcharge fee.

- Load Group 315

Load group 315 is identical to load group 305 with the exception that it supports the logging of cancelled transactions when the cardholder declines the surcharge fee.

- Load Group 354

Load group 354 is required for dial up non-interactive surcharge processing with the BASE24-atm SSB add-on product. This group must be merged with groups 0, 12, and 305. Load group 354 does not support logging of cancelled transactions when the cardholder declines the surcharge fee.

- Load Group 355

Load group 355 is required for dial up non-interactive surcharge processing with the BASE24-atm SSB add-on product. This group must be merged with groups 0, 12, and 315. Load group 355 supports logging of cancelled transactions when the cardholder declines the surcharge fee.

- Load Group 364

Load group 364 is required for dial up non-interactive surcharge processing and Split Deposit support with the Base Application of the BASE24-atm SSB add-on product. This group must be merged with groups 0, 12, 52, and 305. Load group 364 does not support logging of cancelled transactions when the cardholder declines the surcharge fee.

- Load Group 365

Load group 365 is required for dial up non-interactive surcharge processing and Split Deposit support with the Base Application of the BASE24-atm SSB add-on product. This group must be merged with groups 0, 12, 52, and 315. Load group 365 supports logging of cancelled transactions when the cardholder declines the surcharge fee.

The following screens were modified in conjunction with the Surcharging enhancement:

- Screen 113
- Screen 123

The verbiage on Screen 113 was modified to reflect an expanded transaction set eligible for the assessment of surcharge fees. The position of the fee was moved on overlay screen 123.

NCR 5XXX Configuration (CNFG5070)

The following load groups were modified for the Surcharging enhancement:

- Load Group 304

Load group 304 is the Surcharge Group and was modified to present the Surcharge screen (screen108/118) for deposit transactions in addition to withdrawal transactions. Load group 304 does not support logging of

cancelled transactions when the cardholder declines the surcharge fee.

- Load Group 305

Load group 305 is the group required for dial-up ATM non-interactive surcharge processing. Load group 305 was modified to present the Surcharge screen for deposit transactions in addition to withdrawal transactions. Load group 305 does not support logging of cancelled transactions when the cardholder declines the surcharge fee.

- Load Group 307

Load group 307 is the group required for dial-up ATM interactive surcharge processing. Load group 307 was modified to present the Surcharge screen for deposit transactions in addition to withdrawal transactions. Load group 307 does not support logging of cancelled transactions when the cardholder declines the surcharge fee.

The following load groups were added for the Surcharging enhancement:

- Load Group 308

Load group 308 is required for dial up non-interactive deposit surcharge processing with the BASE24-atm SSB Basic Check Depository add-on product. Load Group 308 must be merged with groups 0, 106, 706*, and 305. (*Load group 706 is only required to be merged if Bunch Note Acceptor (BNA) and Check Processing Module (CPM) hardware is being utilized.) Load group 308 does not support logging of cancelled transactions when the cardholder declines the surcharge fee.

- Load Group 309

Load group 309 is required for dial up non-interactive deposit surcharge processing and Split Deposit support with the BASE24-atm SSB Basic Check Depository add-on product. Load Group 309 must be merged with groups 0, 106, 110, 706*, and 305. (*Load group 706 is only required to be merged if Bunch Note Acceptor (BNA) and Check Processing Module (CPM) hardware is being utilized.) Load group 309 does not support logging of cancelled transactions when the cardholder declines the surcharge fee.

- Load Group 314

Load group 314 is identical to load group 304 with the exception that it supports logging of cancelled transactions when the cardholder declines the surcharge fee.

- Load Group 315

Load group 315 is identical to load group 305 with the exception that it

supports logging of cancelled transactions when the cardholder declines the surcharge fee.

- Load Group 317

Load group 317 is identical to load group 307 with the exception that it supports logging of cancelled transactions when the cardholder declines the surcharge fee.

- Load Group 318

Load group 318 is identical to load group 308 with the exception that it supports the logging of cancelled transactions when the cardholder declines the surcharge fee. Load Group 318 must be merged with groups 0, 106, 110, 706*, and 315. (*Load group 706 is only required to be merged if Bunch Note Acceptor (BNA) and Check Processing Module (CPM) hardware is being utilized.) Load group 318 supports logging of cancelled transactions when the cardholder declines the surcharge fee.

- Load Group 319

Load group 319 is identical to load group 309 with the exception that it supports the logging of cancelled transactions when the cardholder declines the surcharge fee. Load Group 319 must be merged with groups 0, 106, 706*, and 315. (*Load group 706 is only required to be merged if Bunch Note Acceptor (BNA) and Check Processing Module (CPM) hardware is being utilized.) Load group 319 supports logging of cancelled transactions when the cardholder declines the surcharge fee.

The following screens were modified in conjunction with the Surcharging enhancement:

- Screen 108
- Screen 118

The verbiage on these screens was modified to reflect an expanded transaction set eligible for the assessment of surcharge fees.

Mega Table (MEGATBL) Changes

The TSRF requester name was added to MEGATBL file, since RQMEGA communicates with the TSRF requesters using the extended message format (i.e., use USER-CONTEXT-EXT instead of USER-CONTEXT).

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Entries were added to the SECTBL to support the screens for the TSRF requester/server.

PATHCONF Changes

The TSRF Server object is bound into the Classic Combined AFM Link Object (i.e., CCAFMLO).

CRT Access Security Changes

Access to the new TSRF requester screens must be added through the SEC requester for the appropriate users.

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Surcharge File (SURF)

The SURF conversion program must be run to convert the SURF. Running the conversion program is required for any BASE24-atm Surcharging user. Users must create the SURF file using the BADDLFUP file.

The SURF conversion program on the BA60UC04 subvolume can be used to convert the SURF records from the release 5.3 format to the release 6.0 format. The SURF conversion program on the BA60UC07 subvolume can be used to convert the SURF records from the release 6.0 version 3 format to the release 6.0 version 4 format. If modifications to the conversion program are necessary, the program can be modified by editing the CNVSURFS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#cnvsurf.t,name $srfm,nowait/ cnvsurfm –  
force –list –listreplace
```


Release 5.3 users should update the CNVSURF obey file on the BA60UC04 subvolume with the appropriate information. Release 6.0 version 3 users should update the CNVSURF obey file on the BA60UC07 subvolume with the appropriate information. The CNVSURF should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREADME file on the BA60UC04 subvolume for release 5.3 customers and the BA60UC07 subvolume for release 6.0 version 3 customers.

Host/Co-Network Impacts

Host/Co-Network impacts will be incurred by those BASE24 users implementing the BASE24-atm Surcharging enhancement. Modifications to existing files and processing will result in impacts to the external interfaces and Super Extract.

External Interfaces

The host or co-network may require modification to detect transaction fees associated with the expanded transaction set and to recognize and process new values within the Surcharge Data Token. The new data will be passed in the ISO message from the BASE24-atm ISO Host Interface (HISO) or BASE24 Interchange (BIC) ISO Interface.

Super Extract Process

Host code changes are required for all BASE24-atm users that utilize the OMF Extract Process to accept and process the new information in the SURF.

Host code changes are required for all BASE24-atm users that utilize the OMF Extract Process to accept and process information in the TSRF.

If the Surcharge Data Token is included in the TLF and ILF Extracts, BASE24-atm users must consider the new data within the token with regards to any host programs that use the data from the log files.

User/Operational Impacts

Disk Impacts

With the introduction of this enhancement, the Surcharge File (SURF) will increase significantly in size. BASE24 customers must examine their current disk capacity and determine whether the introduction of this enhancement into their system will require the purchase of additional disk space.

The Transaction Code/Subtype Relationship File (TSRF) is optional, but if introduced into the BASE24-atm database will impact disk space.

Operational Impacts

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate remedy for a given message. Refer to the event message documentation files on \$<volume>.BA60LOGM.

System Limitations

Existing BASE24-atm Surcharge users should not experience a significant change in messaging or logging to the ILF or TLF.

BASE24 users that introduce BASE24-atm Surcharging in conjunction with this enhancement should analyze their system for the possibility of exceeding internal system limitations for messaging and logging to the ILF and TLF. There are system limitations regarding the amount of data that can be passed through the BASE24 system, logged to a log file, and passed to the host or co-network in an ISO External Message. BASE24 users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens may cause them to exceed these limitations. The **BASE24 Tokens Manual** provides information on assessing the impacts of tokens on log file disk space requirements.

Vendor Impacts

BASE24-atm Surcharge users with NCR 5xxx devices will not be able to take advantage of any check related processing for dial up devices running APTRA Advance NDC. The Check Processing Module (CPM) is only supported in an APTRA Advance NDC environment, which does not provide dial up support. The Document Processing Module does provide check related processing and dial up support; but, is no longer being manufactured by NCR.

NDC+ Release 4.0 (or higher) is a requirement for BASE24-atm Surcharge users that choose to implement the new surcharge feature of logging cardholder cancelled transactions. The use of an Extension State for the Preset Operations Code Buffer State is required to signal the device handler that the cardholder cancelled transaction must be logged.

Reference Documents

Not applicable

Automated Key Distribution System

This section describes the implementation requirements associated with the Automated Key Distribution System (AKDS) enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Automated Key Distribution System enhancement.

A standard BASE24-atm module has been modified to support this enhancement. That module is the Diebold 10XX/478X device handler.

Prerequisite: The Transaction Security Services enhancement must be installed prior to implementation of the Automated Key Distribution System. For more information refer to the ***BASE24 Release 6.0 Implementation Guide (Version 2)***.

Software Impacts

All BASE24 customers:

- Changes to CUSTMACS and FLGSDOC to include the atm_dh_c1000_akd_on flag
- Inline changes to ACI Utilities (BA10UTIL.UTILS)
- Inline changes to the TSS Command Message DDL file (BA60DDL.DDLGTSS)
- Inline changes to the Security Utilities (BA60SRC.SECUTILS)
- Updated log message manual for the Security Utilities

All BASE24-atm customers:

- Changes to Make files to include the atm_dh_c1000_akd_on flag (AT60SRC.ATMFM, AT60SRC.ATMMM)

All customers with the Diebold 10XX/478X device handler:

- Hooks in the main device handler module to AKDS procedures (AT601000.C1000S)
- Minor inline changes in the main module and global file (AT601000.C1000G).
- Minor inline changes in the DCT screen and requester files (AT601000.SCND10XX and AT601000.RQD10XXS)
- Additional procedures in the dummy file (AT601000.C1000D)
- Make file changes (AT601000.AT1000MM and AT601000.C1000M)
- Minor inline changes to the dial-up extension (AT60P100.C100DLUS)
- Updated event (log) message manual with new events (BA60LOGM.AT1000)

Only customers taking the new AKDS functionality:

- New extension subvolume with files containing AKDS functionality (AT60A100)

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD07.SCNDKDS.

CUSTMACS Changes

The following Make flag has been added to the CUSTMACS file.

Set the following flag to true to bind the AKDS module to the Diebold 10XX/478X device handler.

- atm_dh_c1000_akd_on

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement. Note that these modifications are provided in the LCONFAT file located on the BA60MISC subvolume.

ATM-DH-AKDS-TSS-TIMER

This param is used by the Diebold 10XX/478X device handler. It specifies the number of seconds the device handler waits for a reply from Transaction Security Services before aborting the load. This is an optional param. Valid values are 10 through 600. If the param is not found or is invalid, the default value of 600 is used.

ATM-DH-1000-AKDS-SERIAL-NUM-VRFY

This param is used by the Diebold 10XX/478X device handler. It specifies whether or not Transaction Security Services will verify the Encrypting PIN Pad's (EPP's) serial number during the exchange of public keys between the EPP and BASE24. This is an optional param. Valid values are "Y" (yes, verify the serial number) and "N" (no, do not verify the serial number). If the param is not found or is invalid, the default value of "N" is used.

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Initial Registration and Certification for the BASE24 System

Initial registration and certification procedures must be carried out before an Authentication and Public Key Exchange sequence or a Master Key Download sequence can take place between BASE24 and an ATM. A public/private key pair must be generated for the BASE24 system using a host security module. Once the key pair is generated, Diebold's Certificate Authority—Digital Signature Trust Co. (DST)—must verify the BASE24 system's public key, create a signature for it, and package the key and signature in an X.509-format certificate. The Certificate Authority also creates a certificate containing its own public key. The certificates returned by the Certificate Authority must be entered into the Transaction Security Services database.

For more information about these initial registration and certification procedures, refer to the ***BASE24 Transaction Security Services Processing Guide***.

Diebold 10XX/478X Device Handler – DCT Commands

The following commands have been added to the Diebold 10XX Commands Screen (screen 40) of the Device Control Terminal (DCT).

- **LOAD PUBLIC KEY**

When the operator places the cursor on this command and presses function key F1-ENTER, the DCT sends a command to the device handler to initiate an Authentication and Public Key Exchange sequence. During this load sequence, BASE24 and the specified ATM authenticate each other and exchange public keys. These public keys are used for encryption and mutual authentication in all future master key downloads.

The Authentication and Public Key Exchange sequence can also be initiated at the ATM when in supervisor mode.

This command is typically performed only once – when a new ATM with an AKDS-enabled EPP is initially deployed, or when an ATM already in the field is fitted with a new AKDS-enabled EPP.

After this command completes successfully, the LOAD MASTER KEY command can be performed.

- **LOAD MASTER KEY**

When the operator places the cursor on this command and presses function key F1-ENTER, the DCT sends a command to the device handler to initiate a Master Key Download sequence. During this load sequence, BASE24 generates a new master key and downloads it to the specified ATM.

The Master Key Download sequence can also be initiated at the ATM when in supervisor mode.

The Authentication and Public Key Exchange sequence (see LOAD PUBLIC KEY) must complete successfully at least once before this command can be performed.

If the ATM is set to use single-length keys and BASE24 downloads a double-length master key, the ATM will automatically switch to double-length keys. If the ATM is set to use double-length keys and BASE24 downloads a single-length master key, the ATM will automatically switch to single-length keys. There is no need to configure the key length at the ATM prior to remotely transporting the master key.

This command can be performed as often as is necessary to preserve the integrity of the ATM's keys.

After this command completes successfully, the DES encryption keys (e.g. PIN encryption key, MAC key) may be generated and downloaded to the ATM.

Diebold 10XX/478X Device Handler – NCS/NCPCOM Commands

The following 9500 commands are now supported in the Diebold 10XX/478X device handler. They are entered at the BASE24 AFT NCS screen or through NCPCOM.

- **LOADPUBLIC**

This 9500 command is an alternative to the DCT's LOAD PUBLIC KEY command. It causes the device handler to initiate an Authentication and Public Key Exchange sequence. For more information refer to the LOAD PUBLIC KEY command in the preceding subsection (Diebold 10XX/478X Device Handler – DCT Commands).

The syntax of this command is as follows.

DELIVER PROCESS <process name>, "9500LOADPUBLIC <terminal ID>"

<process name>: The symbolic name assigned to the Diebold 10XX/478X device handler process in the XPNET node.

<terminal ID>: The terminal ID defined for the ATM in the ATM Terminal Data file (ATD).

- **LOADMASTER**

This 9500 command is an alternative to the DCT's LOAD MASTER KEY command. It causes the device handler to initiate a Master Key Download sequence. For more information refer to the LOAD MASTER KEY command in the preceding subsection (Diebold 10XX/478X Device Handler – DCT Commands).

The syntax of this command is as follows.

DELIVER PROCESS <process name>, "9500LOADMASTER <terminal ID>"

<process name>: The symbolic name assigned to the Diebold 10XX/478X device handler process in the XPNET node.

<terminal ID>: The terminal ID defined for the ATM in the ATM Terminal Data file (ATD).

Power-Up Considerations

Prior to the Automated Key Distribution System enhancement, the Diebold 10XX/478X device handler automatically performed a full load (or a load of just the PIN encryption and MAC keys) after receiving a power-up message from an ATM. With this enhancement the device handler will continue to do so, but there is an exception: The load will not be automatically performed if the ATM has an AKDS-enabled EPP and a master key has not yet been downloaded to the EPP. Instead, the device handler will generate an event message informing the operator that the master key must be loaded. The master key must be present before a PIN encryption key or MAC key can be loaded.

Financial Institution Table (FIT) Considerations

The local PIN verification key (PEKEY), if used, is encrypted under the master key and stored in the FIT. If a new master key is downloaded to an ATM, every PEKEY in the FIT must be re-encrypted under the new master key and the FIT must be downloaded to the ATM.

Supported Keys

Only the master key (A-key) can be downloaded to an ATM using the AKDS enhancement. AKDS does not support the downloading of the power-up COMM key (B-key) because this key is not currently supported by the Diebold 10XX/478X device

handler. The DES keys (e.g. PIN encryption key, MAC key) will continue to be encrypted under the master key and downloaded to the ATM as per normal procedures prior to this enhancement.

Event Messages

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate remedy for a given message. Refer to the event message documentation files on \$<volume>.BA60LOGM.

Vendor Impacts

Compatible Devices and Software

The following device platforms and software versions are compatible with BASE24's Automated Key Distribution System.

- Diebold model 10XX ATMs running TCS 5.4 (for OS/2) or TCS Plus 1.3.
- Atalla NSP HSM models A8100, A9100 and A10100. Software version AKB 1.20 or later is required. AKB products require the use of a Secure Configuration Terminal (SCT), release 4.0 or later. SCT 4.0 devices support the Atalla AKB products while remaining fully backward compatible with older Atalla releases.
- Thales e-Security (Racal) HSM models RG7100, RG7110, RG7300, RG7310 and RG7400. Software version 6.0 or later and optional DSP plug-in (for RSA functions) are required.

Encrypting PIN Pad

An Encrypting PIN Pad (EPP) must be installed on every ATM that is to implement AKDS. An EPP is a tamper-proof ATM component comprised of a PIN pad, a security module to perform cryptographic processing, and the cryptographic keys. All of the ATM's cryptographic functions are performed by the EPP.

Protocol Considerations

Some ATM communication protocols may not support the increased message lengths required by the AKDS enhancement. Current device and protocol configurations should be discussed with the ATM vendors prior to implementing AKDS.

Reference Documents

- Atalla: Atalla Key Block Banking Command Reference Manual (524130-002, July 2002)
- Diebold: Certificate Management (June 24, 2002)

- Diebold: Remote Key Transport (June 24, 2002)
- Diebold: Triple DES Requirements FIRST Key – 91x Message Formats (June 26, 2002)
- Thales: Host Security Module RG7000 Programmer's Manual (1270A514 Issue 5, July 2002)

Diebold TCS Enhancements Support

This section describes the implementation requirements associated with the BASE24-atm Diebold TCS Enhancements support.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the BASE24-atm Diebold TCS Enhancements support.

Standard BASE24-atm modules have been modified to support this enhancement. These modules include the Diebold Entry Program, the Diebold 10XX/478X device handler and the Receipt requester/server.

Software Impacts

All BASE24-atm customers:

- Changes to the AT60AFT.COBRCPT file to add new receipt field type codes

All Diebold 10XX/478X device handler customers:

- Inline changes to support TCS 4.5 – 5.3 Enhancements (C1000S)
- Minor inline changes to the global files (C1000G)
- DDL changes to the Diebold ATD device dependent data file (C100DDLS)
- Updated Log (Event) Message Manual AT1000

All Diebold 10XX/478X Self Service Banking (SSB) customers :

- Inline code changes in the SSB Device Handler package (C100SSDS)

All BASE24-atm Diebold Entry Program (i.e., AT20IBLD) users:

- DDL changes to the Diebold Entry Program DDL file CNFGDDL
- Minor changes to the Diebold Entry Program Globals file ENTIBLDG
- In line code changes to the following Diebold Entry Program files: ENTFORMS, ENTIBLDS, ENTMISCS, ENTPRNTS and ENTSTATS.

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD07.SCNTCS.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

BASE24-atm Receipt File (RCPT)

Three new field type codes, EQ, ER and ES have been introduced with this enhancement. These field types are only allowed for Diebold 1000 devices (i.e., device type D1000).

The EQ field type code is used for printing a scanned check amount. The ER field type code is used for printing a scanned check image to a thermal consumer printer. The ES field type code is used to select a specific character set for printing.

Additional information on specific field type codes may be found in the ***BASE24-atm Diebold 10xx/478x Device Support Manual*** and the ***BASE24-atm Diebold 10xx/478x SSB Manual***. Information on use of the RCPT file may be found in the ***BASE24-atm Files Maintenance Manual***.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

With this enhancement no changes were made to the Diebold Configuration data file (CNFGIX). Instead support was added to allow customers to take advantage of various TCS 4.5 – 5.3 Enhancements.

Support (via the Diebold Entry Program) was added for three new states: the Buffer Shift State (@N), the Copy Buffer State (@P), and the Enhanced Set Language Bank State (@Y). Customers wanting to take advantage of these states will need to make their own configuration changes.

Changes were also made to the Enhanced Card Reader State (CR) Operation field to allow values of 013, 014, and 015. These values are used to support Dip Smart Card Readers.

The Voice State (;) was enhanced to support the Beeping/Voice Prompting Coordination Flag field. Valid values for this field are 000 (voice prompting commands take precedence, beeping is suppressed) and 001 (voice prompting commands and beeping coexist, both are heard).

Finally, the value range of state numbers has been increased by three values. In releases prior to TCS 5.3, the range was 000 – 250, 256 – 999, and A00 – ZZZ. As of TCS 5.3 the range has been extended to include 251, 252, and 253 (i.e., TCS 5.3 supports 000 – 254 and 256 – ZZZ as valid state numbers).

The Miscellaneous Features 2 field now allows a weighted value of 32. A value of 32 allows the MDS status bytes to be sent to the network in solicited and unsolicited status messages. The MDS status bytes provide additional information regarding a particular device.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

A new field was defined in the Diebold Device Dependent area of the ATD. (Existing USER-FLD-ACI was redefined.) An inline conversion of the ATD will be performed to initialize this field to zero. If a release 5.3 TDF is being converted to release 6.0 version 4, the conversion program will initialize the new field.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

User Impacts

Customers running TCS 5.2 or TCS 5.3 will now be able to limit the number of bills that can be dispensed from an ATM based upon configuration information received from the ATM rather than a hard coded limit. If the ATM is configured to send the Double Expanded Hardware Configuration Status field and bytes 20 and 21 of this field are sent to the device handler, the device handler will use this information when determining the maximum number of bills it can dispense. If this field is not sent to the device handler, it will continue to use the existing hard coded bill limit value.

If the device handler receives a Solicited or Unsolicited Status message from an ATM containing the MDS Status field, the device handler will now display to EMS the MDS Status information. Additional processing of the MDS Status information by the device handler will not be done. (NOTE: The sending of the MDS Status field is controlled via the Write Command III, Miscellaneous Features 2 Message).

Operational Impacts

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new messages and perform the appropriate remedy for a given message. Refer to the Diebold 10XX/478X Event Message documentation file \$<volume>.BA60LOGM.AT1000.

Vendor Impacts

The appropriate level of TCS must be installed on a device in order to take advantage of any of the above TCS 4.5 – 5.3 Enhancements. Customers running prior versions of TCS will still be supported but will not be able to take advantage of any of the new enhancements.

Reference Documents

Diebold 91X Terminal Control Software (TCS), TCS Plus, and 91X TCS CSP
Terminal Programming Manual, version N

Dual Site Support

This section describes the implementation requirements associated with Dual Site Support.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Dual Site Support enhancement.

Sections 4 (Base) and 6 (POS) contain product specific changes relating to the Dual Site Support enhancement.

Dual Site Support involves changes related to the management of a Dual Site BASE24 production environment. This enhancement is to support customers who are planning to implement a Dual Site production environment using GoldenGate to facilitate data replication between the two sites.

A Dual Site BASE24 production environment is where BASE24 is running on two separate NSK machines, with production transactions being processed on both machines. Standard BASE24 files will be held on both systems with updates replicated between the two systems by GoldenGate.

The ATD file has been enhanced to enable the identification of individual terminals as “local” or “remote”, i.e. whether the terminal is normally connected to this system or the alternate system. (Records for each terminal will appear on both copies of the file, with one set to local and one to remote according to the normal connectivity of the terminal.) As records on either system are updated, GoldenGate will replicate the changes to the alternate system and alter the setting of the local / remote flag.

The TLF has been enhanced to enable the identification of individual records as “local” or “remote”, i.e. whether the transaction was authorized on this system or the alternate system. Local records are standard and remote records include a new field, SITE^IND, in the Multiple Logical Network token. (Records for all transactions will appear on both copies of the files, with one copy containing the remote token according to the location of the original transaction.) As records on either system are added, GoldenGate will replicate the records to the alternate system, add the token if not already present, and update the field.

Settlement, Perusal, ATD file access and DCT have been enhanced to support and provide options relating to Dual Site Support. Settlement can be configured to process local only, remote only or all records and terminals during cutover processing and has been enhanced to support broadcast capability in notify lists and process specific TLF assign. TLF perusal has been enhanced to enable the identification of remote (replicated) transactions on the local log file. ATD file access supports a new field, DUAL^SITE^IND, indicating the normal connectivity of the terminal in question,

and DCT enables transparent communication with an individual terminal regardless of its current connectivity.

For more details on using these enhancements and GoldenGate to support a Dual Site production environment, refer to the ***BASE24 Dual Site Processing Guide***.

Software Impacts

For full details on the software impacts for this enhancement see Software Impacts in Section 4 (Base) of this document.

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD07.SCNDUALS.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

This section describes the Assign records that must be added to the LCONF to support this enhancement. These modifications are provided in the LCONFAT file located on the BA60MISC subvolume.

TLF

The fully qualified file name of the Transaction Log File (TLF). This assign is now supported as a process specific assign for BASE24-atm Settlement Initiator process and the Transaction Log File (TLF) Server.

REMOTE-PMON

PPD of Pathmon process on alternate system, enabling the local Device Control Terminal (DCT) Server to communicate with its remote counterpart to support remotely connected terminals.

REMOTE-LCONF

Location of remote LCONF, passed by the local Device Control Terminal (DCT) Server to its remote counterpart to support remotely connected terminals.

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement. These modifications are provided in the LCONFBA file located on the BA60MISC subvolume.

DUAL-SITE-MODE

This param is used by the BASE24-atm Settlement Initiator process and the Device Control Terminal (DCT) Server. It specifies the dual site mode for the process. Valid values are:

L - Local records only to be processed by Settlement / Activates SERVER-DCT Dual Site processing

R - Remote records only to be processed by Settlement / Deactivates SERVER-DCT Dual Site processing

A - All records to be processed by Settlement / Deactivates SERVER-DCT Dual Site processing

DUAL-SITE-DISPLAY

This param is used by the Transaction Log File (TLF) Server. It defines whether the server uses process specific TLF assigns or not. Valid values are Y - Server TLF uses process specific TLF assign and identifies remote records with REMOTE TXN in bottom right corner of detail display and N - Server TLF Dual Site processing deactivated. If the param is not present the default value is N.

BROADCAST-NOTIFY

This param is used by the BASE24-atm Settlement Initiator process. It defines whether the process uses the broadcast facility. Valid values are 0 - Broadcast facility not to be used by Settlement, 1 - Broadcast flag to be set to 1 (broadcast to all service providers) for notify messages sent by Settlement and 2 - Broadcast flag to be set to 2 (broadcast to running service providers) for notify messages sent by Settlement.

BROADCAST-DELAY

This param is used by the BASE24-atm Settlement Initiator process. It defines the number of seconds to delay after file creation before broadcasting to the notify list.

Database Configuration

This section describes the new fields in the files that have been modified to support this enhancement.

Transaction Log File (TLF)

Field Name: SITE INDICATOR
Screen: TRAN LOG LIST Detail Screen 2
DDL Name: MULT^LN^TOKEN.SITE^IND

A value of "p" in the SITE^IND field indicates that the transaction was authorized on the alternate system (and the record replicated by GoldenGate). A value of " " or the absence of the token indicates that the transaction was authorized locally.

This field is maintained by GoldenGate and does not require any action by the customer.

ATM Terminal Definition File (ATD)

Field Name: DUAL SITE INDICATOR
Screen: ATD Screen 1
DDL Name: ATDD1.CORE.DUAL^SITE^IND

A value of "L" or " " (default value) in the DUAL^SITE^IND field indicates that the terminal is typically connected to the system on which the file is located. A value of "R" in the DUAL^SITE^IND field indicates that the terminal is typically connected to the alternate system. Records for each terminal will be present on each copy of the file and the DUAL^SITE^IND flag set by GoldenGate during replication.

This field needs to be set to "L" or "R" prior to Dual Site implementation according to the normal terminal connectivity. After initial configuration the field is maintained by GoldenGate and does not require any action by the customer.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Host/Co-Network impacts will be incurred by those BASE24 users implementing the Dual Site Support enhancement. Modifications to existing files and processing will result in impacts to Super Extract.

Super Extract Process

Host code changes are required for all BASE24-atm users that utilize the OMF Extract Process to accept and process the new information in the ATD.

If the Multiple Logical Network Token is included in the TLF and ILF Extracts, BASE24-atm users must consider the new data within the token with regards to any host programs that use the data from the log files.

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Dynamic Key Management

This section describes the implementation requirements associated with the Dynamic Key Management for BASE24-atm.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Dynamic Key Management enhancement.

Standard BASE24-atm modules have been modified to support this enhancement. Those modules are the Diebold 10XX/478X, NCR 5XXX and Triton device handlers.

Prerequisite: The ATD Infrastructure enhancement must be installed prior to implementing the Dynamic Key Management enhancement in the Diebold 10XX/478X or NCR 5XXX device handlers. In the ATD Infrastructure enhancement, the ATD records were expanded and new fields were added to the Dynamic Core data area to support Dynamic Key Management. For more information refer to the ***BASE24 Release 6.0 Implementation Guide (Version 3)***.

Software Impacts

All BASE24 customers:

- Changes to CUSTMACS for this enhancement that were a part of release 6.0 version 3

All BASE24-atm customers:

- Changes to the AT60SRC.ATMFM and ATMMM files to include the DKM flags
- DDL changes for this enhancement that were a part of release 6.0 version 3

All customers with Diebold 10XX/478X, NCR 5XXX or Triton device handler:

- Hooks in the main device handler module to DKM procs
- Minor inline changes in the main module and global files
- Additional procs in the Dummy files (i.e., AT601000.C1000D, AT60N50.N50D)
- A new dummy file for the Triton subvolume (i.e., AT60TRIT.TRITOND)
- Make file changes that include the DKM flag
- Minor inline changes to the Dial-up product files (i.e., AT60P100.C100DLUS, AT60PN50.N50DLUS)
- New Log (Event) Message Manuals for each device handler with new DKM events

Only customers taking the new DKM functionality :

- New files in the extension subvolume containing DKM functionality (i.e., AT60K000, AT60KN50, AT60KTRT). The files on these subvolumes include

source files, global files and Make files required for the new functionality.

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD07.SCNDKM.

CUSTMACS Changes

The following Make flags have been added to the CUSTMACS file.

Set the following flag to true to bind the DKM module to the Diebold 10XX/478X device handler.

- atm_dh_c1000_dkm_on

Set the following flag to true to bind the DKM module to the NCR 5XXX device handler.

- atm_dh_n50_dkm_on

Set the following flag to true to bind the DKM module to the Triton device handler.

- atm_dh_triton_dkm_on

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement. The params are the same for all three device handlers. Different param records may be specified for each device type or device handler process, if desired.

Note that these modifications are provided in the LCONFAT file located on the BA60MISC subvolume.

ATM-DH-CONS-MAC-ERR-LMT

The number of consecutive message authentication code (MAC) errors that can occur before the device handler process downloads a new MAC and PIN key to the ATM. Valid values are 0 through 32000. A value of 0 indicates that the device handler process does not reload keys regardless of the number of consecutive MAC errors.

ATM-DH-DKM-KEY-CHNG-INTRVL

This param specifies the maximum time allowed between key changes before the device handler downloads a new MAC and PIN key to the ATM. The value is in

hours. Valid values are 0 through 32000. A value of 0 indicates that the device handler process does not reload keys regardless of the length of time since the last key change.

ATM-DH-DKM-MAC-ERR-LMT

This param specifies the number of message authentication code (MAC) errors that can occur before the device handler process downloads a new MAC and PIN key to the ATM. Valid values are 0 through 32000. A value of 0 indicates that the device handler process does not reload keys regardless of the number of MAC errors.

ATM-DH-DKM-MAC-TXN-LMT

This param specifies the number of message authentication code (MAC) generations or verifications that can occur before the device handler process downloads a new MAC and PIN key to the ATM. Valid values are 0 through 32000. A value of 0 indicates that the device handler process does not reload keys regardless of the number of MAC generations or verifications.

ATM-DH-DKM-PIN-ERR-LMT

This param specifies the number of personal identification number (PIN) verification errors that can occur before the device handler process downloads a new MAC and PIN key to the ATM. Valid values are 0 through 32000. A value of 0 indicates that the device handler process does not reload keys regardless of the number of PIN verification errors.

ATM-DH-DKM-PIN-TXN-LMT

This param specifies the number of personal identification number (PIN) verifications that can occur before the device handler process downloads a new MAC and PIN key to the ATM. Valid values are 0 through 32000. A value of 0 indicates that the device handler process does not reload keys regardless of the number of PIN verifications performed.

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

The new fields required in the ATD were added as a part of the ATD Infrastructure enhancement. As a part of that enhancement, the ATD records were expanded and the new fields needed for DKM were added to the Dynamic Core data area to support Dynamic Key Management in the ATD. The ATD Infrastructure enhancement must be installed prior to installing the Dynamic Key Management enhancement.

An inline conversion of the ATD and/or TDF will be performed to initialize the new data fields defined for Dynamic Key Management. If a release 5.3 TDF is being converted to release 6.0 version 4, the conversion program will initialize the new DKM data fields in the ATD and/or TDF.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

This enhancement will automatically load new PIN and MAC keys to ATM terminals. Messages will be generated when the keys are loaded. Operators must be aware of this new functionality. Key load commands issued by operators will function as before. With this enhancement in place, the counters for transaction and error counts will be reset when an operator-initiated key load is performed.

The Diebold 10XX/478X and NCR 5XXX device handlers currently track consecutive MAC verification errors. Once the limit for the counter is reached, the ATM is shut down. Operator intervention is required to send a new MAC key and reopen the ATM. This enhancement will modify the functionality so the MAC key will automatically be sent (along with the PIN key) and the ATM reopened.

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate remedy for a given message. Refer to the event message documentation files on \$<volume>.BA60LOGM.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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EMV M/Chip 4 Support

This section describes the implementation requirements associated with the EMV M/Chip 4 Support enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the EMV M/Chip 4 Support enhancement.

For additional detailed information on this enhancement see Section 4 (Base).

The EMV processing has been enhanced to accommodate a new Issuer Application Format. M/Chip 4 Status Field Checking processing has been enhanced.

Software Impacts

For full details on the software impacts for this enhancement see Software Impacts in Section 4 (Base) of this document.

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD07.SCNMCHP4.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

The Card Verification Results field is a different format on an M/Chip 4 transaction. It is a 6 byte field. The decisional bytes of the field are bytes 4, 5 and 6. The

information contained in these bytes is held in the CVRTBLS and is as follows:

Action Index	Byte	Bit	Auth Action	Meaning
"n"	4	7	auth_act_cont_d	Unable to go Online
"n"	4	6	auth_act_cont_d	Offline PIN verification not performed
"n"	4	5	auth_act_cont_d	Offline PIN verification failed
"n"	4	4	auth_act_cont_d	PTL exceeded
"n"	4	3	auth_act_cont_d	International transaction
"n"	4	2	auth_act_cont_d	Domestic transaction
"n"	4	1	auth_act_cont_d	Terminal erroneously considers Offline PIN OK.
"n"	5	8	auth_act_cont_d	Lower consecutive Offline limit exceeded
"n"	5	7	auth_act_cont_d	Upper consecutive Offline limit exceeded
"n"	5	6	auth_act_cont_d	Lower Cumulative Offline limit exceeded
"n"	5	5	auth_act_cont_d	Upper Cumulative Offline limit exceeded
"n"	5	4	auth_act_cont_d	Go online on next transaction was set
"n"	5	3	auth_act_cont_d	Issuer Authentication failed
"n"	5	2	auth_act_cont_d	Script received
"n"	5	1	auth_act_cont_d	Script failed
"n"	6	2	auth_act_cont_d	Match found in additional check table
"n"	6	1	auth_act_cont_d	No match found in additional check table

The placeholder for all implementations is:

"1", 4, 4, "N", auth_act_cont_d,

This is interpreted as Issuer script processing failed on last transaction (VISA/MCHIP2) and PIN Try Limit exceeded (MCHIP4).

An M/Chip 4 Base Test set has been added to the table which satisfies the rules for UK Switch Debit Card Scheme:

"3", 5, 3, auth_act_dcln_return_d, !Issuer Authentication failed on last on-line
!transaction

During migration, 2 separate tables and action indexes will need to be used to cater for both M/Chip 2.1 and M/Chip 4 transactions.

Vendor Impacts

Not applicable

Reference Documents

- MasterCard Europe - M/Chip 4 card application specification, version 4.0, 16th May 2002
- EMVCo - EMV 2000 version 4.0, December 2000

Forced Cutover Audit

This section describes the implementation requirements associated with Forced Cutover Audit (FCA) for BASE24-atm.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Forced Cutover Audit enhancement.

Standard BASE24-atm modules have been modified to support this enhancement. Those modules are ATM Settlement, the Diebold 10XX/478X and NCR 5XXX device handlers.

Software Impacts

All BASE24 customers:

- CUSTMACS changes to add new Make flags

All BASE24-atm customers:

- Changes to the AT60SRC.ATMFM and ATMMM files to include the FCA flags
- DDL changes for this enhancement

All customers with Diebold 10XX/478X or NCR 5XXX device handler:

- Hooks in the main device handler module to FCA procs
- Minor inline changes in the main module and global files
- Additional procs in the Dummy files (i.e., AT601000.C1000D, AT60N50.N50D)
- Make file changes that include the FCA flag
- Minor inline changes to the Dial-up product files (i.e., AT60P100.C100DLUS, AT60PN50.N50DLUS)
- New Log (Event) Message Manuals for each device handler with new DKM events

Only customers taking the new FCA functionality :

- New files in the extension subvolume containing FCA functionality (i.e., AT60F100, AT60FN50). The files on these subvolumes include source files, global files and Make files required for the new functionality.

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD07.SCNFCA.

CUSTMACS Changes

The following Make flags have been added to the CUSTMACS file.

Set the following flag to true to bind the FCA module to ATM Settlement Initiator.

- atm_fca_on

Set the following flag to true to bind the FCA module to the Diebold 10XX/478X device handler.

- atm_dh_c1000_fca_on

Set the following flag to true to bind the FCA module to the NCR 5XXX device handler.

- atm_dh_n50_fca_on

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement.

Note that this modification is provided in the LCONFAT file located on the BA60MISC subvolume.

ATM-FCA-CUTOVER-AUDIT

Setting this param to 'Y' turns on the Forced Cutover Audit functionality.

The param should be set for the entire logical network rather than individual process. ATM Settlement will send 9500FCA messages to all Diebold 10XX/478X and NCR 5XXX terminals if this param in the LCONF is set to Y. This will enable the device handlers that receive the messages to process them properly.

Database Configuration

BASE24-atm Receipt File (RCPT)

The Administrative templates in the RCPT File are used to display the terminal transaction and balancing information. The same receipts used for card balancing a terminal are used to format the required information on the journal printer when a 9500FCA message is received. No changes to this are required.

Additional information on specific field type codes may be found in the **BASE24-atm Diebold 10xx/478x** or **BASE24-atm NCR 5xxx Device Support Manual**. Information on use of the RCPT file may be found in the **BASE24-atm Files Maintenance Manual**.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

The new fields to support FCA have been added to the Terminal Data Dynamic file (ATDD1) for each device type supported using the additional offset method. This

means the data fields will exist in the data file only if FCA functionality is bound in and turned on (using the LCONF param).

ATM Settlement, the Diebold 10XX/478X and NCR 5XXX device handlers have been modified to add the new data fields in the additional offset in records where the data fields do not exist. No separate conversion program is required.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

This enhancement will cause ATM Settlement to send 9500FCA messages to the Diebold 10XX/478X and NCR 5XXX device handlers for each terminal that Settlement force balances. Operators may see these messages in device handler queues.

When a Diebold 10XX/478X and NCR 5XXX device handler receives a Forced Cutover message, it will shutdown the appropriate terminal. Cutover information (hopper totals and terminal counts) will be sent to the terminal to be printed on the journal printer. The device will then be reopened automatically. Operators must be aware of this new functionality since there will be event messages generated by the shutdown and reopening of the terminal. If the terminal is unable to print the cutover information, the hopper totals and terminal counts will be visible as event messages. No operator action is required for these messages.

Other new event messages have been added for this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate remedy for a given message. Refer to the event message documentation files on \$<volume>.BA60LOGM.

Vendor Impacts

Not applicable

Reference Documents

Not applicable.

Fujitsu 7000 Device Handler

BASE24-atm release 6.0 has been enhanced to support Fujitsu 7000 ATMs. The BASE24-atm Fujitsu 7000 Device Handler module provides the interface between the device and Compaq NSK. The Fujitsu 7000 ATMs are defined in the BASE24-atm Terminal Data File (TDF). This device handler was first introduced in BASE24-atm Release 5.1.

Refer to the BASE24-atm Fujitsu 7000 Device Handler manual on the BASE24-atm system (i.e., \$<volume>.AT60FUJ7.FUJ7MAN) for information on implementing this device and device handler into a BASE24-atm user's system.

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Fujitsu 7000 Triple DES

This section describes the implementation requirements associated with the Fujitsu 7000 Triple DES enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Fujitsu 7000 Triple DES enhancement.

The standard Fujitsu 7000 device handler BASE24-atm module has been modified to support Triple DES encryption. In order to support Triple DES, the device uses double length keys. Prior to this enhancement, the security keys were stored in BASE24 and the ATM devices as 16 hex characters or 8 bytes. However, support of double length keys requires BASE24 and the ATM device to store keys as 32 hex characters or 16 bytes. In order to support the double length keys, the Transaction Security Services is required.

Prerequisite: The Transaction Security Services enhancement must be installed prior to implementation of the Fujitsu Triple DES enhancement. For more information refer to the ***BASE24 Release 6.0 Implementation Guide (Version 2)***.

Software Impacts

All customers with the Fujitsu 7000 device handler:

- Inline changes in the main module and global file

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Channel Security Data Source (CSECD)

Transaction Security:	Channel Security Configuration
Tab:	PIN Information
Field:	Key Length of Exchange Key

The Key Length of the Exchange Key for the Fujitsu 7000 CSECD records must be set to Double to utilize Fujitsu 7000 Triple-DES using double-length keys.

Transaction Security:	Channel Security Configuration
Tab:	PIN Information
Field:	Exchange Key (Header, Part 1-4, Check Digits)

The Exchange Key for the Fujitsu 7000 CSECD records must be modified to contain the new double-length key value.

Transaction Security:	Channel Security Configuration
Tab:	PIN Information
Field:	Key Length of Inbound Key

The Key Length of the Inbound Key for the Fujitsu 7000 CSECD records must be set to Double to utilize Fujitsu 7000 Triple-DES using double-length keys.

Transaction Security:	Channel Security Configuration
Tab:	PIN Information
Field:	Inbound Key (Header, Part 1-4, Check Digits)

The Inbound Key for the Fujitsu 7000 CSECD records must be modified to contain the new double-length key value.

Refer to the online help provided with the ACI Desktop user interface for detailed descriptions of this window and its fields.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Fujitsu 7000 devices supporting Triple DES encryption must have new software and hardware configured to use the Triple DES encryption algorithm and double-length keys. Additionally the new double-length Master key must be manually entered at the device and the new working key must be downloaded.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Compatible Devices and Software

The following device platforms and software versions are compatible with BASE24's Fujitsu 7000 Triple DES enhancement.

- Refer to the Fujitsu 7000 vendor documentation to verify device compatibility of the following components to support Fujitsu 7000 Triple-DES:
 - Eprom Version
 - Program Version
 - Table Version
 - Fujitsu 7000 ATM models

Reference Documents

- ***Fujitsu: Host Programmer's Reference Guide for Windows-based ATMs.***
(Part Number 07100-00010A 09/2002)

Interactive Response Notification Fee Processing and Negative Surcharge Support

This section describes the implementation requirements associated with the Interactive Response Notification Fee Processing and Negative Surcharge Support enhancements for the Tidel Device Handler.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Interactive Response Notification Fee Processing and Negative Surcharge Support enhancements.

Software Impacts

All BASE24 customers:

- COBNAMES change to the Tidel Receipt table

All Customers with Tidel device handler:

- Inline changes in the main module and global files
- A new log message manual for the device handler with new messages
- A new Tidel manual

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

For the Interactive Response Notification Fee enhancement:

BASE24-atm Receipt File (RCPT)

The RCPT File can be used to display issuer fee or rebate information on a receipt.

The IF/ENDIF directives use the ISSFEES or NOISSFEES condition values for issuer fee or rebate receipt information.

The S2 field type code is used for the surcharge amount, and the S3 field type code

is used for the issuer fee or rebate amount.

The A1 field type code contains the withdrawal amount plus any surcharge or issuer fees or rebates.

BASE24 Token File (TKN)

The following BASE24 Base token was added with BASE24 Release 6.0 Version 2 for this enhancement:

- Issuer Fee Rebate token (Token ID “30”) contains data relating to the transaction issuer fee or rebate amount assessed. This token is used if issuer fees or rebates are being processed.

For more information on the changes introduced with BASE24 Release 6.0 Version 2 for this enhancement, refer to the **BASE24 Release 6.0 Implementation Guide (Version 2)**.

If the Issuer Fee Rebate token is not to be logged to the Transaction Log File (TLF) or Interchange Log File (ILF), the appropriate token records must be configured in the TKN.

If the Issuer Fee Rebate token is to be extracted with the TLF, sent to the host, or passed through the BIC ISO Interchange Interface process, the appropriate token records must be configured in the TKN.

For the Negative Surcharge enhancement:

BASE24 Surcharge File (SURF)

The following existing fields in the Surcharge File (SURF) must be configured when performing negative surcharges:

Field Name: FLAT FEE
 PERCENT FEE
Screen: SURF screens 2 and 3
DDL Name: SURF.ATM.TRAN.FLAT-FEE
 SURF.ATM.TRAN.PCNT-FEE

Screens 2 and 3 of the SURF contain the surcharge information for transactions routed using the CPF and SPROUTE file, respectively. Either the flat fee, in whole and fractional currency units, can be used as a surcharge, or the percent fee, in whole and fractional units, can be used when calculating the surcharge. A flat fee, percent fee, or both must be defined for each transaction code. When using negative surcharges, a negative value can be entered in these fields. For additional information on these fields, refer to the **BASE24 Base Files Maintenance Manual**.

Interactive Response Notification Fee Processing and Negative Surcharge Support

The interactive response notification of fees can include both issuer fees and acquirer fees (i.e., surcharges). BASE24 maintains information in the Surcharge File (SURF) only for acquirer fees and does not maintain information in its database for issuer fees. The setting of issuer fees must be done at a host system or an interchange whose BASE24 interchange interface module has been enhanced to carry the issuer fee information in the new BASE24 Issuer Fee Rebate token.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

The Tidel terminal must have the Optional Field Group - Issuer Fee Enabled option configured for this enhancement to be invoked.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

External Interfaces

The host or co-network must be modified to accept and process the Issuer Fee Rebate token in the ISO message from the BASE24-atm ISO Host Interface (HISO) or BASE24-atm BIC ISO process.

If negative surcharges are being used, the host or co-network must be modified to accept and process negative amounts in the existing Surcharge Data token in the TLF or ILF.

Super Extract Process

If the new Issuer Fee Rebate token is to be included in the TLF or ILF Extracts, BASE24-atm users must consider this additional data in host programs that use the data from the log files.

If negative surcharges are being used, the host must be modified to accept and process negative amounts in the existing Surcharge Data token in the TLF or ILF.

User/Operational Impacts

Disk Impacts

The TLF and ILF can require additional disk space because the new token for this enhancement (i.e., Issuer Fee Rebate token) can be included in the transaction records logged to these files.

If the TLF and/or ILF is extracted to a disk file, the extract record size is increased. This can increase the disk/tape space required for the extract file.

Interactive Response Notification Fee Processing and Negative Surcharge Support

Operational Impacts

New event messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on \$<volume>.BA60LOGM or the BASE24 Documentation CD-ROM.

System Limitations

BASE24 users installing this enhancement must be aware of their internal system limitations for messaging to the TLF and ILF. There are system limitations regarding the amount of data that can be passed through the BASE24 system, logged to a log file, and passed to the host or co-network in an ISO External Message. BASE24 users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the **BASE24 Tokens Manual** provides information on assessing the impacts of tokens on log file disk space requirements.

Vendor Impacts

The minimum terminal software version level required for this enhancement is release 06.50.

Reference Documents

Not applicable

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Multiple Currency Dispense Exchange Rate Notification

This section describes the implementation requirements associated with the BASE24-atm Multiple Currency Dispense (MCD) Exchange Rate Notification enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the BASE24-atm Multiple Currency Dispense Exchange Rate Notification enhancement.

The Multiple Currency Dispense enhancement allows cardholders to select the currency to dispense cash in for standard BASE24-atm cash withdrawal transactions (withdrawal from checking account, withdrawal from savings account, withdrawal from credit account). A customer may configure from two to four different currencies that a cardholder can select to withdraw cash in, but only a single currency may be selected and dispensed per withdrawal transaction. This enhancement was added in Release 6.0 Version 3. The MCD Exchange Rate Notification enhancement expands the Multiple Currency Dispense logic to display the exchange rate to the cardholder who can accept or deny the transaction based on the information presented.

The MCD Exchange Rate Notification enhancement is supported only in the Diebold 10XX/478X and NCR 5XXX device handlers and requires the Multiple Currency Dispense enhancement. It is compatible with the Multiple Currency Support enhancement, but the Multiple Currency Support enhancement is not required. In other words, the Multiple Currency extensions to the BASE24-atm Authorization module are not required in order for the Multiple Currency Dispense enhancement to function.

Prerequisite: The Multiple Currency Dispense enhancement must be installed prior to implementation of the MCD Exchange Rate Notification enhancement. For more information refer to the ***BASE24 Release 6.0 Implementation Guide (Version 3)***.

Refer to the ***BASE24-atm Diebold 10XX/478X*** or ***BASE24-atm NCR 5XXX Device Support Manual*** for additional information on the MCD Exchange Rate Notification enhancement.

Refer to the ***BASE24-atm Multiple Currency Support Manual*** for additional information on the Multiple Currency Support enhancement.

Software Impacts

All customers with Diebold 10XX/478X or NCR 5XXX device handler:

- DDL changes in the device specific DDLs (i.e., AT601000.C100DDLs, AT60N50.N50DDLs)

- New Config Files for each device handler with new MCD Exchange Rate Notification screens and states (i.e., AT601000.CNFGIX, AT60N50.CNFG5070)
- Hooks in the main device handler module to MCD Exchange Rate Notification procs
- Minor inline changes in the main module and global files
- Additional procs in the Dummy files (i.e., AT601000.C1000D, AT60N50.N50D)
- New Log (Event) Message Manuals for each device handler and the device handler utilities with new MCD Exchange Rate Notification events

All customers with Multiple Currency Support:

- Inline changes in the Base Multiple Currency Utilities (BA60UMCU.BAMCUTLS)
- Inline changes in the Multiple Currency Authorization module (AT60UATH.AUTHMCUS)

All customers with Multiple Currency Dispense:

- Inline changes in the Base Multiple Currency Utilities (BA60UMCU.BAMCUTLS)
- Inline changes in the Multiple Currency Dispense device handler utilities (AT60UMCD.ATMCDHLM and AT60UMCD.ATMCDHLS)
- New procs have been added in the extension modules containing MCD Exchange Rate Notification functionality. The files modified include source files and global files
- Inline changes in the device handler modules and their extensions have been made to correct defects and enhance error processing in the Multiple Currency Dispense code

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD07.SCNMCD.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param records that must be added in the LCONF to support this enhancement. These modifications are provided in the LCONFAT file located on the BA60MISC subvolume.

ATM-DH-XCHG-RATE-NTFY

This param indicates whether or not the MCD Exchange Rate Notification screen is

to be displayed to the cardholder. This is an optional param. Valid values are “Y” (yes, display the exchange rate screen) and “N” (no, do not display the exchange rate screen). If the param is not found or is invalid, the default value of “N” is used.

Database Configuration

Receipt File (RCPT)

The receipt file can be used to display currency and exchange rate information on a cardholder’s receipt. The amount field type codes (A1, A2, A3, AB, LB, S1, S2, BV, CD, CE, CI, CO and CS) can have a monetary symbol that prints on the receipt. This monetary symbol can be entered manually or determined automatically. Having the monetary symbol determined automatically has been added with the Multiple Currency Dispense enhancement.

The monetary symbol that is used can be determined automatically by the use of a table in the C100NAMS file for the Diebold 10XX/478X or N50NAMS file for the NCR 5XXX and by placing “CC\$” following the field type code. The symbol is determined by the currency code table (RCPT^MONEY^SYM^TBL^G). This table contains an entry for each monetary system for which a special symbol is to be printed. Each entry in the table contains the country code, the number of characters required in the sequence to send the special symbol(s) to the receipt printer (including escape codes), the number of characters taken up on the receipt as a result of the sequence, and the 10-character sequence that prints the proper monetary symbol. If a country code is not found in the table, the 3-character ISO currency code is used (for example, USD for US dollars). If neither the country code nor the ISO currency code exists in the table, no symbol is printed.

Additional information on specific field type codes may be found in the **BASE24-atm Diebold 10xx/478x** or **BASE24-atm NCR 5xxx Device Support Manual**. Information on use of the RCPT file may be found in the **BASE24-atm Files Maintenance Manual**.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

The following files were modified to contain new screens and states in the Multiple Currency Dispense load group for this enhancement.

Diebold 10XX/478X Configuration File (CNFGIX)

The Multiple Currency Dispense load group, group 901, was modified to contain states and screens specific to the MCD Exchange Rate Notification enhancement. New screens have been added (screens 075 and 137) and new states have been added (states 137, 138, 139, 163 and 166) for Exchange Rate Notification processing.

NCR 5XXX Configuration File (CNFG5070)

The Multiple Currency Dispense load group, group 901, was modified to contain states and screens specific to the MCD Exchange Rate Notification enhancement. A new screen has been added (screen 091) and new states have been added (states 091, 092, 093, 094 and 187) for Exchange Rate Notification processing.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

User Impacts

When Multiple Currency Dispense is supported, the account balances printed on the receipt and displayed at the ATM will be in the currency of the account. These values may slightly differ from the value in the account. This is due to multiple conversions that must be performed on these balances. When the authorization module processes the transaction, it takes the balances from the account and converts them to the transaction currency. To display the account balances on the receipt and at the ATM, the device handler module must then convert the balances from the transaction currency to the account currency. When multiple conversions are performed on an amount, the final value may be slightly different.

Although the account balances displayed at the ATM may differ slightly from the value in the account, the transaction amount is accurate throughout the life of the transaction. The difference is that the transaction amount is stored in the Multiple Currency token in each of the currencies used during the transaction, but the account balances are not in the Multiple Currency token. The device handler must use the amount 2 and amount 3 fields in the Standard Internal Message (STM) for the account balances. These fields contain the balances in the transaction currency only and, therefore, need to be converted multiple times.

Operational Impacts

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on \$<volume>.BA60LOGM or the BASE24 Documentation CD-ROM.

BASE24-atm
Optional Enhancements
Multiple Currency Dispense Exchange Rate Notification

Vendor Impacts

Not applicable

Reference Documents

Not applicable

NCR 5XXX EMV Dial-Up

This section describes the implementation requirements associated with the NCR 5XXX EMV Dial-Up enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the NCR 5XXX EMV Dial-Up enhancement.

The standard BASE24-atm NCR 5XXX Device Handler Dial-Up Extension has been modified to support this enhancement. In addition, new NCR EMV Dialup load groups have been added to the CNFG5070.

No other NCR 5XXX device handler modifications are required beyond those made to the dial-up extension module, and the CNFG5070.

This enhancement covers the following functionality:

- Standard EMV Dial-Up Processing: Requires existing load groups only.
- EMV Dial-Up Multiple-Account Select: Requires existing load groups only.
- EMV Dial-Up Standard Surcharge: Requires existing, as well as new EMV/Dial-Up load groups.
- EMV Dial-Up Non-Interactive Surcharge: Requires existing, as well as new EMV/Dial-Up load groups.
- EMV Dial-Up Non-Interactive Surcharge with Fast Cash: Requires existing, as well as new EMV/Dial-Up load groups.
- EMV Dialup Non-Currency Dispense: Requires existing, as well as new EMV/Dial-Up load groups.
- EMV Dial-Up Non-Currency Dispense / Non-Interactive Surcharge: Requires existing, as well as new EMV/Dial-Up load groups.

Prerequisite: This enhancement requires the installation of the NCR 5XXX Dial-Up device handler and NCR EMV extension modules. For information relating specifically to installation of the NCR 5XXX Dial-Up and NCR 5XXX EMV enhancements, please refer to the ***BASE24 Release 6.0 Implementation Guide (Version 2)***.

Software Impacts

All customers with the NCR 5XXX device handler:

- New load groups in the configuration file (CNFG5070)

All customers with the NCR 5XXX Dial-Up device handler:

- Inline changes to the dial-up device handler module

CUSTMACS Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

NCR 5XXX Configuration File, CNFG5070

The NCR 5XXX Configuration File, CNFG5070, has been modified to include several new EMV Dial-Up load groups as follows:

- Load Group 610
Load group 610 is the EMV Dial-Up Standard Surcharge load group. This load group requires loading base group 0, standard EMV load group 600, and the EMV load group 600 tables.
- Load Group 611
Load group 611 is the EMV Dial-Up non-Interactive Surcharge load group. This load group requires loading base group 0, standard EMV load group 600, and the EMV load group 600 tables,
- Load Group 612
Load group 612 is the EMV Dial-Up Non-Interactive Surcharge with Fast Cash load group. This load group requires loading base group 0, standard EMV load group 600, the EMV load group 600 tables, the EMV Dial-Up Non-Interactive Surcharge load group 611, and Fast Cash group 1.

- **Load Group 613**

Load group 613 is the EMV Dial-Up Non-Currency Dispense load group. This load group requires loading base group 0, standard EMV load group 600, the EMV load group 600 tables, the EMV non-Currency Dispense group 605, and the Non-Currency Dispense load group 500 (screens only).

- **Load Group 614**

Load group 614 is the EMV Dial-Up Non-Currency Dispense / Non-Interactive Surcharge load group. This load group requires loading base group 0, standard EMV load group 600, the EMV load group 600 tables, the EMV Non-Currency Dispense load group 605, the EMV Dial-Up Non-Interactive Surcharge load group 611, the EMV Dial-Up Non-Interactive Surcharge load group 613, and the Non-Currency Dispense load group 500 (screens only).

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

This enhancement assumes the customer is aware of the hardware differences between leased-line and dial-up solutions. Hardware vendors should be consulted for specifics. In addition, customer EMV-capable ATMs must support NCR NDC+ CAM 2 v2.14 or higher.

Reference Documents

Not applicable

Self-Service Banking Check Enhancements

This section describes the implementation requirements associated with the Self-Service Banking (SSB) Check Enhancements.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Self-Service Banking (SSB) Check Enhancements.

The Self-Service Banking Check Enhancements allows checks and registered checks to have a unique set of limits and percentages for deposit credit. Previously two sets of limits and percentages for deposit credits were allowed. Envelopes and checks used one set of limits/percentages and cash used the second set of limits/percentages.

Software Impacts

All BASE24 customers:

- Inline changes in the Card Prefix File (CPF) User Interface (AFT) module screen, requester, and server
- Inline changes to the CPF conversion program

All BASE24-atm Authorization customers:

- Hook changes in the main ATM Authorization module, global file, and dummy file

All BASE24-atm Self-Service Banking (SSB) Check customers:

- Inline changes to the Self-Service Banking (SSB) Check Authorization module

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD07.SCNSSBC.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes the new fields in the files that have been modified to support this enhancement.

Card Prefix File (CPF)

Field Name: Deposit Credit Percent - Check
Screen: CPF Screen 14
DDL Name: CPF.SSBCCPF.CHK-DEP-CR-PERCENT

The percentage of each non-registered check deposit to use as a deposit credit, subject to the settings in the following field in addition to the NUM-DEP-CR-LMT and the DEP-CR-LMT in the ATMCPF segment. The value in this field is used with the positive Balance Authorization method only.

Field Name: Maximum Credit Per Check Deposit – Check
Screen: CPF Screen 14
DDL Name: CPF.SSBCCPF.CR-PER-CHK-DEP-LMT

The maximum amount, in whole currency units (e.g., dollars), of deposit credit allowed per non-registered check deposit transaction.

Field Name: Deposit Credit Percent – CSF Check
Screen: CPF Screen 14
DDL Name: CPF.SSBCCPF.CSF-CHK-DEP-CR-PERCENT

The percentage of each registered check deposit to use as a deposit credit, subject to the settings in the following field in addition to the NUM-DEP-CR-LMT and the DEP-CR-LMT in the ATMCPF segment. The value in this field is used with the positive Balance Authorization method only.

Field Name: Maximum Credit Per Check Deposit – CSF Check
Screen: CPF Screen 14
DDL Name: CPF.SSBCCPF.CR-PER-csf-CHK-DEP-LMT

The maximum amount, in whole currency units (e.g., dollars), of deposit credit allowed per registered check deposit transaction.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Card Prefix File (CPF)

The CPF conversion program must be run to convert the CPF. Running the conversion program is required for any self-service banking check (SSBC) user. Users must create the CPF file using the BADDLFUP file.

The CPF conversion program on the BA60UC04 subvolume can be used to convert the CPF records from the release 5.3 format to the release 6.0 format. The CPF conversion program on the BA60UC07 subvolume can be used to convert the CPF records from the release 6.0 version 3 format to the release 6.0 version 4 format. If modifications to the conversion program are necessary, the program can be modified by editing the CNVCPFS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#cpfcnv.t,name $cpfm,nowait/ cnvcpfm –  
force –list –listreplace
```

Release 5.3 users should update the CNVCPFR obey file on the BA60UC04 subvolume with the appropriate information. Release 6.0 version 3 users should update the CNVCPFR obey file on the BA60UC07 subvolume with the appropriate information. The CNVCPFR should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREADME file on the BA60UC04 subvolume for release 5.3 customers and the BA60UC07 subvolume for release 6.0 version 3 customers.

Release 6.0 version 1 and release 6.0 version 2 customers should refer to the **Conversion Program Guidelines** within Section 2 of this document.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Disk Impacts

With the introduction of this enhancement, the Card Prefix File (CPF) Self-Service Banking Check (SSBC) segment will increase in size. BASE24 customers must examine their current disk capacity and determine whether the introduction of this enhancement into their system will require the purchase of additional disk space.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Triton Triple DES

This section describes the implementation requirements associated with the Triton Triple DES enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Triton Triple DES enhancement.

The standard Triton device handler BASE24-atm module has been modified to support Triple DES encryption. In order to support Triple DES, the device uses double length keys. Prior to this enhancement, the security keys were stored in BASE24 and the ATM devices as 16 hex characters or 8 bytes. However, support of double length keys requires BASE24 and the ATM device to store keys as 32 hex characters or 16 bytes. In order to support the double length keys, the Transaction Security Services is required.

Prerequisite: The Transaction Security Services enhancement must be installed prior to implementation of the Triton Triple DES enhancement. For more information refer to the ***BASE24 Release 6.0 Implementation Guide (Version 2)***.

Software Impacts

All customers with Triton device handler:

- Inline changes in the main module and global file
- A new Triton manual

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Channel Security Data Source (CSECD)

Transaction Security: Channel Security Configuration
Tab: PIN Information
Field: Key Length of Exchange Key

The Key Length of the Exchange Key for the Triton CSECD records must be set to Double to utilize Triton Triple-DES using double-length keys.

Transaction Security: Channel Security Configuration
Tab: PIN Information
Field: Exchange Key (Header, Part 1-4, Check Digits)

The Exchange Key for the Triton CSECD records must be modified to contain the new double-length key value.

Transaction Security: Channel Security Configuration
Tab: PIN Information
Field: Key Length of Inbound Key

The Key Length of the Inbound Key for the Triton CSECD records must be set to Double to utilize Triton Triple-DES using double-length keys.

Transaction Security: Channel Security Configuration
Tab: PIN Information
Field: Inbound Key (Header, Part 1-4, Check Digits)

The Inbound Key for the Triton CSECD records must be modified to contain the new double-length key value.

Refer to the online help provided with the ACI Desktop user interface for detailed descriptions of this window and its fields.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Triton devices supporting Triple DES encryption must have new software and hardware configured to use the Triple DES encryption algorithm and double-length keys. Additionally the new double-length Master key must be manually entered at the device and the new Working key must be downloaded.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Compatible Devices and Software

The following device platforms and software versions are compatible with BASE24's Triton Triple DES enhancement.

- Refer to the Triton vendor documentation to verify device compatibility of the following components to support Triton Triple-DES:
 - Eprom Version
 - Program Version
 - Table Version
 - Triton ATM models

Reference Documents

- ***Triton: DES KEY ENTRY FOR TRITON ATMS*** (Part Number 07100-00010A 09/2002)
- ***Triton: Triton Terminal Communications Protocol and Message Format Specification*** (Preliminary Version TSCD5.1 June 26, 2003)

Section 6

BASE24-pos

Section 6 discusses the changes and enhancements made to BASE24-pos in release 6.0 version 4, along with the resulting conversion and implementation requirements that existing BASE24-pos users must consider when upgrading from release 6.0 version 3.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24-pos users, along with the associated conversion and implementation requirements that all BASE24-pos users must address.
- Optional enhancements describe the changes that affect only those BASE24-pos users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24-pos users must address to implement the enhancements.

Section 6 must be used in conjunction with Section 4 to identify the changes that BASE24-pos users need to consider when upgrading to release 6.0 version 4.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24-pos users must address in order to upgrade to release 6.0 version 4.

Changes and requirements are identified using a standard format defined in the introduction to Section 4.

- General Changes
- Enhanced Approval/Referral Code Generation
- General CRT Authorization Enhancements
- SPDH Additional Data Fields

These areas are summarized on the following pages.

Section 2 contains a functional overview of the general enhancements described here.

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General Changes

1. Support was added to the ACI Standard POS Device Handler (SPDH) module for the ISO 87 standard for the ISO Response Code. This includes formatting FID 'X' in a response message to the device and mapping a BASE24 response code to an ISO response code. The module previously only supported the ISO 93 Standard for the ISO response code. A new LCONF param was added to control which ISO response code format is support for the device handler process.
2. The ACI Standard POS Device Handler (SPDH) was enhanced to map an ISO response code to a BASE24 response code based on account type.

Software Impacts

All customers with the ACI Standard POS Device Handler:

- Inline changes to ASPDHG, ASPDHS and SPDHNAMS

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support the SPDH ISO 87 format for the ISO response code. Note that these modifications are provided in the LCONFPS file located on the BA60MISC subvolume.

POS-DH-ISO-RESP-CDE-FRMT

This param is used by the SPDH. It defines the ISO response code format used by the SPDH when formatting FID 'X' – ISO Response Code on a transaction response. Valid values are 93 – ISO 93 Format and 87 – ISO 87 Format. The default is 93.

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

The table that controls mapping a BASE24 POS response code to an ISO response code for the ACI Standard POS Device Handler has been enhanced to include a column for account type. To take advantage of this additional mapping feature, customers will have to update the table with an additional entry for the same BASE24 POS response code and different ISO response code and account type. The num_b24_pos_resp_l must be incremented accordingly. For example, if the BASE24 POS response code '000' – Approved with balances was to be mapped to the ISO 93 response code of '000' – approved for a checking account, but mapped to an ISO response code of '001' – honor with identification for a savings account, an entry would be added to the table as follows:

```

!                                     BASE24 POS RESP CODE      !
!                                     |      +--  ISO RESP CODE    !
!                                     |      |      +--  ACCT TYPE    !
!                                     |      |      |                !
!APPROVE - W/ BALANCES              ! "000" , "000", "01",      !
!APPROVE - W/ BALANCES              ! "000" , "001", "11",      !

```

If mapping from a BASE24 POS response code into an ISO response code does not require account type, the account type should remain blank.

Vendor Impacts

ACI Standard POS Message

The following optional FID has been modified in the ACI Standard POS Message.

FID	Picture	Length	Field Name	RQST	RESP
X	PIC X(3)	2-3 bytes	ISO Response Code		✓

Refer to the **ACI Standard POS Device Message Specifications Manual** for additional information.

Reference Documents

Not applicable

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Enhanced Approval/Referral Code Generation

This section describes the implementation requirements associated with the Enhanced Approval/Referral Code Generation.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Enhanced Approval/Referral Code Generation.

This enhancement includes:

- The BASE24-pos Router/Authorization processing has been enhanced to generate a new referral code if the generation of a referral code results in a duplicate key to the PRF until the record can be written to the PRF.
- The BASE24-pos Router/Authorization processing has been enhanced to generate a new approval code if the generation of an approval code results in a value of zeroes until a non-zero value is returned.

Software Impacts

These changes were done to meet requirements for U.K. SWITCH Issuers.

All BASE24-pos customers:

- Inline changes to the PS60RTAU.AUTHLIBS file to enhance the logic used in generating approval codes and referral codes.

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD07.SCNSWCH.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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General CRT Authorization Enhancements

This section describes the implementation requirements associated with the General CRT Authorization Enhancements.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the General CRT Authorization Enhancements.

This enhancement includes:

- The BASE24-pos CRT Authorization processing has been enhanced to display the Fraud Prone Outlet Flag on the PRF screen.
- The BASE24-pos CRT Authorization processing has been enhanced to format and add the Reversal Date and Time Token when a reversal is generated from the OLT Screen. This allows Issuer Interface modules to set fields in the interface process to the actual transaction date and time the reversal was generated.

Software Impacts

These changes were done to meet requirements for U.K. SWITCH Issuers.

All BASE24-bp customers:

- Inline changes to the BP60CRTA.SCRNPRF, BP60CRTA.RQPRFS, and BP60CRTA.SVPRFS files to allow the Fraud Prone Outlet Flag to be displayed on the PRF screen.
- Inline changes to the BP60CRTA.CRTAUDHS file to allow the Reversal Date and Time to be added when a reversal is generated from the OLT Screen.

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD07.SCNSWCH.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes the new fields in the files that have been modified to support this release. The settings of each field depend on the configuration desired by the institution.

BASE24-pos Referral File (PRF)

It will now be possible to see the Fraud rating of the Retailer.

Field Name: FRAUD RATING
Screen: PRF Screen 1
DDL Name: PS51-TKN.FRD-PRN-FLG

The Fraud Prone Flag from the PS51 token. This field displays the Fraud Prone flag value with the associated text describing the fraud rating for the retailer.

Token File (TKN)

The following BASE24 token has been used for this enhancement:

- The Reversal Date and Time token (token ID "BH") contains the actual date and time that a reversal was initiated at a POS terminal.

If the token for this enhancement is not to be logged to the PTLF or ILF, the appropriate token records must be configured in the TKN.

If the token for this enhancement is to be extracted with the PTLF or ILF, sent to the host, or passed through an Interchange Interface, the appropriate TKN records must be configured.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Super Extract Process

If the Reversal Date and Time Token is included in the PTLF and ILF Extracts, BASE24-pos users must consider the new data within the token with regards to any host programs that use the data from the log files.

User/Operational Impacts

Disk Impacts

The PTLF and ILF require additional disk space because the Reversal Date and Time token can be included in the transaction records logged to these files.

Operational Impacts

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on \$<volume>.BA60LOGM.

System Limitations

BASE24-pos users installing this enhancement must be aware of their internal system limitations for messaging to the PTLF and ILF. There are system limitations regarding the amount of data that can be passed through the BASE24-pos system, logged to a log file, and passed to the host or co-network in an ISO External Message. BASE24 users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the ***BASE24 Tokens Manual*** provides information on assessing the impacts of tokens on log file disk space requirements.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

SPDH Additional Data Fields

This section describes the implementation requirements associated with the SPDH Additional Data Fields enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the SPDH Additional Data Fields enhancement.

A standard BASE24-pos module has been modified to support this enhancement. That module is the ACI Standard POS Device Handler (SPDH).

Software Impacts

All BASE24-pos customers:

- Changes to File Maintenance components to support new BASE24-pos Terminal Data Files (PTD) fields (PS60AFT.RQPTDS, PS60AFT.SCRNPTD, PS60AFT.SVPTDTBL and PS60AFT.SVPTDTS)
- Changes to File Maintenance components to support a new BASE24-pos Retailer Definition File (PRDF) field (PS60AFT.RQPRDFS, PS60AFT.SCRNPRDF)
- Changes to DDL files (BA60DDL.DDLPSTKN, PS60DDL.DDLFPRDF and PS60DDL.DDLGPTD)
- Changes to token conversion utilities and a token ID file (BA60SRC.PSTKNCVS and BA60SRC.PSTKNID)

All customers with the ACI Standard POS Device Handler:

- Changes to the main device handler module (PS60SPDH.ASPDHS)
- Updated event (log) message manual with new events (BA60LOGM.PSSPDH)

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD07.SCNSADF.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

The following database configuration changes are optional.

ACI Standard POS Device Handler Configuration File (ACNF)

FID 4 (Industry Data) must be set to "Y" in the Request Field Map or Response Field Map records for the transaction types that Industry Data will be received from the terminal or returned to the terminal.

BASE24-pos Retailer Definition File (PRDF)

The following field has been added to PRDF file maintenance screen 4.

Field Name: EMV CAPABLE
Screen: PRDF Screen 4
DDL Name: PRDF.EMV-CAPABLE

Indicates the EMV capability of retailer. Valid values are:

Blank = Not used
Y = Yes, the retailer is EMV capable
N = No, the retailer is not EMV capable

BASE24-pos Terminal Data Files (PTD)

The following five fields have been added to PTD file maintenance screen 19.

Field Name: TERMINAL ATTENDANCE
Screen: PTD Screen 19
DDL Name: PTDS1.TERM-ATTEND-IND

A code indicating whether the terminal is attended by the card acceptor.
Valid values are:

Blank = Not used
0 = The terminal is attended
1 = The terminal is unattended (for example, a cardholder-activated terminal or a personal computer)
2 = No terminal used (for example, voice or ARU authorization)

Field Name: TERMINAL LOCATION
Screen: PTD Screen 19
DDL Name: PTDS1.TERM-LOC-IND

A code indicating the location of the terminal. Valid values are:

- Blank = Not used
- 0 = The terminal is on the premises of the card acceptor facility
- 1 = The terminal is off the premises of the card acceptor facility
(merchant terminal in a remote location)
- 2 = The terminal is on the premises of the cardholder location
(home personal computer)
- 3 = No terminal was used (voice or ARU authorization)

Field Name: CARD CAPTURE
Screen: PTD Screen 19
DDL Name: PTDS1.CRD-CAPTR-IND

A code indicating whether the terminal has card capture capabilities.
Valid values are:

- Blank = Not used
- 0 = The terminal does not have card capture capabilities
- 1 = The terminal has card capture capabilities

Field Name: CARDHOLDER ACTIVATED TERMINAL
Screen: PTD Screen 19
DDL Name: PTDS1.CRDHLDR-ACTVT-TERM-IND

A code indicating whether the cardholder can activate the terminal with
a card, and if so, the level of security. Valid values are:

- Blank = Not used
- 0 = The transaction is not a CAT transaction
- 1 = Automated dispensing machine with PIN—level 1 security
- 2 = Self-service terminal—level 2 security
- 3 = Limited amount terminal—level 3 security
- 4 = In-flight commerce—level 4 security
- 5 = Script device
- 6 = Electronic commerce
- 7 = Radio frequency device

Field Name: TERMINAL INPUT CAPABILITIES
Screen: PTD Screen 19
DDL Name: PTDS1.TERM-INPUT-CAP-IND

A code indicating the terminal capabilities for transferring the data on
the card to the terminal. Valid values are:

Blank	=	Not used
0	=	Unknown or unspecified
1	=	No terminal used (voice or ARU authorization)
2	=	Magnetic stripe reader
3	=	Bar code
4	=	Optional character recognition
5	=	Magnetic stripe reader and EMV-compatible ICC reader
6	=	Key entry only
7	=	Magnetic stripe reader and key entry
8	=	Magnetic stripe reader, key entry, and EMV-compatible ICC reader
9	=	EMV-compatible ICC reader

Token File (TKN)

The following BASE24-pos token has been added for this enhancement:

- Industry Data Token (Token ID “CK”)

If the Industry Data Token is not to be logged to the PTLF or ILF, the appropriate token record must be configured in the TKN.

If the Industry Data Token is to be extracted with the PTLF or ILF, or sent to a host processor or an interchange, the appropriate token record must be configured in the TKN.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

The new PRDF field utilized by this enhancement (EMV-CAPABLE) was taken from available space in USER-FLD-ACI. As this user field is initialized to blanks, and blank is the default value for EMV-CAPABLE, no file conversion is necessary for this enhancement.

The PTD fields utilized by this enhancement (TERM-ATTEND-IND, TERM-LOC-IND, CRD-CAPTR-IND, CRDHLDR-ACTVT-TERM-IND and TERM-INPUT-CAP-IND) were originally added to the PTDS1 record definition in Release 6.0 Version 3. As these fields are initialized to blanks, and blanks are the default values, no file conversion is necessary for this enhancement.

Host/Co-Network Impacts

Super Extract Process

If the Industry Data Token is to be included in the PTLF and ILF extracts, BASE24-pos users must consider this additional data in host programs that use the data from the log files.

User/Operational Impacts

Operational Impacts

One new field was added to PRDF file maintenance screen 4. That field is EMV Capable. Operators must be trained in the proper use of this field.

Five new fields were added to PTD file maintenance screen 19. Those fields are Terminal Attendance, Terminal Location, Card Capture, Cardholder Activated Terminal and Terminal Input Capabilities. Operators must be trained in the proper use of these fields.

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate remedy for a given message. Refer to the event message documentation files on \$<volume>.BA60LOGM.

System Limitations

BASE24-pos users installing this enhancement must be aware of their internal system limitations for messaging to the PTLF and ILF. There are system limitations regarding the amount of data that can be passed through the BASE24-pos system, logged to a log file, and passed to the host or co-network in an ISO External Message. BASE24 users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the **BASE24 Tokens Manual** provides information on assessing the impacts of tokens on log file disk space requirements.

Vendor Impacts

ACI Standard POS Message

The following optional field identifiers (FIDs) have been added to the ACI Standard POS Message.

FID	Picture	Length	Field Name	RQST	RESP
4	PIC X(171)	156-171 bytes	Industry Data	✓	✓

The following optional sub field identifiers (subFIDs) have been added to the Product subFIDs data field (FID 6) of the ACI Standard POS Message.

SFID	Picture	Length	Field Name	RQST	RESP
U	PIC X(16)	16 bytes	Transaction Subtype Data	✓	
V	PIC X	1 byte	Authentication Collection Indicator		✓
W	PIC X	1 byte	CAVV/AAV Result Code		✓
X	PIC X(6)	6 bytes	Point of Service Data	✓	
Y	PIC X(202)	2-202 bytes	Authentication Data		✓
Z	PIC X	1 byte	Card Verification Flag 2	✓	✓
e	PIC X(21)	21 bytes	Interchange Compliance Data		✓
f	PIC X	1 byte	Response Source or Reason Code		✓
g	PIC X(4)	4 bytes	POS Merchant Data	✓	

The following optional subFIDs have been modified in the ACI Standard POS Message.

SFID	Picture	Length	Field Name	RQST	RESP
F	PIC X(2)	1-2 bytes	Electronic Commerce Data	✓	

Refer to the **ACI Standard POS Device Message Specifications Manual** for additional information.

Reference Documents

- MasterCard IPM Clearing Formats, 05/2003
- Visa BASE II Clearing Interchange Formats, TC 50 to TC 92, 04/2003
- The following table illustrates how the SPDH maps these FIDs, subFIDs, PTD fields and PRDF fields to BASE24-pos Standard Internal Message tokens.

BASE24-pos
General Enhancements
SPDH Additional Data Fields

Token Source/Destination		SPDH FID/SFID		Rqst	Resp	PTD	PRDF
ID	Field	FID, SFID, Posn	Description				
BM	TXN-SUBTYP	FID 6, SFID U, posn 1-4	Transaction Subtype	X			
BM	ACQ-PROC-CDE	FID 6, SFID U, posn 5-10	Acquirer Processing Code	X			
BM	ISS-PROC-CDE	FID 6, SFID U, posn 11-16	Issuer Processing Code	X			
C0	AUTHN-COLL-IND	FID 6, SFID V	Authentication Collection Indicator		X		
C0	CAVV-AAV-RSLT-CDE	FID 6, SFID W	CAVV/AAV Result Code		X		
C4	TERM-ATTEND-IND			X		X	
C4	TERM-LOC-IND			X		X	
C4	CRDHLD-PRESENT-IND	FID 6, SFID X, posn 1	Cardholder Present Indicator	X			
C4	CRD-PRESENT-IND	FID 6, SFID X, posn 2	Card Present Indicator	X			
C4	CRD-CAPTR-IND			X		X	
C4	TXN-STAT-IND	FID 6, SFID X, posn 3	Transaction Status Indicator	X			
C4	TXN-SEC-IND	FID 6, SFID X, posn 4	Transaction Security Indicator	X			
C4	CRDHLD-ACTVT-TERM-IND	FID 6, SFID X, posn 5	Cardholder Activated Terminal Indicator	X		X	
C4	TERM-INPUT-CAP-IND			X		X	
C4	CRDHLD-ID-METHOD	FID 6, SFID X, posn 6	Cardholder ID Method	X			
CE	AUTHN-IND-FLG	FID 6, SFID Y, posn 1-2	Authentication Indicator Flag		X		
CE	AUTHN-IND-DATA	FID 6, SFID Y, posn 3-202	Authentication Indicator Data		X		
CH	RESP-SRC-RSN-CDE	FID 6, SFID f	Response Source or Reason Code		X		
CH	CRD-VRFY-FLG2	FID 6, SFID Z	Card Verification Flag 2	X	X		
CH	EMV-CAPABLE-OUTLET			X			X
CH	RECUR-PMNT-IND	FID 6, SFID F, posn 2	Recurring Payment Indicator	X			
CI	E-COMM-GOODS-IND	FID 6, SFID g, posn 1	E-Commerce Goods Indicator	X			
CI	EXISTING-DEBT-IND	FID 6, SFID g, posn 2	Existing Debt Indicator	X			
CI	DEFERRED-BILLING-IND	FID 6, SFID g, posn 3	Deferred Billing Indicator	X			
CI	RELN- PARTICIPANT-IND	FID 6, SFID g, posn 4	Relationship Participant Indicator	X			
CK	(ALL)	FID 4, posn 1-171	Industry Data	X	X		
20	TRACE-ID	FID 6, SFID e, posn 1-15	Trace ID		X		
20	VALID-CDE	FID 6, SFID e, posn 16-19	Valid Code		X		
20	MONITORING-STAT	FID 6, SFID e, posn 20	Monitoring Status		X		
20	ERR-IND	FID 6, SFID e, posn 21	Error Indicator		X		

Refer to the **ACI Standard POS Device Message Specifications Manual** and the **BASE24 Tokens Manual** for additional information.

Optional Enhancements

This section identifies the BASE24-pos enhancements that can be enabled or disabled with release 6.0 version 4, along with the changes, conversion requirements, and implementation requirements associated with each. These enhancements include the following:

- Configurable Approval Code Length
- Dual Site Support
- DUKPT MACing
- EMV M/Chip 4 Support
- Optional CRT Authorization Enhancements
- POS Balance Inquiry Service
- Secure Internet Validation
- SPDH DUKPT Support via TSS
- SPDH Enhanced Multi-Threading
- SPDH EMV Additional Support
- Triple DES for BASE24 SPDH

BASE24-pos users can choose whether or not to implement each of the above enhancements. Users who plan to implement a particular enhancement must address the corresponding release 6.0 version 4 changes, conversion requirements, and implementation requirements. Users who do not plan to implement a particular enhancement can simply review the corresponding release 6.0 version 4 changes, conversion requirements, and implementation requirements for future reference.

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Configurable Approval Code Length

This section describes the implementation requirements associated with the configurable approval code length enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Configurable Approval Code Length Enhancement.

This enhancement allows the generation of an approval code with the approval code length being the lesser of the value sent from the terminal and the value specified at the card prefix level. The approval code length must be a value between 2 and 6 inclusive.

The Card Prefix File has been enhanced so that the required approval code length can be configured at the card prefix level.

The Card Prefix File has also been enhanced to include fields required for future product developments. These fields are described in this section, but are not currently used in transaction processing by BASE24.

The POS Router/Authorization processing has been enhanced to compare the approval code length sent from the terminal (in the PSTM) to the length of the approval code configured in the CPF record. The length of the approval code generated will be the lesser of the two values.

Software Impacts

These changes were done to meet requirements for U.K. SWITCH Issuers.

All BASE24 customers:

- Inline changes to the BA60AFT.SCRNCPF, BA60AFT.RQCPFS, and BA60AFT.SVCPFS files to provide AFT access to the new CPF fields.
- Inline changes to the BA60EXT.OMFXS file to allow the new CPF fields to be extracted in EBCDIC, if required.
- Inline changes to the CPF conversion program

All BASE24-pos customers:

- Inline changes to the PS60RTAU.RTAUG, PS60RTAU.ROUTERS, and PS60RTAU.AUTHLIBS files to enhance the logic used in calculating the approval code length.

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD07.SCNSWCH.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes the new and enhanced fields in the files that are required to support this enhancement.

Card Prefix File (CPF)

Field Name: APPROVAL CODE LENGTH

Screen: CPF Screen 8

DDL Name: CPF.POSCPF.APPRV-CDE-LGTH

The length of the approval code generated by BASE24-pos.

For additional information on the usage of this field, refer to the ***BASE24 Base Files Maintenance Manual***.

The following fields have also been added as part of this enhancement, but they are not currently used in transaction processing by BASE24:

Card Prefix File (CPF)

Field Name: CHECK IF HOST ONLINE – SERVICE CODE

Screen: CPF Screen 3

DDL Name: CPF.CPFBASE.SRVC-CDE-CHK-FLG

The field indicates whether service code checking should be performed for this prefix.

Field Name: SERVICE CODE CHECKING ACTION INDEX

Screen: CPF Screen 3

DDL Name: CPF.CPFBASE.SVC-CDE-ACT-TBL-IDX

The action table index is used to identify which of the preset rules for service code checking are to be used for this prefix. The decision arrays which define the action to be taken based on the service code value are pre-configured in EDIT files, sourced in to the

Router/Authorisation modules at compile time. Four different arrays are supported to cater for the processing requirements of different card schemes. This field is used to select the appropriate array for the card scheme. Value values are 1 to 4. The default is 1.

Field Name: FORCE ONLINE COUNT
Screen: CPF Screen 8
DDL Name: CPF.POSCPF.FORCE-ONL-CNT

Indicates the maximum number of consecutive transactions that could be authorized below the AST limit before a transaction is forced online to the issuer.

Field Name: DRAFT CAPTURE ISSUER
Screen: CPF Screen 8
DDL Name: CPF.POSCPF.DFT-CAPTR-ISS

Overrides the existing ISSUER field (Screen 7) on draft capture transactions, allowing a different AST entry to be accessed. Valid values are "00"- "99" with the default value being "00".

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

An entry has been added to the SECTBL to support the new screen 8 for the CPF Requester and Server.

PATHCONF Changes

Not applicable

CRT Access Security Changes

This enhancement provides customers with the capability to give BASE24 AFT users appropriate security access to the new CPF screen 8.

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Card Prefix File (CPF)

If the BASE24-pos user wishes to enable Configurable Approval Code Length processing, the user must update the new fields in the CPF with the appropriate values. Otherwise, the new fields in the CPF will default to not perform the additional approval code length processing.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Card Prefix File (CPF)

This enhancement provides customers with the capability to enter an Approval Code Length for the card prefix on Screen 8.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Dual Site Support

This section describes the implementation requirements associated with Dual Site Support.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Dual Site Support enhancement.

Sections 4 (Base) and 5 (ATM) contain product specific changes relating to the Dual Site Support enhancement.

Dual Site Support involves changes related to the management of a Dual Site BASE24 production environment. This enhancement is to support customers who are planning to implement a Dual Site production environment using GoldenGate to facilitate data replication between the two sites.

A Dual Site BASE24 production environment is where BASE24 is running on two separate NSK machines, with production transactions being processed on both machines. Standard BASE24 files will be held on both systems with updates replicated between the two systems by GoldenGate.

The PTD file has been enhanced to enable the identification of individual terminals as “local” or “remote”, i.e. whether the terminal is normally connected to this system or the alternate system. (Records for each terminal will appear on both copies of the file, with one set to local and one to remote according to the normal connectivity of the terminal.) As records on either system are updated, GoldenGate will replicate the changes to the alternate system and alter the setting of the local / remote flag.

The PTLF has been enhanced to enable the identification of individual records as “local” or “remote”, i.e. whether the transaction was authorized on this system or the alternate system. Local records are standard and remote records include a new field, SITE^IND, in the Multiple Logical Network token. (Records for all transactions will appear on both copies of the files, with one copy containing the remote token according to the location of the original transaction.) As records on either system are added, GoldenGate will replicate the records to the alternate system, add the token if not already present, and update the field.

Settlement, Perusal and PTD file access have been enhanced to support and provide options relating to Dual Site Support. Settlement can be configured to process local only, remote only or all records and terminals during cutover processing and has been enhanced to support broadcast capability in notify lists and module specific PTLF assign. PTLF perusal has been enhanced to enable the identification of

remote (replicated) transactions on the local log file. PTD file access supports a new field, DUAL^SITE^IND, indicating the normal connectivity of the terminal in question.

For more details on using these enhancements and GoldenGate to support a Dual Site production environment, refer to the ***BASE24 Dual Site Processing Guide***.

Software Impacts

For full details on the software impacts for this enhancement see Software Impacts in Section 4 (Base) of this document.

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD07.SCNDUALS.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

This section describes the Assign records that must be added to the LCONF to support this enhancement. These modifications are provided in the LCONFPS file located on the BA60MISC subvolume.

POS-PTLF

The fully qualified file name of the POS Transaction Log File (PTLF). This assign is now supported as a process specific assign for BASE24-pos Settlement Initiator process and the POS Transaction Log File (PTLF) Server.

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement. These modifications are provided in the LCONFBA file located on the BA60MISC subvolume.

DUAL-SITE-MODE

This param is used by the BASE24-pos Settlement Initiator process. It specifies the dual site mode for the process. Valid values are:

L - Local records only to be processed by Settlement

R - Remote records only to be processed by Settlement

A - All records to be processed by Settlement

DUAL-SITE-DISPLAY

This param is used by the POS Transaction Log File (PTLF) Server. It defines whether the server uses process specific PTLF assigns or not. Valid values are Y - Server PTLF uses process specific PTLF assign and identifies remote records with REMOTE TXN in bottom right corner of detail display and N - Server PTLF Dual Site processing deactivated. If the param is not present the default value is N.

BROADCAST-NOTIFY

This param is used by the BASE24-pos Settlement Initiator process. It defines whether the process uses the broadcast facility. Valid values are 0 - Broadcast facility not to be used by Settlement, 1 - Broadcast flag to be set to 1 (broadcast to all service providers) for notify messages sent by Settlement and 2 - Broadcast flag to be set to 2 (broadcast to running service providers) for notify messages sent by Settlement.

BROADCAST-DELAY

This param is used by the BASE24-pos Settlement Initiator process. It defines the number of seconds to delay after file creation before broadcasting to the notify list.

Database Configuration

This section describes the new fields in the files that have been modified to support this enhancement.

POS Transaction Log File (PTLF)

Field Name: SITE INDICATOR
Screen: PTLF Detail Screen 4
DDL Name: MULT^LN^TOKEN.SITE^IND

A value of "p" in the SITE^IND field indicates that the transaction was authorised on the alternate system (and the record replicated by GoldenGate). A value of " " or the absence of the token indicates that the transaction was authorised locally.

This field is maintained by GoldenGate and does not require any action by the customer.

POS Terminal Definition File (PTD)

Field Name: DUAL SITE INDICATOR
Screen: PTD Screen 1
DDL Name: PTDD1.CORE.DUAL^SITE^IND

A value of “L” or “ ” (default value) in the DUAL^SITE^IND field indicates that the terminal is typically connected to the system on which the file is located. A value of “R” in the DUAL^SITE^IND field indicates that the terminal is typically connected to the alternate system. Records for each terminal will be present on each copy of the file and the DUAL^SITE^IND flag set by GoldenGate during replication.

This field needs to be set to “L” or “R” prior to Dual Site implementation according to the normal terminal connectivity. After initial configuration the field is maintained by GoldenGate and does not require any action by the customer.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Host/Co-Network impacts will be incurred by those BASE24 users implementing the Dual Site Support enhancement. Modifications to existing files and processing will result in impacts to Super Extract.

Super Extract Process

Host code changes are required for all BASE24-pos users that utilize the OMF Extract Process to accept and process the new information in the PTD.

If the Multiple Logical Network Token is included in the PTLF and ILF Extracts, BASE24-pos users must consider the new data within the token with regards to any host programs that use the data from the log files.

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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DUKPT MACing

This section describes the implementation requirements associated with the DUKPT MACing enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the DUKPT MACing enhancement.

The SPDH supports DUKPT MACing for transactions originating at the terminals directly connected to BASE24-pos.

Using the DUKPT method, each terminal security module (TSM), or PIN pad, uses a derived unique key for each transaction. However, the TSM never contains any information that could lead to the determination of any past MAC keys used by the device, or of any past or future keys used by any other TSM. The originating TSM must derive the current MAC key from an initial key, loaded into the TSM on initialization. The receiving BASE24-pos security module (Atalla or Thales) then determines the current MAC key by using a key held on BASE24-pos, and non-secret information contained in the transaction message.

The PTD MAC-ENCRYPT-TYP field supports new values of 07 (DUKPT MACing and PIN Translate in one command) and 08 (DUKPT MACing and PIN Translate in different commands). The Derivation Key File (KEYD) stores the key locator in the derivation key field indicating the record in the Derivation Key Data Source file (DRVKD) used by TSS containing the derivation keys. The BASE24 AFT facility is used to maintain the data in the KEYD file and the ACI Desktop is used to maintain the data in the DRVKD file.

The Standard POS Device Handler (SPDH) message definition can optionally include FID 6 sub-FID T, which is used to transmit the Key Serial Number (KSN) and KSN descriptor from the terminal to BASE24. This KSN is used for PIN verification using DUKPT but it also is used for DUKPT MACing.

Upon receipt of a terminal request message, the Standard POS Device Handler (SPDH) accesses the BASE24-pos Terminal Data file (PTD). The SPDH uses the PTD MAC-ENCRYPT-TYP field to determine whether the terminal is using DUKPT MACing. The PTD RETAILER-ID, KEYD-GROUP and TERM-ID fields are used as the key to access the KEYD.

As with standard MACing, the SPDH module will verify the MAC on the request and generate a MAC on the response using the Security Utilities (SECUTILS) module. The SPDH supports DUKPT PIN processing in addition to DUKPT MACing. If DUKPT PIN processing and DUKPT MACing are configured, the SPDH determines whether to perform the PIN translation (from DUKPT to DES) and DUKPT MAC

verification with one call to the security module. This is based on the MAC-ENCRYPT-TYP field in the PTD. If the MAC-ENCRYPT-TYP is set to “07”, the PIN translation and MAC verification will be done with one call to the security module. If the MAC-ENCRYPT-TYP is set to “08”, the PIN translation and MAC verification are done with separate calls to the security module.

Prerequisites: The Transaction Security Services, Derived Unique Key Per Transaction (DUKPT) and SPDH DUKPT Support via TSS enhancements must be installed prior to implementation of the DUKPT MACing enhancement. For more information on the Transaction Security Services and DUKPT enhancements refer to the ***BASE24 Release 6.0 Implementation Guide (Version 2)***. For more information on the SPDH DUKPT Support via TSS enhancement refer to the SPDH DUKPT Support via TSS section in this document.

Software Impacts

All BASE24 customers:

- Inline changes to SECUTILS.
- DDL changes in BA60DDL.DDLGTSS.

All BASE24-pos customers:

- Changes in BA60DDL.DDLPCNST.
- Inline changes to SPDH (ASPDHS) and the PTD server (SVPTDTS, SVPTDTBL and SCRNPDT).

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD07.SCNDKMAC.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

ACI Standard Device Configuration File (ACNF)

FID G (MAC) should be included in the request and response Field Map records of the ACNF for DUKPT driven terminals when DUKPT MACing is to be performed.

BASE24-pos Terminal Data files (PTD)

Field Name: MAC Type
Screen: PTD Screen 7
DDL Name: PTDD1.[ptdd1-core].MAC-DATA.MAC-ENCRYPT-TYP

Indicates the MAC encryption type supported for the terminal. Two values were added with this enhancement: 07 (DUKPT MAC verification with DUKPT PIN Translation) and 08 (DUKPT MAC verification without DUKPT PIN Translation). Note: If 07 or 08 is chosen, the only values allowed in the PIN encryption type field are 00 (None) and 07 (DUKPT). If the PIN encryption type is set to 00 (None), it is recommended that the MAC encryption type be set to 08 (DUKPT MAC verification without DUKPT PIN translation). This will ensure the MAC is verified as soon as possible within the device handler.

Field Name: MAC Master Key
Screen: PTD Screen 7
DDL Name: PTDD1.[ptdd1-core].MAC-DATA.ENCR-KEYS.MAS-KEY

Contains the key locator to the CSECD record for the terminal.

Channel Security Data Source (CSECD)

Transaction Security: Channel Security Configuration
Tab: MAC Information
Field: Key Variant

If using an Atalla non-AKB security device (this field only applies to Atalla non-AKB devices), this field must be set to the appropriate value based on the key variant that was applied when encrypting the MAC key exchange key.

Transaction Security: Channel Security Configuration
Tab: MAC Information
Field: Key Length of Exchange Key

The Key Length of the Exchange Key for the SPDH CSECD records must be set to the appropriate length depending on whether single, double or triple-length keys are being used.

Transaction Security: Channel Security Configuration
Tab: MAC Information
Field: Exchange Key (Header, Part 1-4, Check Digits)

The Exchange Key for the SPDH CSECD records must be modified to contain the new key value.

Transaction Security: Channel Security Configuration
Tab: MAC Information
Field: Key Length of Inbound Key

The Key Length of the Inbound Key for the SPDH CSECD records must be set to the appropriate length depending on whether single, double or triple-length keys are being used.

Transaction Security: Channel Security Configuration
Tab: MAC Information
Field: Inbound Key (Header, Part 1-4, Check Digits)

The Inbound Key for the SPDH CSECD records must be modified to contain the new key value.

Refer to the online help provided with the ACI Desktop user interface for detailed descriptions of this window and its fields.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

The device must have new software/hardware configured to use DUKPT MACing.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

If DUKPT MACing is configured, access to the keys stored in the DRVKD and CSECD that used to be stored in the KEYD and PTD, respectively, is through the new ACI Desktop Graphical User Interface that is set up separately from the BASE24 Pathway CRT access system. Refer to the ***BASE24 Classic 6.0 Installation Guide*** for further information.

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Performance Impacts

When configured to perform DUKPT PIN verification and DUKPT MACing: If the MAC encryption type field on screen 7 of the PTD is set to "07" (DUKPT MAC verification/PIN translation), one I/O is required to the security device to translate the PIN from DUKPT encryption to using master/session encryption and to verify the MAC. If the MAC encryption type field on screen 7 of the PTD is set to "08" (DUKPT MAC verification/no PIN translation), two I/Os are required to the security device: one for the PIN translation and one for the MAC verification.

Vendor Impacts

The terminal vendor must create the MAC using DUKPT encryption and single-, double- or triple-length keys. This is passed in FID G. The clear derivation key used in the terminal must be encrypted and placed into the corresponding DRVKD record instead of the KEYD.

Reference Documents

Not applicable

EMV M/Chip 4 Support

This section describes the implementation requirements associated with the EMV M/Chip 4 Support enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the EMV M/Chip 4 Support enhancement.

For additional detailed information on this enhancement see Section 4 (Base).

The EMV processing has been enhanced to accommodate a new Issuer Application Format. M/Chip 4 Status Field Checking processing has been enhanced.

Software Impacts

For full details for this enhancement see Software Impacts in Section 4 (Base) in this document.

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD07.SCNMCHP4.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

The Card Verification Results field is a different format on an M/Chip 4 transaction. It is a 6 byte field. The decisional bytes of the field are bytes 4, 5 and 6. The

information contained in these bytes is held in the CVRTBLS and is as follows:

Action Index	Byte	Bit	Force Online	Auth Action	Meaning
"n"	4	7	"Y"	auth_act_cont_d	Unable to go Online
"n"	4	6	"Y"	auth_act_cont_d	Offline PIN verification not performed
"n"	4	5	"Y"	auth_act_cont_d	Offline PIN verification failed
"n"	4	4	"Y"	auth_act_cont_d	PTL exceeded
"n"	4	3	"Y"	auth_act_cont_d	International transaction
"n"	4	2	"Y"	auth_act_cont_d	Domestic transaction
"n"	4	1	"Y"	auth_act_cont_d	Terminal erroneously considers Offline PIN OK.
"n"	5	8	"Y"	auth_act_cont_d	Lower consecutive Offline limit exceeded
"n"	5	7	"Y"	auth_act_cont_d	Upper consecutive Offline limit exceeded
"n"	5	6	"Y"	auth_act_cont_d	Lower Cumulative Offline limit exceeded
"n"	5	5	"Y"	auth_act_cont_d	Upper Cumulative Offline limit exceeded
"n"	5	4	"Y"	auth_act_cont_d	Go online on next transaction was set
"n"	5	3	"Y"	auth_act_cont_d	Issuer Authentication failed
"n"	5	2	"Y"	auth_act_cont_d	Script received
"n"	5	1	"Y"	auth_act_cont_d	Script failed
"n"	6	2	"Y"	auth_act_cont_d	Match found in additional check table
"n"	6	1	"Y"	auth_act_cont_d	No match found in additional check table

The placeholder for all implementations is:

"1", 4, 4, "N", auth_act_cont_d,

This is interpreted as Issuer script processing failed on last transaction (VISA/MCHIP2) and PIN Try Limit exceeded (MCHIP4).

A base test set for M/Chip 4 has been added. This satisfies the rules for the UK Switch Debit Card scheme.

Unable to go online

"3", 4, 7, "Y", auth_act_cont_d,

Go online on next transaction was set

"3", 5, 4, "Y", auth_act_cont_d,

Issuer Authentication failed

"3", 5, 3, "Y", auth_act_ref_d,

Script failed

"3", 5, 1, "Y", auth_act_cont_d,

During migration two separate tables and action indexes will need to be used to cater for both M/Chip 2.1 and M/Chip 4 transactions.

Vendor Impacts

Not applicable

Reference Documents

- MasterCard Europe - M/Chip 4 card application specification, version 4.0, 16th May 2002
- EMVCo - EMV 2000 version 4.0, December 2000

Optional CRT Authorization Enhancements

This section describes the implementation requirements associated with the Optional CRT Authorization Enhancements.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Optional CRT Authorization Enhancements.

This enhancement supports the following:

- The Online Transaction Screen (OLT) has been enhanced to support Purchase with Cash Back transactions, with the exception of adjustment and Force Post adjustment messages.
- The BASE24-pos CRT Authorization processing has been enhanced to support the entering of an Override Approval Code in the POS Referral File (PRF).

Software Impacts

These changes were done to meet requirements for U.K. SWITCH Issuers.

All BASE24-bp customers:

- Inline changes to the BP60CRTA.RQOLTS, BP60CRTA.SCRNCSEC, and BP60CRTA.RQCSECS files to allow Purchase with Cash Back transactions to be entered on the OLT screen.
- Inline changes to the BP60CRTA.SCRNPRF, BP60CRTA.RQPRFS, and BP60CRTA.SVPRFS files to display the cash amount in a referred Purchase with Cash Back transaction and to allow the Override Approval Code to be entered on the PRF screen.
- Inline changes to the BP60CDDL.CRTDHG and BP60CRTA.CRTAUDHS files to process the Override Approval Code if entered on the PRF Screen.

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD07.SCNWCH.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

CRT Authorization Security File (CSEC)

The enhancement provides the capability to allow Purchase with Cash Back transactions as a valid CRT transaction (Screen 1).

Online Transaction Screen (OLT)

The enhancement provides the capability to use a transaction code of "PC" for Purchase with Cash Back transactions (Screen 1).

Support for Purchase with Cash Back transactions for Normal and Forced functions have been enabled. Normal functions allow an operator to send a particular transaction type for authorization and cancel, adjust, or reverse that transaction from the Online Transaction screen. Forced functions allow an operator to force post and adjust, reverse, or cancel the force post the corresponding transaction type from the Online Transaction screen (for those transactions that can be force posted).

Purchase with Cash Back Adjustment and Force Post Adjustment transactions are not supported.

Field Name: TRAN CODE
Screen: OLT Screen 1
DDL Name: PTLF.AUTH.TRAN-CDE.TC

A code identifying the type of transaction being attempted. This field can accept "PC" Purchase with Cash Back except for adjustment and Forced Post adjustment transactions. Valid values are as follows:

NP = Normal purchase
CA = Cash advance
MO = Mail/phone order
MR = Merchandise return
CV = Card verification
PC = Purchase with cash back

BASE24-pos Referral File (PRF)

The enhancement provides the capability to enter an Override Approval Code, and to view the cash back amount for referred Purchase with Cash Back transactions.

When using the PRF screen and an Override Approval Code has been entered and

the Override Approval Code entered fails the validation checks, the following message will be displayed on the message line of the PRF screen:

“CRTDH REPLY = INVALID APPROVAL CODE”

Field Name: CASH AMOUNT
Screen: PRF Screen 1
DDL Name: WS-AMT2-X

This field is only displayed if the transaction is purchase with cash back. Otherwise, this screen field is not visible.

Field Name: OVERRIDE APPROVAL CODE
Screen: PRF Screen 1
DDL Name: PTLF.AUTH.APPRV-CDE

The approval code associated with this transaction. This field can be entered by the user and must be the same number of characters as the referral code, contains either alpha or numeric characters and does not have embedded spaces.

For configuration information on the CSEC/OLT/PRF fields added for this enhancement, refer to the ***BASE24-pos CRT Authorization and CRT Administration Manual***.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Operational Impacts

The Tran Code field was modified on the OLT file maintenance screen 1. Operators must be trained in the proper use of this field.

Two new fields were added to the PRF file maintenance screen 1. Those fields are Cash Amount and Override Approval Code. Operators must be trained in the proper use of these fields

Vendor Impacts

Not applicable

Reference Documents

Not applicable

ACI Worldwide Inc.

POS Balance Inquiry Service

This section describes the implementation requirements associated with the POS Balance Inquiry Service enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the POS Balance Inquiry Service enhancement.

The POS Balance Inquiry Service enhancement allows acquirer and issuers to return balances on purchase transactions in compliance with Visa rules for Visa prepaid products. Configuration at the card prefix level will allow BASE24 customers to specify when balances are returned on the Purchase transaction for various Visa prepaid products. Balances can be returned via purchase, purchase with cash back, preauthorization, and balance inquiry transactions.

Participation in the POS Balance Inquiry Service is mandatory for issuers and issuer processors of specific Prepaid products.

Software Impacts

All BASE24 customers:

- DDL changes to DDLFCPF and DDLPSTKN
- Inline changes to the POS Token Conversion Utility
- Inline changes to OMF Extract
- Inline changes to the CPF Screen, Requester, and Server
- Inline changes to the CPF conversion program

All BASE24-pos customers:

- Minor inline changes to the Router/Authorization module
- New Log (Event) Message Manuals for the Router/Authorization module with the new POS Balance Inquiry Service events.

All BASE24-pos customers with the SPDH:

- Minor inline changes in the main module for the Standard POS Device Handler (SPDH).

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD07.SCNVISA.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes the new fields in the files that have been modified to support this enhancement.

Card Prefix File (CPF)

Field Name: RETURN BALANCES
Screen: CPF Screen 8
DDL Name: CPF.RTRN-BAL

The RTRN-BAL field signifies whether account balances should be returned on designated transactions. Valid values are:

- '0' = Balances are not returned for purchase transactions (default)
- '1' = Balances are returned for purchase transactions
- '2' = Balances are returned for purchase transactions, based on terminal configuration (PTDS1-CORE.BAL-RTRN-FLG)

Token File (TKN)

The POS Balances Token, (Token ID "CB") already existed, and will now be used to send balances back to the acquirer. BASE24-pos customers utilizing the POS Balance Inquiry Service should review their TKN record.

If the BASE24-pos token for this enhancement is not to be logged to the PTLF or ILF, the appropriate token record must be configured in the TKN.

If the token for this enhancement is to be extracted with the PTLF or ILF, sent to the host, or passed through BIC ISO Interchange Interface, the appropriate TKN records must be configured.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Acquirers and Acquirer Processors may need to work with merchants to make changes to terminal software that supports purchase and stand-alone balance inquiries to ensure that balance data can be printed on cardholder receipts.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Card Prefix File (CPF)

If the BASE24-pos user wishes to enable POS Balance Inquiry Service processing, the user must update the new Return Balances field in the CPF with the appropriate value. Otherwise, the new field in the CPF will default to indicate that Balances are not to be returned.

Host/Co-Network Impacts

Super Extract Process

If the token (i.e., POS Balances Token) for this enhancement is to be included in the PTLF and ILF Extracts, BASE24 users must consider this additional data in host programs that use the data from the log files.

Host code changes are required for all BASE24 users that use OMF Extract to accept and process information for the change to the CPF records.

ISO Host Interface Process

If the token for this enhancement (i.e., POS Balances Token) is included in the ISO message to the host, this change to the message needs to be handled by the host programs.

User/Operational Impacts

Disk Impacts

The PTLF and ILF will require additional disk space as the token for this enhancement (i.e., POS Balances Token) can be included in the transaction records logged to these files.

Operational Impacts

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on \$<volume>.BA60LOGM.

System Limitations

BASE24 users installing this enhancement must be aware of their internal system limitations for messaging to the PTLF and ILF. There are system limitations regarding the amount of data that can be passed through the BASE24 system and logged to a log file. BASE24 users should be aware of these limitations and be cognizant that

support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the ***BASE24 Tokens Manual*** provides information on assessing the impacts of tokens on log file disk space requirements.

Vendor Impacts

Not applicable

Reference Documents

- Visa “October 2003 Visanet Business Enhancements Technical Letter – Section 1.13”, Version 3, April 11, 2003

ACI Worldwide Inc.

Secure Internet Validation

This section describes the implementation requirements associated with the Secure Internet Validation enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Secure Internet Validation enhancement.

ACI's E-Commerce products support the ability for a cardholder to register for either the Visa (3-D Secure) or MasterCard (SPA UCAF) Internet security products and to perform cardholder authentication during an online purchase. These are supported through cardholder registration and cardholder authentication during a purchase with an Access Control Server (ACS) that is compliant with Visa or MasterCard's rules.

The Secure Internet Validation (SIV) enhancement extends secure internet processing to BASE24-pos. BASE24-pos will participate in the initial authentication via a card verification transaction and in the actual purchase via a purchase, preauthorization request, or preauthorization completion transaction.

Note: Preauthorization processing for a lesser amount is not supported in the Secure Internet Validation enhancement.

If Virtual Primary Account Numbers (VPANs) are being used, the currency code and transaction amount in the card verification are based on the issuer's currency. In addition, the currency code and amount in the actual purchase are based on the issuer's currency. Thus, the BASE24-pos router/authorization process uses the currency code and transaction amount fields in the PSTM (i.e., PSTM.ORIG-CRNCY-CDE and PSTM.TRAN.AMT-1) for SIV processing.

If VPANs are not being used, the currency code and transaction amount in the card verification are based on the merchant's currency. In addition, the currency code and amount in the actual purchase are based on the merchant's currency. Thus, the BASE24-pos router/authorization process uses the currency code and transaction amount fields in the Original Currency 6.0 Token (i.e., ORIG-CRNCY-60-TKN.CRNCY-CDE and ORIG-CRNCY-60-TKN.AMT-1) for SIV processing. If the token is not available, the currency code and transaction amount fields in the PSTM are used.

Software Impacts

All BASE24 customers:

- DDL changes to DDLFCPF
- CUSTMACS changes for this enhancement that were a part of release 6.0 version 3.
- Hooks in the security utilities source file (BA60SRC.SECUTILS)
- A new dummy file for the security utilities (BA60SRC.SECUTILD)

All BASE24-pos customers:

- Changes to the PS60SRC.POSFM and POSMM files to include the SIV flags.
- Minor inline changes in the main module and global files for Router/Authorization.
- Additional procs in the Dummy files (PS60RTAU.RTAUD)
- Make file changes that include the SIV flag.
- New Log (Event) Message Manuals for the Router/Authorization module with the new SIV events.

Only customers taking the new SIV functionality :

- New files in the extension subvolumes containing SIV functionality (i.e., BA60VSR, PS60VRTA). The files on these subvolumes include source files, global files and Make files required for the new functionality.

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD07.SCNSIV.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

This section describes the Assign records that must be added to the LCONF to support this enhancement. These modifications are provided in the LCONFPS file located on the BA60MISC subvolume.

POS-EAF

The fully qualified file name of the E-Commerce Authentication File. This is required for Secure Internet Validation customers using VPANs, validating CAVVs without a host security module, or validating AAVs.

POS-EAF-PREV

The fully qualified file name of the previous E-Commerce Authentication File. This is required for Secure Internet Validation customers using VPANs, validating CAVVs

without a host security module, or validating AAVs.

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement. These modifications are provided in the LCONFPS file located on the BA60MISC subvolume.

POS-EAF-NOTIFY-INTERVAL

The number of writes made to the E-Commerce Authentication File (EAF) between calculating the percentage of file full. Each time a write is made to the EAF, a counter is increased. When the counter is equal to the value in this param, the Router module calculates how full the EAF is. The counter is then reset to zero. When the file reaches the percent full specified in the POS-EAF-NOTIFY-PERCENTAGE param, the Router module generates an event message indicating that the file is becoming full. Until more file space is made available, the Router module continues to generate event messages indicating that the file is becoming full at the interval specified in this param. Valid values are 5 to 32766, and the default is 5.

POS-EAF-NOTIFY-PERCENTAGE

The percentage of file full at which the Router module of the BASE24-pos Device Handler/Router/Authorization process generates an event message indicating that the E-Commerce Authentication File (EAF) is becoming full. The Router module calculates the percentage of file full and continues to calculate the percentage at the interval indicated in the POS-EAF-NOTIFY-INTERVAL param as described in the following paragraphs until more file space is made available. If the percentage has increased by one percent or more since the preceding calculation, the Router module generates an event message indicating that the file is becoming full. Valid values are 10 through 99, and the default is 70.

POS-SETL-EAF-NOTIFY*identifier*

The symbolic name of a BASE24-pos Device Handler/Router/Authorization process that the Main Settlement process must notify after the new EAF file has been created. The Main Settlement process uses this param. A maximum of 500 processes can be notified.

Each POS-SETL-EAF-NOTIFY param contains the name of one process. When multiple processes must be notified, this param can include an optional *identifier* to specify params used for different processes in the same logical network. The optional *identifier* is usually a number but can be any unique set of characters that has not already been declared by another POS-SETL-EAF-NOTIFY*identifier* param. For example, the POS-SETL-EAF-NOTIFY01 param could contain the P1A^RTAU01 process and the POS-SETL-EAF-NOTIFY02 param could contain the P1A^SPDH01 process.

POS-SETL-EAF-RETENTION-PERIOD

The number of days the EAF and the previous EAF should be retained before the BASE24-pos settlement process cuts over the file.

Database Configuration

This section describes the new fields in the files that have been modified to support this enhancement.

Card Prefix File (CPF)

Field Name: SIV CHECK TYPE
Screen: CPF Screen 6
DDL Name: CPF.POS.SIV-CHK-TYP

Indicates whether SIV card verification is to be performed.

Field Name: SIV KEYA GROUP
Screen: CPF Screen 6
DDL Name: CPF.POS.SIV-KEYA-GRP

An identifier used to show the card verification group to which this record belongs on the Key Authorization file (KEYA).

Field Name: CHECK IF HOST ONLINE SIV
Screen: CPF Screen 6
DDL Name: CPF.POS.SIV-CHK

Indicates whether a pre-screen SIV card verification check should be performed by the Router/Authorization processes before sending a transaction to the host.

Field Name: BAD SIV ACTION
Screen: CPF Screen 6
DDL Name: CPF.POS.SIV-BAD-DISP

The action that the Router/Authorization processes will take when a cardholder uses a card that contains invalid verification data.

Field Name: SIV NOT PRESENT ACTION
Screen: CPF Screen 6
DDL Name: CPF.POS.SIV-NOT-PRSN-DISP

The action that the Router/Authorization processes will take when a Cardholder Authentication Verification Value (CAVV) or Accountholder Authentication Value (AAV) was not present in the Secure Internet Validation transaction.

Field Name: SIV ATTEMPT ACTION
Screen: CPF Screen 6
DDL Name: CPF.POS.SIV-ATTEMPT-DISP

The action that the Router/Authorization processes will take when the interchange informs BASE24 that a pass-validation-attempt situation occurred on the Cardholder Authentication Verification Value (CAVV) or the Accountholder Authentication Value (AAV). This field is only used if the SIV CHECK-TYPE field is not equal to either zeroes or spaces.

External Message File (EMF)

Data elements P-12 and P-13 must be included in the ISO message that is sent to the E-Commerce product on a BASE24-pos 0430 response. Thus, the EMF record used on outbound BASE24-pos 0430 messages must be updated or added to include data elements P-12 and P-13.

Host Configuration File (HCF)

When the E-Commerce product sends a reversal for BASE24-pos card verification transactions, E-Commerce requires a response. Thus, the ACK TO DPC field on screen 1 of the HCF must be sent to Y for the ISO Host Interface configured for the E-Commerce product.

Key Authorization file (KEYA)

KEYA records may be added for a SIV KEYA group, since the SIV KEYA group may be different than the CARD VERIFICATION KEYA group. In addition, BASE24 user may not have performed card verification, but the user is configuring SIV card verification.

BASE24-pos Transaction Log File (PTLF)

Field Name: VIRTUAL PAN

Screen: PTLF Screen 4

DDL Name: PSEUDO-CRD-NUM-TKN.PSEUDO-CRD-NUM

The Virtual PAN data from the Pseudo Card Number Token. This field is only displayed when the Pseudo Card Number Token is present and the transaction is a purchase, preauthorization purchase, preauthorization completion, mail phone order, or card verification. Otherwise, this new screen field is not visible.

Field Name: EXP DAT

Screen: PTLF Screen 4

DDL Name: PSEUDO-CRD-NUM-TKN.EXP-DAT

The expiration date of the Virtual PAN from the Pseudo Card Number Token. This field is only displayed when the Pseudo Card Number Token is present and the transaction is a purchase, preauthorization purchase, preauthorization completion, mail phone order, or card verification. Otherwise, this new screen field is not visible.

Token File (TKN)

The following BASE24-pos token has been added for the Secure Internet Validation enhancement:

- The Pseudo Card Number Token (Token ID “BL”) contains the Virtual PAN data needed for transaction processing.

If the BASE24-pos token for this enhancement is not to be logged to the PTLF or ILF, the appropriate token record must be configured in the TKN.

If the token for this enhancement is to be extracted with the PTLF or ILF, sent to the host, or passed through BIC ISO Interchange Interface, the appropriate TKN records must be configured.

New Data Files

E-Commerce Authentication File (EAF)

The E-Commerce Authentication File (EAF) is used to store E-Commerce transactions for transaction validation. An EAF file (EAF) and a previous EAF file (EAFPREV) will exist for each logical network. These two files together must be kept for the maximum life of a transaction. Due to potential recurring payments, it is recommended that the files must be kept for 300 days, and therefore the EAF file must be kept for at least 150 days as well as the EAFPREV file.

A new record will be written to the EAF file or another transaction entry will be added to an EAF record on an E-Commerce card verification transaction. A transaction entry in either the EAF and EAFPREV files will be updated on a purchase, preauthorization purchase, preauthorization purchase completion, or mail phone order transaction.

The EAF and EAFPREV records are referenced by the Virtual PAN (VPAN) and real PAN, so that a VPAN can be linked to a real PAN. There is not BASE24 AFT subsystem access to the EAF or EAFPREV files.

Template Changes

CUSTCNST

The constant CUST-EAF-REC-TXN-MAX-L from the CUSTCNST file on the PS60DDL subvolume needs to be modified if the Secure Internet Validation enhancement is being used. The value of this constant must be set according to the customer's configuration.

After the modification of this constant, the BASE24-pos DDLs and Secure Internet Validation enhancement code must be recompiled. The value in this constant dictates the maximum number of E-commerce authentication transactions outstanding that a customer's given logical network will contain in the EAF and EAFPREV files. It is imperative that a customer consider the number of transactions outstanding for one cardholder based on their VPAN or possibly their real PAN in the EAF and EAFREV files.

The use of this constant allows a customer to tailor the size of the EAF and EAFPREV files to its actual customer usage, thereby eliminating some possibly wasted space in the EAF and EAFPREV files. The value in this constant should never be set to more than 46, as that is the maximum number of E-commerce authentication transactions outstanding.

EAF

There is one new template for the EAF and EAFPREV files. The template must be created before starting BASE24-pos settlement. When settlement is started, the EAF and EAFPREV files are created using the new EAF template file.

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

The BASE24-pos user must consider the new pre-screening checks defined in the POS segment of the CPF for this enhancement.

Network Configuration

An ISO Host Interface (Acquiring) process may be needed to be configured and maintained between the issuer's ACS or SPA Wallet Server, since the issuer ACS or SPA Wallet Server must send a card verification transaction into BASE24 using the ISO Host Interface process. This must be received and approved prior to allowing the cardholder to purchase goods such that the EAF can be populated prior to receiving the purchase transaction.

File Conversions

Card Prefix File (CPF)

If the BASE24-pos user wishes to enable SIV processing, the user must update the new fields in the POS segment of the CPF with appropriate values. Otherwise, these new fields in the POS segment of the CPF indicate that SIV processing is disabled.

Host/Co-Network Impacts

Super Extract Process

If the token (i.e., Pseudo Card Number Token) for this enhancement is to be included in the PTLF and ILF Extracts, BASE24 users must consider this additional data in host programs that use the data from the log files.

Host code changes are required for all BASE24 users that use OMF Extract to accept and process information for the changes to the CPF records.

The PTLF record for the card verification portion of a SIV transaction will contain a non-zero transaction amount. Host code changes may be required for this situation.

ISO Host Interface Process

If the token for this enhancement (i.e., Pseudo Card Number Token) is included in the ISO message to the host, this change to the message needs to be handled by the host programs.

Host code changes may be required for those BASE24-pos users that wish to specify an expiration date other than the YYMM format for a Virtual PAN during the card verification portion of a SIV transaction. Data element 126 of the ISO message is used to specify whether the SIV transaction expires in x minutes, y hours, or z days.

User/Operational Impacts

Disk Impacts

The new EAF and EAFPREV impacts disk requirements. The total impact varies for each BASE24 user depending on the number of EAF and EAFPREV records defined in the BASE24 system.

The KEYA may require additional disk space when new KEYA records are added for a SIV KEYA group. The same KEYA record may be used when the SIV KEYA group matches the CARD VERIFICATION KEYA group.

The PTLF and ILF require additional disk space as the new token for this enhancement (i.e., Pseudo Card Number Token) can be included in the transaction records logged to these files.

Operational Impacts

PTLF Perusal support the data defined in the new Pseudo Card Number Token. Operators should be trained to recognize the new information on the screen.

BASE24-pos response code 077 will be returned by the Router/Authorization processes when the cardholder already had the maximum number of E-Commerce authentication transactions outstanding. Additional SIV card verification transactions will continue to be denied until at least one of the authentication transactions expires or is removed from the EAF. Previously, this response code was only returned when the cardholder already had the maximum number of preauthorization transactions outstanding. The maximum number of preauthorizations outstanding is specified on CPF screen 7.

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on \$<volume>.BA60LOGM.

System Limitations

BASE24 users installing this enhancement must be aware of their internal system limitations for messaging to the PTLF and ILF. There are system limitations regarding the amount of data that can be passed through the BASE24 system and logged to a log file. BASE24 users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the **BASE24 Tokens Manual** provides information on assessing the impacts of tokens on log file disk space requirements.

Performance Impacts

E-Commerce transactions that are written to the EAF require additional disk I/O.

For those BASE24-pos users that configure the CAVV validation with a host security module (i.e., HSM) an Inter-Process Communication (IPC) message is added for each call to the HSM.

User Impacts

BASE24 users utilizing the Secure Internet Validation enhancement must enter the appropriate information in the PS60DDL.CUSTCNST file.

Vendor Impacts

Not applicable

Reference Documents

- Electronic-Commerce Initiatives BASE24 and ACI Enterprise Payment System Support, December 31, 2001
- Europay “Maestro Implementation Guide for e-Commerce, Dual Message”, January 2002
- MasterCard International “SPA and UCAF Functional Specification”, Version, 1.0, April 3, 2001
- MasterCard International “SPA and UCAF Detailed Specification”, Version, 1.00, June 30, 2001
- Visa “3-D Secure Functional Requirements Access Control Server”, Version 1.0, August 3, 2001, (Ref. No. 70002-01)
- Visa “3-D Secure Functional Requirements Access Control Server”, Version 1.02, July 16, 2002, (Ref. No. 70002-01)
- Visa “3-D Secure Introduction”, Version 1.01, September 28, 2001 (Ref. No. 70001-01)
- Visa “3-D Secure Protocol Specification Core Functions”, Version 1.02, July 16, 2002, (Ref. No. 70000-01)
- Visa “April 2002 VisaNet Business Enhancements Technical Letter, (80007-01)
- Visa “April 2002 VisaNet Business Enhancements Technical Letter Update Bulletin #1, (Ref. No. 80007-02)
- Visa “October 2002 VisaNet Business Enhancements Technical Letter Update Bulletin #2, (Ref. No. 80011-03)
- Visa “Visa Payer Authentication Service Technical Requirements”, Version 1.14, January 2, 2001
- Visa “V.I.P. System SMS POS Technical Specifications (U.S.),” January 30, 1999

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SPDH DUKPT Support via TSS

This section describes the implementation requirements associated with the SPDH DUKPT Support via TSS enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the SPDH DUKPT Support via TSS enhancement.

The SPDH supports DUKPT for the encryption of PIN blocks for transactions originating at the terminals directly connected to BASE24-pos. DUKPT support was added in Release 6.0 Version 2. This enhancement expands support for DUKPT in three ways:

- through Transaction Security Services
- using double-length or triple-length keys
- performing additional security checks - comparing the KSN IDs and transaction counters from two consecutive transactions.

Prerequisites: The Derived Unique Key Per Transaction (DUKPT) and Transaction Security Services enhancements must be installed prior to implementation of the SPDH DUKPT Support via TSS enhancement. For more information refer to the ***BASE24 Release 6.0 Implementation Guide (Version 2)***.

Software Impacts

All BASE24 customers:

- Inline changes to SECUTILS and the KEYD server (SVKEYDS).
- New files for the KEYD conversion program (TSSKEYDM, TSSKEYDR, TSSKEYDS).
- DDL changes in BA60DDL.DDLGTSS.
- DDL changes in BA60UC05.DDLFTSS.
- Changes to the BA60UC05.BAUC05M and BAUC05MM files to include the KEYD conversion program.

All BASE24-pos customers:

- Inline changes to the PTD Server (SVPTDTS)

All customers with the ACI Standard POS Device Handler:

- Inline changes to SPDH (ASPDHS and ASPDHG).
- New Log (Event) Message Manuals for the SPDH module with the new DUKPT events.

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD07.SCNDKTSS.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement. Note that these modifications are provided in the LCONFPS file located on the BA60MISC subvolume.

POS-DH-KSN-MAX-DIFFERENCE

This param is used by the SPDH device handler. This param indicates the largest allowable difference in the transaction counter between two consecutive transactions from the same terminal. This is an optional param. Valid values are 0 (zero) through 4096. The value 0 (zero) indicates that the transaction counter will not be checked. If the param is not found or is invalid, the default value of 4096 is used.

POS-DH-KSN-APPRV-OPT

This param is used by the SPDH device handler. This param indicates the transaction processing options when the KSN ID in the message from the device does not match the KSN in the PTD (the KSN from the previous transaction). This is an optional param. Valid values are 1 (deny the transaction), 2 (approve the transaction, do not save the KSN) and 3 (approve the transaction, save the KSN). If the param is not found or is invalid, the default value of 3 is used.

POS-DH-KSN-CNTR-APPRV-OPT

This param is used by the SPDH device handler. This param indicates the transaction processing options when the transaction counter in the message from the device has not been incremented or when the comparison between the transaction counter in the message from the device and the transaction counter stored in the PTD exceeds the maximum difference allowed (as specified in the POS-DH-KSN-MAX-DIFFERENCE param). This is an optional param. Valid values are 1 (deny the transaction), 2 (approve the transaction, do not save the KSN) and 3 (approve the transaction, save the KSN). If the param is not found or is invalid, the default value of 3 is used.

Database Configuration

Channel Security Data Source (CSECD)

Transaction Security: Channel Security Configuration
Tab: PIN Information
Field: Key Length of Exchange Key

The Key Length of the Exchange Key for the SPDH CSECD records must be set to the appropriate length depending on whether single, double or triple-length keys are being used.

Transaction Security: Channel Security Configuration
Tab: PIN Information
Field: Exchange Key (Header, Part 1-4, Check Digits)

The Exchange Key for the SPDH CSECD records must be modified to contain the new key value.

Transaction Security: Channel Security Configuration
Tab: PIN Information
Field: Key Length of Inbound Key

The Key Length of the Inbound Key for the SPDH CSECD records must be set to the appropriate length depending on whether single, double or triple-length keys are being used.

Transaction Security: Channel Security Configuration
Tab: PIN Information
Field: Inbound Key (Header, Part 1-4, Check Digits)

The Inbound Key for the SPDH CSECD records must be modified to contain the new key value.

Refer to the online help provided with the ACI Desktop user interface for detailed descriptions of this window and its fields.

KEYD

The derivation key field in the KEYD records must be updated to contain key locators. The key locators are used by Transaction Security Services to read DRVKD (Derivation Key Data Source) records to retrieve the actual derivation key used by the POS device.

Refer to the File Conversions subsection of this enhancement to see how the key field in the KEYD can be updated programmatically and corresponding DRVKD records created.

If the conversion program is not used, the derivation key fields in the KEYD records can be updated manually using the KEYD requester/server.

DRVKD

A DRVKD record must exist for every set of derivation keys used by POS devices in the BASE24 network. There must be one DRVKD record for every KEYD group defined in the network.

Refer to the File Conversions subsection of this enhancement to see how the DRVKD record for each KEYD record can be created programmatically.

If the conversion program is not used, the DRVKD records can be added manually using the ACI Desktop.

New Data Files

A new data file is required that contains derivation keys referenced by key locators during transaction processing. This file can be created by running the conversion program or records can be added, deleted or modified manually using the ACI Desktop.

- DRVKD Derivation Key Data Source

The following table shows the relationship of the existing BASE24 Base database file to the corresponding Transaction Security Services database file. The first column lists the existing BASE24 file and the second column lists the file accessed by the Transaction Security Services process to get the real derivation keys used during transaction processing.

BASE24 Database Files	Transaction Security Services Database Files
KEYD	DRVKD

Template Changes

Not applicable

Device Configuration

The device must have new software/hardware configured to use DUKPT in general, and the DUKPT Triple DES algorithm and double-length or triple-length keys, if necessary.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

If all PINS will be encrypted in DUKPT, access to the keys stored in the DRVKD and CSECD that used to be stored in the KEYD and PTD, respectively, is through the new ACI Desktop Graphical User Interface that is set up separately from the BASE24 Pathway CRT access system. Refer to the ***BASE24 Classic 6.0 Installation Guide*** for further information.

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

A file conversion for the following BASE24 Base file is supplied with this enhancement. There are no structural changes in any files, but Transaction Security Services files can be created and data populated from BASE24 files. BASE24 files will be updated with locators to point BASE24 processes to the correct Transaction Security Services records. Transaction Security Services data can be entered manually using the ACI Desktop Graphical User Interface instead of using the conversion programs.

- KEYD

BASE24 Base Derivation Key file (KEYD)

The KEYD conversion program can be run to create DRVKD records while updating KEYD records with key locators that point to the appropriate DRVKD records. Running the conversion program is not required. DRVKD records can be entered manually using the ACI Desktop Graphical User Interface instead. Key fields in corresponding KEYD records would have to be updated manually as well.

The KEYD conversion program is on the BA60UC05 subvolume. If modifications to the conversion program are necessary, the program can be modified by editing the TSSKEYDS file.

To compile the conversion program, enter the following command at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#keydtss.t,name $keydm,nowait/  
tsskeydm -force -list -listreplace
```

Users must update the TSSKEYDR obey file on the BA60UC05 subvolume with the appropriate information. TSSKEYDR must then be obeyed to execute the conversion program.

For more information on this file conversion, refer to the AAREAD05 and TSSKEYDR files on the BA60UC05 subvolume.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Operational Impacts

When a new PIN Pad is installed in the terminal, the KSN in the PTD must be cleared if the POS-DH-KSN-APPRV-OPT param in the LCONF is set to “1” (deny the transaction) or “2” (approve the transaction, do not save the KSN). To clear the KSN in the PTD, perform these steps:

1. Set the POS-DH-KSN-APPRV-OPT param to a value of “3” (approve the transaction, save the KSN).
2. Warmboot the device handler.
3. Run a transaction.
4. Reset the POS-DH-KSN-APPRV-OPT param to the desired value.
5. Warmboot the device handler.

If batch, shift and day totals have been cleared, the PTD record can be deleted and readded to clear the KSN.

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on \$<volume>.BA60LOGM.

Vendor Impacts

The terminal vendor must encrypt the PIN block using DUKPT encryption and single-, double- or triple-length keys. This is passed in FID b or FID c (admin). The clear derivation key used in the terminal must be encrypted and placed into the corresponding DRVKD record instead of the KEYD.

Reference Documents

Not applicable

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SPDH EMV Additional Support

This section describes the implementation requirements associated with the SPDH EMV Additional Support enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the SPDH EMV Additional Support enhancement.

The SPDH EMV Additional Support enhancement involves the changes related to processing of:

- Authorization Response Code
- Pre Auth Request Data
- CVM Results Data
- Additional Discretionary Data Elements

These requirements are discussed in more detail below.

Authorization Response Code

The Issuer generates the Issuer Authentication Data - IAD. The Issuer Data is specific to the issuer and may contain the Authorization Response Code – ARC. Currently BASE24 SPDH does not process ARC as a separate field. The complete IAD, if present, is sent to the Terminal in sub-FID 6Q (EMV Response Data).

Clarify case #357017 highlighted that some cards exist, where the issuer does not include ARC as part of IAD. BASE24 SPDH has been enhanced so that a separate ARC (BASE24 generated) is always sent to the Terminal in extended sub-FID 6Q.

This enhancement allows an ARC value to be sent to the Terminal regardless of it being received in the IAD or not. The ARC value that will be sent to the terminal, is a mapping of the BASE24-pos 3 digit response code to a 2-digit ARC code according to the following mapping:

BASE24-pos Response Code	Authorization Response Code
000 – 049	00 (approve)
100 – 149	01 (refer)
900 – 949	04 (pick-up)
All others	05 (decline)

The format of sub-FID 6Q has been modified to include the 2-digit ARC value according to the above mapping.

Pre Auth Request Data

At present EMV pre-authorization transactions include a cryptogram, which is calculated using the amount field plus other EMV Request Data fields sent in the request message. This cryptogram is verified successfully within BASE24. The completion message may include the original cryptogram, but if the amount sent in the message is different (in most cases they would be different) from the original pre-authorization message, the cryptogram verification fails.

This enhancement required the cryptogram in pre-authorization completion messages to be verified successfully.

BASE24 SPDH has been enhanced so that the transaction currency code and amount used to create the cryptogram are new mandatory fields in pre-authorization and completion request messages received from the Terminal. This will allow BASE24 to include the cryptogram in both the request messages sent to issuers. The new fields, cryptogram amount and cryptogram currency code should be included in the extended subFID 6O sent from the terminal to SPDH Device Handler.

The cryptogram amount field and cryptogram currency code will be moved to the AMT_AUTH and TRAN_CRNCY_CDE field in EMV Request Token respectively. EMV Request Token Data is subsequently used to verify the cryptogram.

The pre-authorization completion request messages are downgraded to non EMV so that the cryptogram is not verified by BASE24. BASE24 SPDH has been enhanced so that the transaction currency code and amount used to create the cryptogram are new mandatory fields in pre-authorization and completion request messages received from the Terminal. This will allow BASE24 to include the cryptogram in both the request messages sent to issuers. The new fields, cryptogram amount and cryptogram currency code should be included in the extended Sub-FID 6O sent from the terminal to SPDH Device Handler.

CVM Results Data

The CVM Results field is not one of the 13 mandatory data elements in EMV request messages. The advent of offline PIN processing means that for these transactions, acquirers and issuers cannot determine whether offline PIN verification was performed (and successful) unless the CVM Results data element is included in the POS transaction request.

BASE24 SPDH has been enhanced so that the CVM Results data element (6 bytes) can be included in the extended subFID 6P from the terminal to the SPDH. The CVM Results data will also be added to the EMV Status and EMV Discretionary Data tokens.

The EMV Status token field DEV-INFO has been redefined to hold the CVM Results data.

The BASE24-pos APACS 30/40 Device Handler has also been enhanced to add CVM results data to the EMV Status token.

Additional Discretionary Data Elements

Within BASE24 the EMV Discretionary Token contains a number of fields that cannot be used for messages received in the SPDH. While the enhancements were being made to cater for CVM Results etc as described in the above sections, BASE24 SPDH has been enhanced so that the Application Version Number and Dedicated File Name which are fields in EMV Discretionary Token, can be included in the extended subFID 6P from the terminal.

The subFID 6P has been extended to include 4 bytes of Application Version Number and a maximum of 32 bytes of Dedicated File Name. The new data is moved to APPL-VER-NUM and DF-NAME fields in the EMV Discretionary Data Token. The Issuer interfaces will then be able to access this data and send to the issuer if required.

Software Impacts

All BASE24-pos customers with the SPDH EMV extension:

- Inline changes to SPDHDDL, SPDHEMVG and SPDHEMVS.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

The EMV enhancements to SPDH detailed in above are backwards compatible with terminals not modified to support the new message formats. This enhancement uses the smart card scheme in sub-FIDs 6O, 6P and 6Q to identify terminals that have been updated to use the new message formats. Existing terminals only allow a value “00” in this field. Upgraded terminals that conform to this enhancement will need to send a value of “01” for the card scheme in sub-FIDs 6O and 6P. The SPDH device handler will also send “01” for the card scheme field in a response message to the terminal in sub-FID 6Q, if it had received “01” in the request message sub-FID 6O.

Reference Documents

Not applicable

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SPDH Enhanced Multi-Threading

This section describes the implementation requirements associated with the SPDH Enhanced Multi-Threading enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the SPDH Enhanced Multi-Threading enhancement.

A standard BASE24-pos module has been modified to support this enhancement. That module is the ACI Standard POS Device Handler (SPDH).

Software Impacts

All BASE24 customers:

- Changes to BA60MAKE.CUSTMACS to include the pos_dh_spdh_mt_on flag

All BASE24-pos customers:

- Changes to Make files to include the pos_dh_spdh_mt_on flag (PS60SRC.POSFM and PS60SRC.POSMM)
- Changes to File Maintenance components to support new BASE24-pos Terminal Data Files (PTD) fields and function keys (OK60AFM.AFMSVTG, PS60AFT.RQPTDS, PS60AFT.SCRNPTD, PS60AFT.SVPTDTBL and PS60AFT.SVPTDTS)
- Changes to DDL files (PS60DDL.DDLFPTD and PS60DDL.DDLGPTD)

All customers with the ACI Standard POS Device Handler:

- Hooks in the main device handler module to SPDH Enhanced Multi-Threading procedures (PS60SPDH.ASPDHS)
- Additional procedures in the dummy file (PS60SPDH.SPDHD)
- Make file changes (PS60SPDH.ASPDHM)
- Updated event (log) message manual with new events (BA60LOGM.PSSPDH)

Only customers taking the new SPDH Enhanced Multi-Threading functionality:

- New extension subvolume with files containing SPDH Enhanced Multi-Threading functionality (PS60MSPD)

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD07.SCNSEMT.

CUSTMACS Changes

The following Make flag has been added to the CUSTMACS file.

Set the following flag to true to bind the SPDH Enhanced Multi-Threading module to the ACI Standard POS Device Handler.

- pos_dh_spdh_mt_on

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

The following database configuration changes should be made if this enhancement is implemented.

BASE24-pos Terminal Data Files (PTD)

The following two fields have been added to PTD file maintenance screen 1 in support of this enhancement.

Field Name: SITE ID

Screen: PTD Screen 1

DDL Name: PTDD1.MULT-THREAD-SPDH.SITE-ID

This field contains a user-defined identifier for the site. It is used to group the PTD records of the terminals at the site. To implement the SPDH Enhanced Multi-Threading enhancement, this field must contain a non-blank value.

Field Name: RECORD TYPE

Screen: PTD Screen 1

DDL Name: PTDD1.MULT-THREAD-SPDH.REC-TYP

This field further defines the site's PTD records. Valid values are:

P = Primary record
S = Secondary record
Blank = Not applicable

To implement the SPDH Enhanced Multi-Threading enhancement, this field must contain a non-blank value.

For the site's primary PTD record (Record Type = "P"), the Terminal Cutover field on screen 3 should be set to "3" (Network forced-cutover by the Settlement Initiator process). This will instruct the Settlement process to force-cutover the site's primary PTD record at the end of the retailer cutover window. During forced-cutover, the Settlement process will clear the site's aggregate transaction totals, which are contained in the site's primary PTD record.

To keep the terminals' totals in sync with the site's totals, it is recommended that the Terminal Cutover field of the site's secondary PTD records (Record Type = "S") also be set to "3".

The card types allowed at a given terminal are defined on PTD screens 8, 9 and 10. Note the allowed card types for every terminal belonging to a given site. All of those card types should be defined on screens 8, 9 and 10 of the site's primary PTD record (Record Type = "P"). This will allow the SPDH to properly maintain the site's aggregate card totals, a.k.a. service totals.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

The PTD fields utilized by this enhancement (SITE-ID and REC-TYP) were originally added to the PTDD1 record definition in Release 6.0 Version 2. As these fields are initialized to blanks, and blanks are the default values, no file conversion is necessary for this enhancement.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Disk Impacts

This enhancement introduces a new alternate key to the BASE24-pos Terminal Data Dynamic File—general data (PTDD1) file. This alternate key, keytag "MT", contains the Site ID and Record Type. If this enhancement is implemented, a new alternate key file (PTDD11) will be created. This alternate key file will contain one 36-byte record for every record in the PTDD1. BASE24-pos customers implementing this enhancement should examine their current disk capacity and determine this enhancement's impacts.

Operational Impacts

Two new fields were added to PTD file maintenance screen 1. Those fields are Site Id and Record Type. Operators must be trained in the proper use of these fields.

Three new function keys were added to the PTD file maintenance screen. Those function keys are SF2 – Read Site Primary Record, SF5 – Update Site Secondary Records and SF6 – Read Next Record by Site ID. Operators must be trained in the proper use of these function keys.

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate remedy for a given message. Refer to the event message documentation files on \$<volume>.BA60LOGM.

Performance Impacts

The SPDH will perform an extra read/write on the PTDD1 file for every financial transaction originating from every terminal defined as "Secondary" (PTD Record Type = "S"). This extra read/write updates the site's aggregate transaction totals in the primary PTD record.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Triple DES for BASE24 SPDH

This section describes the implementation requirements associated with the Triple DES for BASE24 SPDH enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Triple DES for BASE24 SPDH enhancement.

The Triple DES for BASE24 SPDH enhancement modified the BASE24-pos SPDH module to support Transaction Security Services (TSS) and triple DES encryption/decryption.

Prerequisite: The Transaction Security Services enhancement must be installed prior to implementation of the Triple DES for BASE24 SPDH enhancement. For more information refer to the ***BASE24 Release 6.0 Implementation Guide (Version 2)***.

Software Impacts

All BASE24 customers:

- Modifications are inline in the DDLBCNST file.

All customers with the ACI Standard POS Device Handler:

- Modifications are inline in the SPDH (ASPDHS and ASPDHG) and SPDH Loaders (SPDHLDERS).
- New Log (Event) Message Manuals for the SPDH module with the new Triple DES events.

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD07.SCNSPTD.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Channel Security Data Source (CSECD)

Transaction Security: Channel Security Configuration
Tab: PIN Information
Field: Key Length of Exchange Key

The Key Length of the Exchange Key for the SPDH CSECD records must be set to Double or Triple to utilize SPDH Triple DES using double-length or triple-length keys.

Transaction Security: Channel Security Configuration
Tab: PIN Information
Field: Exchange Key (Header, Part 1-4, Check Digits)

The Exchange Key for the SPDH CSECD records must be modified to contain new double-length or triple-length key value.

Transaction Security: Channel Security Configuration
Tab: PIN Information
Field: Key Length of Inbound Key

The Key Length of the Inbound Key for the SPDH CSECD records must be set to Double or Triple to utilize SPDH Triple DES using double-length or triple-length keys.

Transaction Security: Channel Security Configuration
Tab: PIN Information
Field: Inbound Key (Header, Part 1-4, Check Digits)

The Inbound Key for the SPDH CSECD records must be modified to contain new double-length or triple-length key value.

Transaction Security: Channel Security Configuration
Tab: MAC Information
Field: Key Variant

If using an Atalla non-AKB security device (this field only applies to Atalla non-AKB devices), this field must be set to the appropriate value based on the key variant that was applied when encrypting the MAC key exchange key.

Transaction Security: Channel Security Configuration
Tab: MAC Information
Field: Key Length of MAC Key

The Key Length of the Exchange Key for the SPDH CSECD records must be set to Double or Triple to utilize SPDH Triple DES using double-length or triple-length keys.

Transaction Security: Channel Security Configuration
Tab: MAC Information
Field: Exchange Key (Header, Part 1-4, Check Digits)

The Exchange Key for the SPDH CSECD records must be modified to contain new double-length or triple-length key value.

Transaction Security: Channel Security Configuration
Tab: MAC Information
Field: Key Length of Inbound Key

The Key Length of the Inbound Key for the SPDH CSECD records must be set to Double or Triple to utilize SPDH Triple DES using double-length or triple-length keys.

Transaction Security: Channel Security Configuration
Tab: MAC Information
Field: Inbound Key (Header, Part 1-4, Check Digits)

The Inbound Key for the SPDH CSECD records must be modified to contain new double-length or triple-length key value.

Refer to the online help provided with the ACI Desktop user interface for detailed descriptions of this window and its fields.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

SPDH devices supporting Triple DES encryption must have new software/hardware configured to use the Triple DES encryption algorithm and double-length or triple-length keys. Additionally the new double-length or triple-length master key must be manually entered at the device and the new working key must be downloaded.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on \$<volume>.BA60LOGM.

Vendor Impacts

ACI Standard POS Message

The following optional field identifiers (FIDs) have been modified in the ACI Standard POS Message.

FID	Picture	Length	Field Name	RQST	RESP
H	PIC X(74)	16-74 bytes	Authentication Key		✓
M	PIC X(74)	16-74 bytes	PIN Communications Key		✓

The following optional download field identifiers (DIDs) have been modified in the ACI Standard POS Message.

DID	Picture	Length	Field Name
g	PIC X(74)	16-74 bytes	PIN Encryption Key
h	PIC X(74)	16-74 bytes	MAC Key

Reference Documents

Not applicable

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Section 7

BASE24-teller

Section 7 discusses the changes and enhancements made to BASE24-teller in release 6.0 version 4, along with the resulting conversion and implementation requirements that existing BASE24-teller users must consider when upgrading from release 6.0 version 3.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24-teller users, along with the associated conversion and implementation requirements that all BASE24-teller users must address.
- Optional enhancements describe the changes that affect only those BASE24-teller users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24-teller users must address to implement the enhancements.

Section 7 must be used in conjunction with Section 4 to identify the changes that BASE24-teller users need to consider when upgrading to release 6.0 version 4.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24-teller users must address in order to upgrade to release 6.0 version 4.

There are no general enhancements inherent to BASE24-teller.

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Optional Enhancements

This section identifies the BASE24-teller enhancements that can be enabled or disabled with release 6.0 version 4, along with the changes, conversion requirements, and implementation requirements associated with each. These enhancements include the following:

There are no optional enhancements inherent to BASE24-teller.

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Section 8

BASE24 Kernel/IEP

Section 8 discusses the changes and enhancements made to BASE24 Kernel/IEP in release 6.0 version 4, along with the resulting conversion and implementation requirements that existing BASE24 Kernel/IEP users must consider when upgrading from release 6.0 version 3.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24 Kernel/IEP users, along with the associated conversion and implementation requirements that all BASE24 Kernel/IEP users must address.
- Optional enhancements describe the changes that affect only those BASE24 Kernel/IEP users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24 users must address to implement the enhancements.

Section 8 must be used in conjunction with Section 4 to identify the changes that BASE24 Kernel/IEP users need to consider when upgrading to release 6.0 version 4.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24 Kernel/IEP users must address in order to upgrade to release 6.0 version 4.

There are no general enhancements inherent to BASE24 Kernel/IEP.

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Optional Enhancements

This section identifies the BASE24 Kernel/IEP enhancements that can be enabled or disabled with release 6.0 version 4, along with the changes, conversion requirements, and implementation requirements associated with each.

There are no optional enhancements inherent to BASE24 Kernel/IEP.

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Section 9

BASE24-telebanking

Section 9 discusses the changes and enhancements made to BASE24-telebanking in release 6.0 version 4, along with the resulting conversion and implementation requirements that existing BASE24-telebanking users must consider when upgrading from release 6.0 version 3.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24-telebanking users, along with the associated conversion and implementation requirements that all BASE24-telebanking users must address.
- Optional enhancements describe the changes that affect only those BASE24-telebanking users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24-telebanking users must address to implement the enhancements.

Section 9 must be used in conjunction with Sections 4 and 8 to identify the changes that BASE24-telebanking users need to consider when upgrading to release 6.0 version 4. Enhanced Telebanking users must also use Section 10, BASE24-billpay, to identify all changes, as Enhanced Telebanking shares modules with BASE24-billpay.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24-telebanking users must address in order to upgrade to release 6.0 version 4.

There are no general enhancements inherent to BASE24-telebanking.

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Optional Enhancements

This section identifies the BASE24-telebanking enhancements that can be enabled or disabled with release 6.0 version 4, along with the changes, conversion requirements, and implementation requirements associated with each.

There are no optional enhancements inherent to BASE24-telebanking.

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Section 10

BASE24-billpay

Section 10 discusses the changes and enhancements made to BASE24-billpay in release 6.0 version 4, along with the resulting conversion and implementation requirements that existing BASE24-billpay users must consider when upgrading from release 6.0 version 3.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24-billpay users, along with the associated conversion and implementation requirements that all BASE24-billpay users must address.
- Optional enhancements describe the changes that affect only those BASE24-billpay users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24-billpay users must address to implement the enhancements.

Section 10 must be used in conjunction with Sections 4, 8 and 9 to identify the changes that BASE24-billpay users need to consider when upgrading to release 6.0 version 4.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24-billpay users must address in order to upgrade to release 6.0 version 4.

There are no general enhancements inherent to BASE24-billpay.

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Optional Enhancements

This section identifies the BASE24-billpay enhancements that can be enabled or disabled with release 6.0 version 4, along with the changes, conversion requirements, and implementation requirements associated with each.

There are no optional enhancements inherent to BASE24-billpay.

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Section 11

BASE24-customer service

Section 11 discusses the changes and enhancements made to BASE24-customer service in release 6.0 version 4, along with the resulting conversion and implementation requirements that existing BASE24-customer service users must consider when upgrading from release 6.0 version 3.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24-customer service users, along with the associated conversion and implementation requirements that all BASE24-customer service users must address.
- Optional enhancements describe the changes that affect only those BASE24-customer service users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24-customer service users must address to implement the enhancements.

Section 11 must be used in conjunction with Sections 4 and 8 to identify the changes that BASE24-customer service users need to consider when upgrading to release 6.0 version 4.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24-customer service users must address in order to upgrade to release 6.0 version 4.

There are no general enhancements inherent to BASE24-customer service.

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Optional Enhancements

This section identifies the BASE24-customer service enhancements that can be enabled or disabled with release 6.0 version 4, along with the changes, conversion requirements, and implementation requirements associated with each.

There are no optional enhancements inherent to BASE24-customer service.

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Section 12

BASE24 Interchange Interfaces

Section 12 discusses the changes and enhancements made to BASE24 Interchange Interfaces in release 6.0 version 4, along with the resulting conversion and implementation requirements that existing BASE24 Interchange Interface users must consider when upgrading from release 6.0 version 3.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24 Interchange Interface users, along with the associated conversion and implementation requirements that all BASE24 Interchange Interface users must address.
- Optional enhancements describe the changes that affect only those BASE24 Interchange Interface users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24 Interchange Interface users must address to implement the enhancements.

Section 12 must be used in conjunction with Section 4 to identify the changes that BASE24 Interchange Interface users need to consider when upgrading to release 6.0 version 4.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24 Interchange Interface users must address in order to upgrade to release 6.0 version 4.

There are no general enhancements inherent to BASE24 Interchange Interfaces.

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Optional Enhancements

This section identifies the BASE24 Interchange Interface enhancements that can be enabled or disabled with release 6.0 version 4, along with the changes, conversion requirements, and implementation requirements associated with each. These enhancements include the following:

- BIC ISO Interchange Interface Enhancements

BASE24 Interchange Interface users can choose whether or not to implement each of the above enhancements. Users who plan to implement a particular enhancement must address the corresponding release 6.0 version 4 changes, conversion requirements, and implementation requirements. Users who do not plan to implement a particular enhancement can simply review the corresponding release 6.0 version 4 changes, conversion requirements, and implementation requirements for future reference.

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BIC ISO Interchange Interface Enhancements

In the Republic of Ireland, electronic transactions (both ATM and POS) are passed between the acquiring and issuing banks using a message format based closely on the ISO 8583 (1987) Standard. The format is very similar to the BASE24 external message, and the BASE24 users amongst the Irish banks use the BIC ISO Interchange Interface module to support this message processing.

This section describes the enhancements to the BIC ISO Interchange Interface to support the Irish processing requirements, e.g. for the Irish “Chip and PIN” initiative. Although the changes have been applied predominantly to support the requirements of the Irish market, other BASE24 users of the BIC ISO Interchange Interface can take advantage of the enhanced functionality.

The following enhancements to the BIC ISO Interchange Interface have been identified:

- Enhance response code mappings in BIC ISO for EMV-specific failure conditions.
- Allow BASE24-originated reconciliation messages to be logged to the Interchange Log File (ILF) by BIC ISO.
- Increase flexibility in the setting of institution ids in BIC ISO.
- Rationalize currency conversion processing in BIC ISO.

These enhancements are described in detail below.

Enhance Response Code Mappings For EMV-Specific Failure Conditions

When EMV support was implemented in Release 6.0, new response codes were added to both BASE24-atm and BASE24-pos to identify EMV-specific failure conditions, e.g. bad ARQC. When returning responses to acquiring Host or Interchange systems, BIC ISO currently maps these new codes to generic non-approved ISO response codes, e.g. '05' (do not honor). This means that the acquiring system often cannot identify the specific reason for an EMV transaction being declined.

New unique ISO response codes (from the range of ISO values “reserved for private use”) will be used to identify specific EMV failure conditions, e.g. bad ARQC. The BIC ISO module on the issuer’s system will map the EMV-specific BASE24 response codes to these new ISO codes, and the BIC ISO module on the acquirer’s system will map the new ISO codes to the corresponding EMV-specific BASE24 response codes. Similarly, special processing will be implemented for PIN Management transactions to ensure that a BASE24-atm response code of ‘71’ (contact card issuer) is used on the acquirer’s system, if a more specific failure condition does not apply.

The change will be configurable, because this enhancement will be controlled by a

new LCONF parameter, and the processing can therefore be disabled.

Allow BASE24-Originated Reconciliation Messages To Be Logged To The ILF

The BIC ISO module may receive reconciliation totals from an interchange and/or send reconciliation totals to an interchange in 05xx messages. This processing is described in more detail in section 6 of the BIC ISO Interface manual. For 05xx messages received from the interchange, BIC ISO logs the totals to the ILF, but for 05xx messages sent to the interchange, BIC ISO does not log the totals to the ILF.

The BIC ISO reconciliation message processing will be enhanced to write reconciliation totals to the ILF under the following conditions:

- An 0510 reconciliation response message is received from the interchange;
- An 0520 reconciliation advice message is sent to the interchange.

The impact of this enhancement on existing BASE24 customers will be minimal, i.e. a maximum of two additional records per day written to the ILF, so the implementation of this functionality will not be configurable.

Increase Flexibility In The Setting Of Institution Ids

In the 1987 version of the ISO 8583 Standard on which the BIC ISO message protocol is based), there are effectively three institution ids defined - the acquiring institution id, the forwarding institution id, and the receiving institution id. In the BIC ISO interface, these ids are mapped between the Standard External Message (SEM) and the BASE24-atm Standard Internal Message (STM) or the BASE24-pos Standard Internal Message (PSTM).

Some banks in Northern Ireland are members of both the U.K. national debit card scheme (Switch) and the Irish national debit card scheme (Laser). These banks require an override for one or more of these institution ids because a bank's institution id is different for the Laser and Switch card schemes.

The Base segment of the Enhanced Interchange Configuration File (ICFE) already contains fields that can be used as an override for these institution ids (SWI-ID and DFLT-ACQ-ID-NUM), neither of which is currently used by BIC ISO. A new Logical Network Configuration File (LCONF) parameter will control the use of these fields. When formatting messages to be sent to the interchange, BIC ISO will use the override institution id value for the appropriate field, if configured to do so by the new LCONF parameter. When formatting messages received from the interchange, BIC ISO will never use the override institution id value for the forwarding institution id, because either the ATM Authorisation module or the POS Router/Authorisation module will assign a default value. However, BIC ISO will use the override institution id value for the acquiring institution id or the receiving institution id, but only if the corresponding field has not already been assigned a value in the external or internal

message.

The change will be configurable, because this enhancement will be controlled by the new LCONF parameter, and the processing can therefore be disabled.

Rationalise Currency Conversion Processing

The BIC ISO module may perform currency conversion on amount fields in messages received from an interchange and/or messages sent to an interchange. This processing is performed only if the BASE24 Multiple Currencies product is not configured. This processing is described in more detail in section 1 of the BIC ISO Interface manual.

Some of this currency conversion logic is incomplete, because it is not possible to pass certain amounts in the BASE24 Standard External Message (SEM) in more than one currency. Changes to the SEM layout to include additional amount fields will ensure that full currency data can be passed between BASE24 and an interchange.

Support will also be implemented for the Original Currency 6.0 Token, which is already added by a number of BASE24 interfaces to international interchanges, e.g. EPS-Net, VisaNet. On an issuer system, the BIC ISO module will add the Original Currency 6.0 Token (containing fields relating to the currency of the transaction) if the co-network has converted the transaction amount(s) to the settlement currency. The Original Currency 6.0 Token can then be logged to the ILF and the (P)TLF, and used in settlement processing with the co-network

The change will be configurable, because this enhancement will be controlled by a new LCONF parameter, and the processing can therefore be disabled.

Software Impacts

All BASE24 customers:

- Changes to the BA60DDL.DDLGISO file to include additional data elements in existing message fields.

All BASE24 customers who have licensed the BIC ISO Interchange Interface:

- Inline changes to the SW60BABI.BICIDDL file to increase flexibility in the setting of institution ids.
- Inline changes to the SW60BABI.BABICIG and SW60BABI.BABICIS files to allow BASE24-originated reconciliation messages to be logged to the ILF.

All BASE24-atm customers who have licensed the BIC ISO Interchange Interface:

- Inline changes to the SW60ATBI.ATBICIS file to increase flexibility in the setting of institution ids, to rationalize currency conversion processing, and to enhance response code mappings for EMV-specific failure conditions.

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD07.SCNBICI.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement.

ENHANCED-EMV-RC-MAPPING

This new LCONF parameter will indicate whether response code mappings for EMV-specific failure conditions should be enhanced, and may contain one of four values:

- 0 = Do not perform enhanced EMV response code mapping.
- 1 = Perform enhanced EMV response code mapping for ATM transactions only.
- 2 = Perform enhanced EMV response code mapping for POS transactions only.
- 3 = Perform enhanced EMV response code mapping for ATM and POS transactions.

If the parameter is missing or invalid, then the default value of '0' will be used.

OVERRD-INST-ID-FLDS

This new LCONF parameter will indicate whether the flexibility in the setting of Institution Ids should be increased, and may contain one of eight values:

- 0 = Do not override institution id values.
- 1 = Override acquiring institution id only.
- 2 = Override forwarding institution id only.
- 3 = Override acquiring and forwarding institution id.
- 4 = Override receiving institution id only.
- 5 = Override acquiring and receiving institution id.
- 6 = Override forwarding and receiving institution id.
- 7 = Override acquiring, forwarding and receiving institution id.

If the parameter is missing or invalid, then the default value of '0' will be used.

ENHANCED-CRNCY-CONV

This new LCONF parameter will indicate whether enhanced currency conversion processing should be performed, and may contain one of four values:

0 = Do not perform enhanced currency conversion processing.

1 = Perform enhanced currency conversion processing for ATM transactions only.

2 = Perform enhanced currency conversion processing for POS transactions only.

3 = Perform enhanced currency conversion processing for ATM and POS transactions.

If the parameter is missing or invalid, then the default value of '0' will be used.

Database Configuration

BASE24 Enhanced Interchange Configuration File (ICFE)

To implement support for increased flexibility in the setting of Institution Ids, the following ICFE fields must be set as specified:

Field Name: SWITCH ID
Screen: ICFE Screen 1
DDL Name: ICFE.SWI-ID

This field must be set to the ID of the BASE24 system.

Field Name: DEFAULT ACQUIRER ID NUM
Screen: ICFE Screen 1
DDL Name: ICFE.DFLT-ACQ-ID-NUM

This field must be set to the ID of co-network.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

BASE24 users implementing this enhancement will incur some impact to the SEM.

Standard External Message

The following SEM fields must be enhanced to define an additional amount:

- Additional Amounts (field P-54);
- ATM Deposit Credit Amount (field S-123).

In addition, BIC ISO does not currently use the SETL-AMT sub-field in the SEM

Replacement Amounts (field S-95).

BASE24 users implementing enhanced EMV response code mapping must ensure that the co-network supports the new ISO response codes in field P-39.

User/Operational Impacts

Disk Impacts

For the enhancement to allow BASE24-originated reconciliation messages to be logged to the BIC ISO ILF, the maximum impact on existing BIC ISO customers is that two additional records per day will be logged to the ILF.

The BIC ISO ILF requires additional disk space if the Original Currency Token is configured to be included in the transaction records logged to the file. The total impact varies for each BASE24 user depending on the transaction volume in the BASE24 system.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Section 13

BASE24 Value-Added Products

Section 13 discusses the changes and enhancements made to the BASE24 value-added products in release 6.0 version 4, along with the resulting conversion and implementation requirements that existing users of these products must consider when upgrading from release 6.0 version 3.

The value-added products discussed in this section include the following:

- BASE24-InfoBase
- BASE24 Retail Products
- BASE24-card
- ACI Host Link
- ACI Plastic Card Control System
- ACI Claims Manager
- ACI Proactive Risk Manager
- ACI Settlement Manager
- ACI Payments Manager
- ACI Smart Chip Manager

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24 value-added product users, along with the associated conversion and implementation requirements that all BASE24 value-added product users must address.
- Optional enhancements describe the changes that affect only those BASE24 value-added product users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24 value-added product users must address to implement the enhancements.

Section 13 must be used in conjunction with Sections 4 through 12 to identify the changes that BASE24 value-added product users need to consider when upgrading to release 6.0 version 4.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24 value-added product users must address in order to upgrade to release 6.0 version 4.

Changes and requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the general enhancements described here.

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BASE24-InfoBase

This section describes the enhancements that impact the BASE24-InfoBase value-added product and the conversion and implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements affecting BASE24-InfoBase users.

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BASE24 Retail Products

This section describes the enhancements that impact the value-added BASE24 Retail Products and the conversion and implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements affecting the users of the BASE24 Retail Products.

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BASE24-card

This section describes the enhancements that impact the BASE24-card value-added product and the conversion and implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements affecting BASE24-card users.

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ACI Host Link

This section describes the enhancements that impact the ACI Host Link value-added product and the conversion and implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements affecting the ACI Host Link users.

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ACI Plastic Card Control System

This section describes the enhancements that impact the ACI Plastic Card Control System value-added product and the conversion and implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements affecting the ACI Plastic Card Control System users.

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ACI Claims Manager

This section describes the enhancements that impact the ACI Claims Manager value-added product and the conversion and implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements affecting the ACI Claims Manager users.

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ACI Proactive Risk Manager

This section describes the enhancements that impact the ACI Proactive Risk Manager value-added product and the conversion and implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements affecting the ACI Proactive Risk Manager users.

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ACI Settlement Manager

This section describes the enhancements that impact the ACI Settlement Manager value-added product and the conversion and implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements affecting the ACI Settlement Manager users.

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ACI Payments Manager

This section describes the enhancements that impact the ACI Payments Manager value-added product and the conversion and implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements affecting the ACI Payments Manager users.

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ACI Smart Chip Manager

This section describes the enhancements that impact the ACI Smart Chip Manager value-added product and the conversion and implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements affecting the ACI Smart Chip Manager users.

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Optional Enhancements

This section identifies the BASE24 enhancements that can be enabled or disabled with release 6.0 version 4 that have an impact on the following BASE24 value-added products, along with the changes, conversion requirements, and implementation requirements associated with each.

- BASE24-InfoBase
- BASE24 Retail Products
- BASE24-card
- ACI Host Link
- ACI Plastic Card Control System
- ACI Claims Manager
- ACI Proactive Risk Manager
- ACI Payments Manager
- ACI Settlement Manager
- ACI Smart Chip Manager

BASE24 value-added product users can choose whether or not to implement the enhancements described in this section. Users who plan to implement a particular enhancement must address the corresponding release 6.0 version 4 changes, conversion requirements, and implementation requirements. Users who do not plan to implement a particular enhancement can simply review the corresponding release 6.0 version 4 changes, conversion requirements, and implementation requirements for future reference.

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BASE24-InfoBase

This section describes the enhancements that impact the BASE24-InfoBase value-added product.

There are no optional enhancements inherent to BASE24-InfoBase.

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BASE24 Retail Products

This section describes the enhancements that impact the value-added BASE24 Retail Products.

There are no optional enhancements inherent to BASE24 Retail Products.

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BASE24-card

This section describes the enhancements that impact the BASE24-card value-added product.

There are no optional enhancements inherent to BASE24-card.

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ACI Host Link

This section describes the enhancements that impact the ACI Host Link value-added product.

There are no optional enhancements inherent to ACI Host Link.

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ACI Plastic Card Control System

This section describes the enhancements that impact the ACI Plastic Card Control System value-added product.

There are no optional enhancements inherent to ACI Plastic Card Control System.

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ACI Claims Manager

This section describes the enhancements that impact the ACI Claims Manager value-added product.

There are no optional enhancements inherent to ACI Claims Manager.

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ACI Proactive Risk Manager

This section describes the enhancements that impact the ACI Proactive Risk Manager value-added product.

There are no optional enhancements inherent to ACI Proactive Risk Manager.

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ACI Payments Manager

This section describes the enhancements that impact the ACI Payments Manager value-added product.

There are no optional enhancements inherent to ACI Payments Manager.

ACI Worldwide Inc.

ACI Settlement Manager

This section describes the enhancements that impact the ACI Settlement Manager value-added product.

There are no optional enhancements inherent to ACI Settlement Manager.

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ACI Smart Chip Manager

This section describes the enhancements that impact the ACI Smart Chip Manager value-added product.

There are no optional enhancements inherent to the ACI Smart Chip Manager.

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Appendix A

Logical Network Configuration File Changes

Appendix A contains the Logical Network Configuration File (LCONF) changes.

This information includes assigns and params that have been added, deleted or modified with release 6.0 version 4 along with their associated page numbers. Refer to the specified page numbers for explanations of how these assigns and params can or must be set.

Refer to the ***BASE24 Logical Network Configuration File (LCONF) Manual*** for instructions on adding them to the LCONF.

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ASSIGNS

The following assigns were added or modified for release 6.0 version 4. Refer to the specified page numbers for explanations of how these assigns can or must be set.

Dual Site Support:

POS-PTLF	4-23
	6-34
REMOTE-LCONF	5-32
REMOTE-PMON	5-32
TLF	4-22
	5-32

Secure Internet Validation:

POS-EAF	6-62
POS-EAF-PREV	6-62

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PARAMS

The following params were added or modified for release 6.0 version 4. Refer to the specified page numbers for explanations of how these params can or must be set.

Automated Key Distribution System:

ATM-DH-AKDS-TSS-TIMER	5-20
ATM-DH-1000-AKDS-SERIAL-NUM-VERFY	5-20

BIC ISO Interchange Interface Enhancements:

ENHANCED-EMV-RC-MAPPING	12-10
OVRRD-INST-ID-FLDS	12-10
ENHANCED-CRNCY-CONV	12-11

Dual Site Support:

BROADCAST-DELAY	4-23
	5-33
	6-35
BROADCAST-NOTIFY	4-23
	5-33
	6-35
DUAL-SITE-DISPLAY	5-33
	6-35
DUAL-SITE-MODE	4-23
	5-33
	6-34

Forced Cutover Audit:

ATM-FCA-CUTOVER-AUDIT	5-48
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Multiple Currency Dispense Exchange Rate Notification:

ATM-DH-XCHG-RATE-NTFY	5-64
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POS General Changes:

POS-DH-ISO-RESP-CDE-FRMT	6-5
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Secure Internet Validation:

POS-EAF-NOTIFY-INTERVAL	6-63
POS-EAF-NOTIFY-PERCENTAGE	6-63
POS-SETL-EAF-NOTIFYidentifier	6-63
POS-SETL-EAF-RETENTION-PERIOD	6-64

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Appendix B

Host/CO-Network Impacts

Appendix B contains a list of release 6.0 version 4 enhancements that have host/co-network impacts, along with their associated page numbers. Refer to the specified page numbers for more information about these enhancements.

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Enhancements

The following enhancements have host/co-network impacts. Refer to the specified page numbers for more information about these enhancements.

Acquirer Surcharging	5-17
BIC ISO Interchange Interface Enhancements	12-12
Dual Site Support.....	4-24
	5-35
	6-37
EMV M/Chip 4 Support	4-35
Enhanced Expiration Date Checking	4-42
General CRT Authorization Enhancements	6-15
Interactive Response Notification Fee Processing and Negative Surcharge Support	5-60
POS Balance Inquiry Service.....	6-58
Secure Internet Validation.....	6-69
SPDH Additional Data Fields	6-21
TSS Key Index Support	4-46

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Appendix C

CUSTMACS File Changes

Appendix C contains a list of release 6.0 version 4 enhancements that have CUSTMACS changes, along with their associated page numbers. Refer to the specified page numbers for more information about these enhancements.

BASE24 users should review these changes to determine how the CUSTMACS file should be set for their BASE24 environment before running MAKE to compile their system. More information on all MAKE flags used within BASE24 can be found in the FLGSDOC file located on the BA60MAKE subvolume delivered with BASE24 Release 6.0 Version 4 code.

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Enhancements

The following enhancements have CUSTMACS file changes:

Automated Key Distribution System	5-20
Base General Changes.....	4-8
Dynamic Key Management.....	5-37
Forced Cutover Audit.....	5-47
Multiple Currency Dispense	5-63
SPDH Enhanced Multi-Threading	6-88
TSS Key Index Support	4-44

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Appendix D

Re-Sequenced Files and New Files

Appendix D consists of the following tables:

- A table illustrating the new files for the BASE24 General Changes.
- A table illustrating the new files for the optional BASE24 Enhanced Expiration Date Checking enhancement.
- A table illustrating the new files for the optional BASE24-atm Acquirer Surcharging enhancement.
- A table illustrating the new files for the optional BASE24-atm Automated Key Distribution System enhancement.
- A table illustrating the new files for the optional BASE24-atm Dynamic Key Management enhancement.
- A table illustrating the new files for the optional BASE24-atm Forced Cutover Audit enhancement.
- A table illustrating the new files for the optional BASE24-atm Fujitsu 7000 Device Handler.
- A table illustrating the new files for the optional BASE24-pos POS Balance Inquiry Service enhancement.
- A table illustrating the new files for the optional BASE24-pos Secure Internet Validation enhancement.
- A table illustrating the new files for the optional BASE24-pos Self-Service Banking Check enhancement.
- A table illustrating the new files for the optional BASE24-pos SPDH DUKPT Support via TSS enhancement.
- A table illustrating the new files for the optional BASE24-pos SPDH Enhanced Multi-Threading enhancement.
- A table illustrating the new files for the optional BASE24-pos SWITCH Auth Issuer Functions enhancement.

No files were re-sequenced in release 6.0 version 4.

New Files for the BASE24 General Changes

General Changes	
New File Location and File Name	File Language Type
at60aft.cobrcpt	COBOL Tables
at60rpt.arcoutls	COBOL
at60rpt.cobatrpt	COBOL
ba60aft.cobtkn	COBOL Tables

New Files for the Optional BASE24 Enhanced Expiration Date Checking
Enhancement Table

Enhanced Expiration Date Checking New File Location and File Name	File Language Type
ba60uc07.aaread07	TGAL
ba60uc07.bauc07fm	MAKE
ba60uc07.bauc07m	MAKE
ba60uc07.bauc07mm	MAKE
ba60uc07.cnvcpf	Generated during a compile
ba60uc07.cnvcpfe	Generated during a compile
ba60uc07.cnvcpfm	MAKE
ba60uc07.cnvcpfo	Generated during a compile
ba60uc07.cnvcpfr	TACL
ba60uc07.cnvcpfs	Generated during a compile

New Files for the Optional BASE24-atm Acquirer Surcharging Enhancement Table

Acquirer Surcharging New File Location and File Name	File Language Type
ba60aft.rqtsrfss	ACIAID
ba60aft.rqtsrfxm	MAKE
ba60aft.rqtsrfxo	Generated during a compile
ba60aft.rqtsrfxs	SCOBOL
ba60aft.svtsrfte	Generated during a compile
ba60aft.svtsrftg	TAL
ba60aft.svtsrftm	MAKE
ba60aft.svtsrfto	Generated during a compile
ba60aft.svtsrfts	TAL
ba60ddl.ddlftsr	DDL
ba60srch.cobsurf	COBOL
ba60uc07.aaread07	TGAL
ba60uc07.bauc07fm	MAKE
ba60uc07.bauc07m	MAKE
ba60uc07.bauc07mm	MAKE
ba60uc07.cnvsurf	Generated during a compile
ba60uc07.cnvsurfe	Generated during a compile
ba60uc07.cnvsurfm	MAKE
ba60uc07.cnvsurfo	Generated during a compile
ba60uc07.cnvsurfr	TACL
ba60uc07.cnvsurfs	Generated during a compile

New Files for the Optional BASE24-atm Automated Key Distribution System
Enhancement Table

Automated Key Distribution System New File Location and File Name	File Language Type
at60a100.ata100fm	MAKE
at60a100.ata100mm	MAKE
at60a100.c100akde	Generated during a compile
at60a100.c100akdg	TAL
at60a100.c100akdm	MAKE
at60a100.c100akdo	Generated during a compile
at60a100.c100akds	TAL

New Files for the Optional BASE24-atm Dynamic Key Management Enhancement Table

Dynamic Key Management New File Location and File Name	File Language Type
Diebold 10xx/478x Device Handler	
at60k100.atk100fm	MAKE
at60k100.atk100mm	MAKE
at60k100.c100dkme	Generated during a compile
at60k100.c100dkmg	TAL
at60k100.c100dkmm	MAKE
at60k100.c100dkmo	Generated during a compile
at60k100.c100dkms	TAL
NCR 5xxx Device Handler	
at60kn50.atkn50fm	MAKE
at60kn50.atkn50mm	MAKE
at60kn50.n50dkme	Generated during a compile
at60kn50.n50dkmg	TAL
at60kn50.n50dkmm	MAKE
at60kn50.n50dkmo	Generated during a compile
at60kn50n50dkms	TAL
Triton Device Handler	
at60ktrt.atkn50fm	MAKE
at60ktrt.atkn50mm	MAKE
at60ktrt.tritdkme	Generated during a compile
at60ktrt.tritdkmg	TAL
at60ktrt.tritdkmm	MAKE
at60ktrt.tritdkmo	Generated during a compile
at60ktrt.tritdkms	TAL

New Files for the Optional BASE24-atm Forced Cutover Audit Enhancement Table

Forced Cutover Audit New File Location and File Name	File Language Type
Diebold 10xx/478x Device Handler	
at60f100.atf100fm	MAKE
at60f100.atf100mm	MAKE
at60f100.c100fcae	Generated during a compile
at60f100.c100fcag	TAL
at60f100.c100fcam	MAKE
at60f100.c100fcao	Generated during a compile
at60f100.c100fcas	TAL
NCR 5xxx Device Handler	
at60fn50.atfn50fm	MAKE
at60fn50.atfn50mm	MAKE
at60fn50.n50fcae	Generated during a compile
at60fn50.n50fcag	TAL
at60fn50.n50fcam	MAKE
at60fn50.n50fcao	Generated during a compile
at60fn50n50fcas	TAL
Triton Device Handler	
at60ftrt.atfn50fm	MAKE
at60ftrt.atfn50mm	MAKE
at60ftrt.tritfcae	Generated during a compile
at60ftrt.tritfcag	TAL
at60ftrt.tritfcam	MAKE
at60ftrt.tritfcao	Generated during a compile
at60ftrt.tritfcas	TAL

New Files for the Optional BASE24-atm Fujitsu 7000 Device Handler Table

Fujitsu 7000	
New File Location and File Name	File Language Type
at60fuj7.atfuj7fm	MAKE
at60fuj7.atfuj7m	MAKE
at60fuj7.atfuj7mm	MAKE
at60fuj7.cnfgfuj7	Config File
at60fuj7.cnfgdoc	DOC
at60fuj7.entfuj7	Entry Program
at60fuj7.fuj7	Generated during a compile
at60fuj7.fuj7cob	Generated during a compile
at60fuj7.fuj7csee	Generated during a compile
at60fuj7.fuj7cses	TAL
at60fuj7.fuj7ddl	MAKE
at60fuj7.fuj7ddls	DDL
at60fuj7.fuj7e	Generated during a compile
at60fuj7.fuj7fup	Generated during a compile
at60fuj7.fuj7g	TAL
at60fuj7.fuj7m	MAKE
at60fuj7.fuj7man	DOC
at60fuj7.fuj7nams	TAL
at60fuj7.fuj7o	Generated during a compile
at60fuj7.fuj7omxe	Generated during a compile
at60fuj7.fuj7omxg	TAL
at60fuj7.fuj7omxm	MAKE
at60fuj7.fuj7omxo	Generated during a compile
at60fuj7.fuj7omxs	TAL
at60fuj7.fuj7s	TAL
at60fuj7.fuj7tal	Generated during a compile
at60fuj7.rqdfuim	MAKE
at60fuj7.rqdfuji	Generated during a compile
at60fuj7.rqdfujis	SCOBOL
at60fuj7.rqtufuj7m	MAKE
at60fuj7.rqtufuj7o	Generated during a compile
at60fuj7.rqtufuj7s	SCOBOL
at60fuj7.scrnfuji	ACIAID
at60fuj7.scrnfuj7	ACIAID
at60fuj7.svdfuj7m	MAKE

Fujitsu 7000	
New File Location and File Name	File Language Type
at60fuj7.svdfuj7o	Generated during a compile
at60fuj7.svdfuj7s	TAL
at60fuj7.svfuj7te	Generated during a compile
at60fuj7.svfuj7tg	TAL
at60fuj7.svfuj7tm	MAKE
at60fuj7.svfuj7to	Generated during a compile
at60fuj7.svfuj7ts	TAL

New Files for the Optional BASE24-pos POS Balance Inquiry Service Enhancement Table

POS Balance Inquiry Service New File Location and File Name	File Language Type
ba60uc07.aaread07	TGAL
ba60uc07.bauc07fm	MAKE
ba60uc07.bauc07m	MAKE
ba60uc07.bauc07mm	MAKE
ba60uc07.cnvcpf	Generated during a compile
ba60uc07.cnvcpfe	Generated during a compile
ba60uc07.cnvcpfm	MAKE
ba60uc07.cnvcpfo	Generated during a compile
ba60uc07.cnvcpfr	TACL
ba60uc07.cnvcpfs	Generated during a compile

New Files for the Optional BASE24-pos Secure Internet Validation Enhancement Table

Secure Internet Validation New File Location and File Name	File Language Type
ba60src.secutild	TAL
ba60src.secutilo	Generated during a compile
Ba60src.secutilm	MAKE
ba60uc07.aaread07	TGAL
ba60uc07.bauc07fm	MAKE
ba60uc07.bauc07m	MAKE
ba60uc07.bauc07mm	MAKE
ba60uc07.cnvcpf	Generated during a compile
ba60uc07.cnvcpfe	Generated during a compile
ba60uc07.cnvcpfm	MAKE
ba60uc07.cnvcpfo	Generated during a compile
ba60uc07.cnvcpfr	TACL
ba60uc07.cnvcpfs	Generated during a compile
ba60vsrc.bavsrcfm	MAKE
ba60vsrc.bavsrcm	MAKE
ba60vsrc.basrcmm	MAKE
ba60vsrc.secsive	Generated during a compile
ba60vsrc.secsivm	MAKE
ba60vsrc.secsivo	Generated during a compile
ba60vsrc.secsivs	TAL
ps60ddl.custcnst	DDL
ps60ddl.ddlfeaf	DDL
ps60rtau.rtaud	TAL
ps60vrta.psvrtafm	MAKE
ps60vrta.psvrtamm	MAKE
ps60vrta.rtausiv	Generated during a compile
ps60vrta.rtausive	Generated during a compile
ps60vrta.rtausivg	TAL
ps60vrta.rtausivm	MAKE
ps60vrta.rtausivo	Generated during a compile
ps60vrta.rtausivs	TAL

New Files for the Optional BASE24-atm Self-Service Banking Check Enhancement Table

Self-Service Banking Check New File Location and File Name	File Language Type
ba60uc07.aaread07	TGAL
ba60uc07.bauc07fm	MAKE
ba60uc07.bauc07m	MAKE
ba60uc07.bauc07mm	MAKE
ba60uc07.cnvcpf	Generated during a compile
ba60uc07.cnvcpfe	Generated during a compile
ba60uc07.cnvcpfm	MAKE
ba60uc07.cnvcpfo	Generated during a compile
ba60uc07.cnvcpfr	TACL
ba60uc07.cnvcpfs	Generated during a compile

New Files for the Optional BASE24-pos SPDH DUKPT Support via TSS
Enhancement Table

SPDH DUKPT Support via TSS	
New File Location and File Name	File Language Type
ba60uc05.tsskeydm	MAKE
ba60uc05.tsskeydr	TACL
ba60uc05.tsskeyds	TAL

New Files for the Optional BASE24-pos SPDH Enhanced Multi-Threading
Enhancement Table

SPDH Enhanced Multi-Threading New File Location and File Name	File Language Type
ps60mspd.psmspdfm	MAKE
ps60mspd.psmspdm	MAKE
ps60mspd.psmspdmm	MAKE
ps60mspd.spdhmte	Generated during a compile
ps60mspd.spdhmtg	TAL
ps60mspd.spdhmtm	MAKE
ps60mspd.spdhmto	Generated during a compile
ps60mspd.spdhmts	TAL

New Files for the Optional BASE24-pos SWITCH Auth Issuer Functions
Enhancement Table

SWITCH Auth Issuer Functions	
New File Location and File Name	File Language Type
ba60uc07.aaread07	TGAL
ba60uc07.bauc07fm	MAKE
ba60uc07.bauc07m	MAKE
ba60uc07.bauc07mm	MAKE
ba60uc07.cnvcpf	Generated during a compile
ba60uc07.cnvcpfe	Generated during a compile
ba60uc07.cnvcpfm	MAKE
ba60uc07.cnvcpfo	Generated during a compile
ba60uc07.cnvcpfr	TACL
ba60uc07.cnvcpfs	Generated during a compile

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Appendix E

Moved, Renamed, Deleted Files

Appendix E contains a list of files that were Moved, Renamed, or Deleted in BASE24 Release 6.0 Version 4.

No files were moved, renamed or deleted in release 6.0 version 4.

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Appendix F

Regional Specific Information

Appendix F contains the information that is required to implement the changes for release 6.0 version 4 that are specific to each region.

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